

INFORMATION AND COMMUNICATION TECHNOLOGY

S5

STUDENT BOOK

EXPERIMENTAL VERSION

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FOREWORD

Dear Student,

Rwanda Basic Education Board is honored to present to you this ICT Student's Book for Senior five which serves as a guide to competence-based teaching and learning to ensure consistency and coherence in the learning of ICT. The Rwandan educational philosophy is to ensure that you achieve full potential at every level of education which will prepare you to be well integrated in society and exploit employment opportunities.

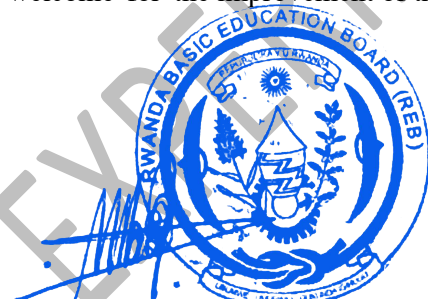
The government of Rwanda emphasizes the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn and the competences you acquire. Those factors include the instructional materials available among others. In this book, special attention was paid to the activities that facilitate the learning process in which you can develop your ideas and make new discoveries during concrete activities carried out individually or with peers.

In competence-based curriculum, learning is considered as a process of active building by the learner where concepts are mainly introduced by an activity, a situation or a scenario that helps the learner to construct knowledge, develop skills and acquire positive attitudes and values.

For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of knowledge and skills;
- Share relevant information with other learners through presentations, discussions, group work and other active learning techniques such as field visit, laboratory work, investigation, role play, case studies and research in the library, from the Internet or from your community;
- Participate and take responsibility for your own learning;
- Draw conclusions based on the findings from the learning activities.

I wish to sincerely extend my appreciation to REB staff who organized the editing process of this book. Special gratitude goes to the lecturers, teachers, illustrators and designers who diligently worked to successful completion of this book. Any comment or contribution would be welcome for the improvement of this textbook for the next edition.



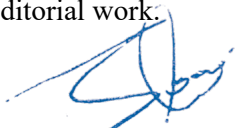
Dr. MBARUSHIMANA Nelson
Director General, REB

ACKNOWLEDGEMENT

I wish to express my appreciation to all the people who played a major role in editing process of ICT Student's Book for Senior Five. It would not have been successful without their active participation.

Special thanks are given to those who gave their time to read and refine this textbook to meet the needs of competence-based curriculum. I owe gratitude to different Universities and secondary schools in Rwanda that allowed their staff to work with REB to edit this book. I therefore, wish to extend my sincere gratitude to lecturers, teachers, illustrators, designers and all other individuals whose efforts in one way or the other contributed to the completion of this edition.

Finally, my word of gratitude goes to the Rwanda Basic Education Board staff particularly those from Curriculum, Teaching and Learning Resources Department who were involved in the whole process of editorial work.



Joan MURUNGI,

Head of Department of Curriculum, Teaching and Learning Resources

Contents

FOREWORD	i
ACKNOWLEDGEMENT	i
UNIT 1: GOOGLE SHEET	1
1.0 Introduction	1
1.1 Creating and sharing a Google Sheet link	2
1.1.1 Logging into Google workspace	2
1.1.2 Naming a sheet	4
1.1.3 Google Sheets permission.....	4
1.2 Text formatting	6
1.2.1 Bold, italics, and underline.....	7
1.2.2 Font size, color, alignment.....	7
1.2.3 Cell formatting feature	8
1.3 Calculations.....	10
1.3.1 SUM function	11
1.3.2 AVERAGE function.....	11
1.3.3 IF function	12
1.4 Creating charts	13
1.4.1 Formatting charts.....	14
UNIT 2: DATABASE QUERIES	21
2.0 Introduction	22
2.1 Introduction to SQL.....	22
2.1.1 Database queries	23
2.1.2 MySQL installation	23
2.1.3 SQL languages	27
2.2 Data Definition Language.....	28
2.2.1 Create.....	28
2.2.2 Alter.....	31
2.2.3 Drop	33
2.3 Data Manipulation Language	34
2.3.1. Operation	35
2.3.2. Inserting data into table.....	35
2.3.3. Selecting data from the table	36

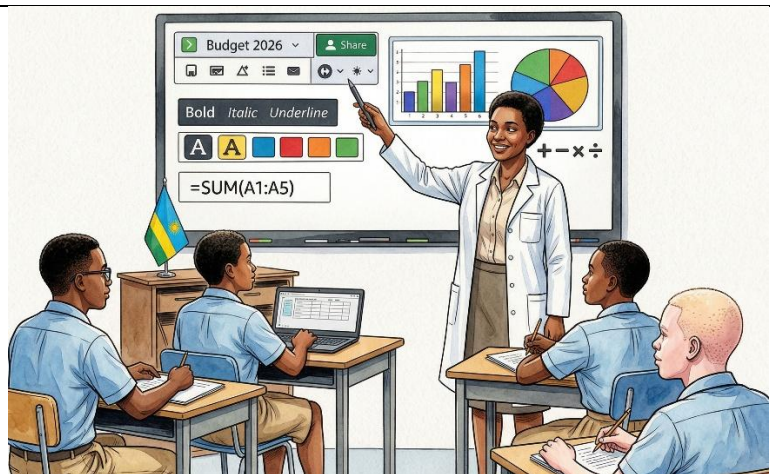
2.4 Updating and deleting data in a table.....	38
2.4.1 Updating data with UPDATE	38
2.4.2 Deleting data with DELETE.....	39
2.5 SQL constraints	41
2.5.1 Primary key	42
2.5.2 NOT NULL.....	42
2.5.3 Foreign key.....	42
2.6 Aggregate Functions	43
2.6.1 COUNT function	44
2.6.2 SUM function	45
2.6.3 AVG function	45
2.6.4 MAX function	46
2.6.5 MIN function	46
2.7 Filtering and sorting data	48
2.7. 1 Filtering data	48
2.7.2 Sorting data	49
UNIT 3: PYTHON PROGRAMMING II	54
3.0 Introduction	54
3.1. Functions in Python	55
3.1.1. Creating and calling a function	56
3.1.2. Local and global variables	58
3.2. Arrays in Python.....	60
3.2.1 Importance of list.....	60
3.2.2 Creating and accessing lists.....	61
3.2.3. Modifying list elements.....	62
3.2.4 Iterating through lists using loops	63
3.2.5. Common list methods.....	64
3.3 Introduction to Object-Oriented Programming (OOP)	66
3.3.1 Concept of object and class	67
3.3.2 Creating a class in Python	68
3.3.3 Creating and using objects.....	68
UNIT 4: MACHINE LEARNING	72
4.0 Introduction	73

4.1	Data fundamentals.....	74
4.1.1	Data, information, and knowledge	75
4.1.2	Data types and structures	75
4.1.3	Data formats	76
4.1.4	Data sources.....	77
4.1.5	Characteristics of quality data	77
4.2	Data collection, preparation, and cleaning	79
4.2.1.	Methods of data acquisition	79
4.2.2.	Data preprocessing techniques.....	80
4.2.3	Handling missing data	81
4.2.4	Data ethics in collection and storage	81
4.3	Data analysis and representation	82
4.3.1	Data organization in spreadsheets and databases	82
4.3.2	Data visualization techniques	83
4.3.3	Descriptive and inferential statistics.....	84
4.4	Machine Learning foundation.....	85
4.4.2	AI, Machine Learning, and Deep Learning	85
4.4.3	Types of ML algorithms.....	86
4.4.4	Machine Learning applications	88
4.5	ML workflow and model building	89
4.5.1	ML pipeline.....	90
4.5.2	Datasets	90
4.5.3	Model evaluation metrics	91
4.5.4	Concept of overfitting and underfitting.....	92
4.6	Tools for Machine Learning	93
4.6.1	Google Teachable Machine.....	94
4.6.2.	Scratch ML extensions	97
4.6.3.	Python based notebooks	98
4.7.	End unit assessment	99
4.7.1.	Project activity	99
4.7.2.	Assessment questions.....	100
UNIT 5: WEB DATABASE CONNECTIVITY.....		102
5.0	introduction	103

5.1 Python in web development.....	103
5.1.1 Components of web application	103
5.1.2 Front-end to back-end connection	105
5.2 Frontend and backend design.....	110
5.2.1 Design an HTML form.....	111
5.2. 2 Design databases	115
5.3 Connecting to a database	117
5.4 Inserting and retrieving data.....	120
5.4.1 Inserting inputs into a database	120
5.4.2 Displaying stored data from a database	121
5.5 Updating and deleting data	123
5.5.1 Update data into database	123
5.5.2 Delete data from database	125
UNIT 6: COMPUTER NETWORKING.....	129
6.0. Introduction	129
6.1 Introduction to computer networks	130
6.1.1 Definition of a computer network	130
6.1.2 Importance and benefits of networking.....	131
6.1.3 Types of network connections.....	132
6.2 Types of networks.....	135
6.2.1 Personal Area Network.....	135
6.2.2 Local Area Network.....	136
6.2.3 Metropolitan Area Network	137
6.3 Network devices.....	138
6.3.1 Types of networks devices	139
6.3.2 Connecting network devices using Packet Tracer	141
6.4 Network configuration.....	145
6.4.1. Network addresses	146
6.4.2 Subnet mask and default gate way.....	147
6.4.3 Configuration of devices in Packet Tracer	150
6.5 Network troubleshooting.....	152
6.5.1 Testing connectivity	153
6.5.3 Verifying connection	153

6.5.4 Correcting errors.....	154
UNIT 7. ARTIFICIAL NEURAL NETWORK	159
7.0 Introduction to neural networks.....	159
7.1 Introduction to Artificial Neural Networks	160
7.1.1 Concepts in ANN	160
7.1.2 Importance of ANN in AI	161
7.1.2. Relationship between AI, Machine Learning, ANN and Deep Learning.....	161
7.2 Network architecture.....	163
7.2.1 Input, hidden, and output layers	163
7.2.2 Training of Neural Networks.....	164
7.3 Types of ANN.....	165
7.3.1 Deep Neural Networks.....	165
7.3.2 Convolutional Neural Networks.....	166
7.3.4 Transfer learning in ANN.....	167
7.4 Applications of ANN	168
7.4.1 Image recognition	169
7.4.2 Speech recognition	169
7.4.3 Sample Python program implementing ANN.....	170
7.5 Ethics in Neural Networks.....	174
REFERENCES.....	178

UNIT 1: GOOGLE SHEET



Key unit competence: Create Google Sheets to solve real-life problems and organize information.

Learning Objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Explain how to access and manage Google Sheets in Google Workspace.
- Describe text formatting options and basic spreadsheet calculations.
- Identify different types of charts and explain their uses in presenting and visualizing data.

Skills

- Use Google Sheets effectively by accessing it through Google Workspace and sharing sheets with appropriate permissions.
- Apply spreadsheet formatting tools such as text formatting, fill color, merging cells, and text wrapping to organize data.
- Perform calculations by using arithmetic operations and functions.

Attitude and values

- Appreciate the importance of spreadsheets in organizing, analyzing, and presenting data efficiently.
- Demonstrate responsibility and ethical behavior when sharing and managing digital documents.

1.0 Introduction

Google Sheets is a web-based application that enables users to create, update and modify spreadsheets and share the data online in real time from your computer, Android phone or tablet.

Google Sheets also lets multiple users in different parts of the world work on a spreadsheet at the same time and chat using a built-in instant messaging program.

Users can upload spreadsheets from their computer or mobile device. Every change is saved automatically, and users can see other users' changes as they are being made.

Introductory Activity

Our class is planning a field trip to the nearby science museum. As the event coordinator, you are responsible for organizing all the necessary information to make the trip successful. You will need to list all students attending and track their permission forms, record transportation costs and snacks for everyone, calculate the total budget required for the trip, and share this information with your teacher and classmates.

- 1) How could you calculate totals and other useful information automatically?
- 2) What would be the best way to share your work with others and keep it updated?
- 3) How will you visually show the data such as those who brought snacks?

1.1 Creating and sharing a Google Sheet link

Learning activity 1.1

At one school the management of students' information is not computerized. The school's headteacher is putting in place a solution so that the database of student's information can be collected and received automatically and hence speeding up the production of information when they are requested.

- 1) What do you understand by online Google Sheet?
- 2) If you are given the task of collecting that information, how are you going to proceed?

1.1.1 Logging into Google workspace

Logging into Google workspace is the first step toward accessing a wide range of powerful tools designed to support communication, collaboration, and productivity. Through a single account, users can easily access applications such as email, cloud storage, document creation, and virtual meetings.

Follow steps below to log into your Google workspace account:

Step 1: Open a web browser such as Google Chrome, Microsoft Edge or Mozilla Firefox

Step 2: Go to Google then click on "Sign in"

Step 3: Enter your email address and a valid Password

Step 4: Access Google Workspace

Step 5: Click on the **Google Apps icon (9 dots)** on the top right corner of your gmail account.

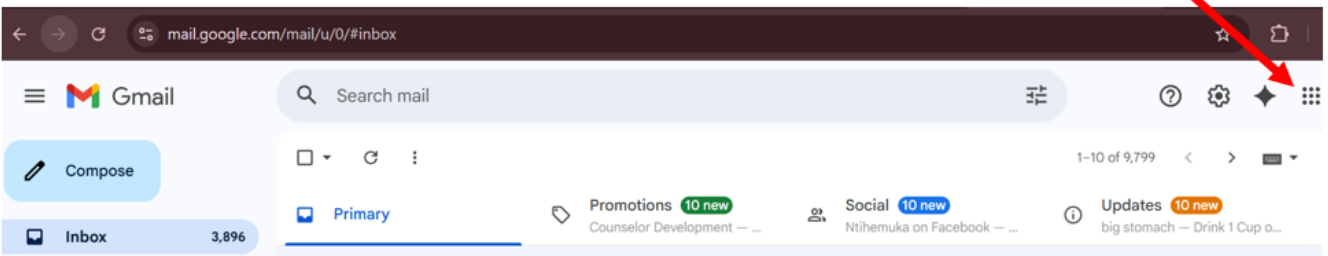


Figure 1. 1. Gmail inbox interface showing the Google apps

Step 6: Select the workspace you want to use. In this case, click on **Google Sheet**.

Step 7: Click on **Blank spreadsheet**.

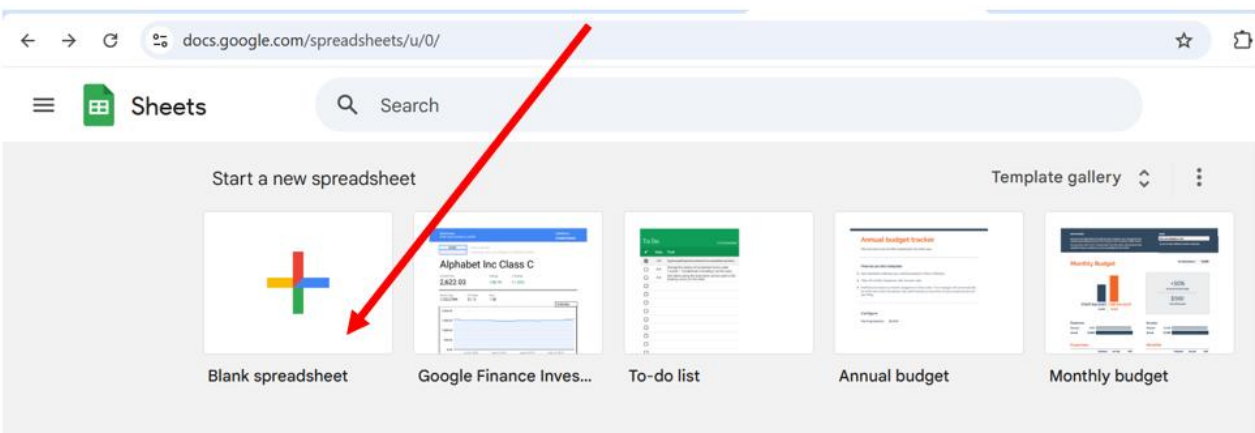


Figure 1. 2. Option for accessing a new blank spreadsheet

Step 8: Google sheet working environment gets opened as in the screen below.

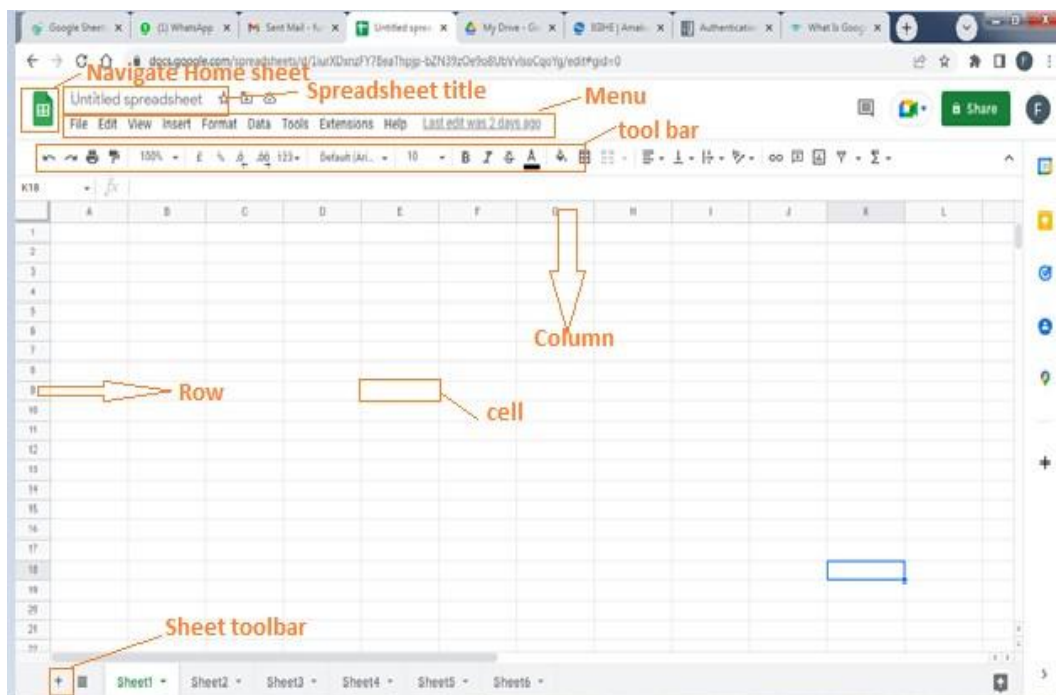


Figure 1. 3. Google sheet working environment

1.1.2 Naming a sheet

Google Sheets allows users to create multiple worksheets (tabs) in one file. Each worksheet is called a **sheet** and giving it a clear name helps users easily identify and organize their data.

Steps to name a sheet:

- Step 1: Open your Google Sheets document.
- Step 2: Look at the bottom of the screen where sheet tabs are located. You can see Sheet1, Sheet2, etc.
- Step 3: Double-click on the sheet name you want to change.
- Step 4: Type the new name such as “Students Marks”.
- Step 5: Press **Enter** on your keyboard.
- Step 6: The sheet will now display the new name.

1.1.3 Google Sheets permission.

Google Sheets allows users to work together by sharing spreadsheets with other people. When a spreadsheet is shared, the owner can control what other users are allowed to do with the file, such as viewing, commenting, or editing. This makes it easy for people to collaborate and work on the same spreadsheet at the same time.

a. Sharing the file through email

Step 1. Click the **Share** button in the top-right corner of the Google Sheets window.

Step 2. Enter the email address of the person you want to share the file with,

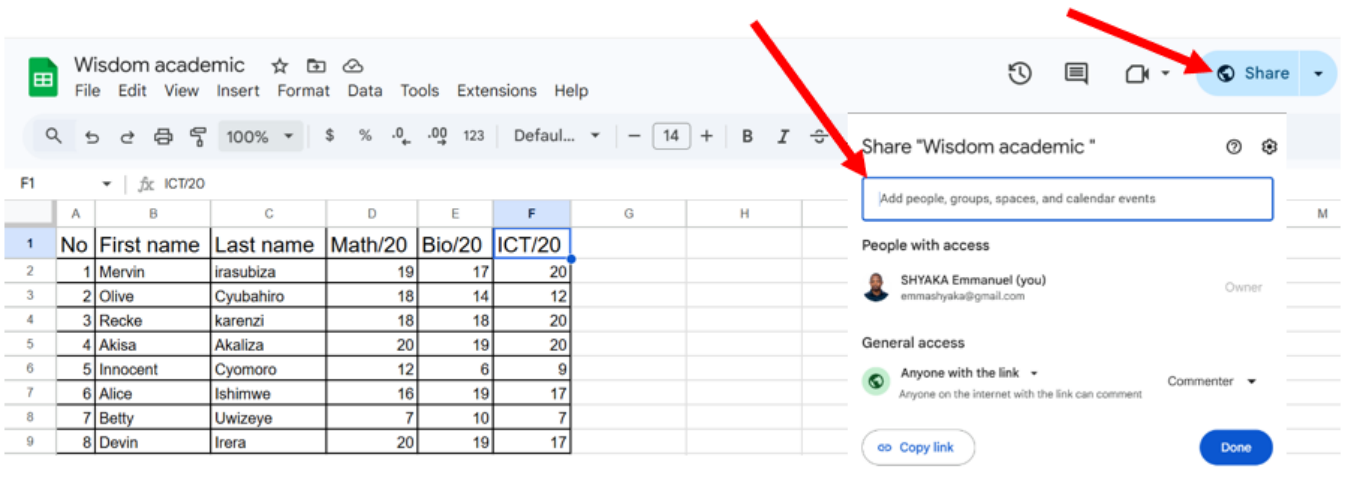


Figure 1. 4. Google Sheets sharing options highlighted.

Step 3. Choose the permission level you want to give the person in the list. Those permissions are Viewer, Commenter and Editor.

The viewer can see the data but cannot make changes, the commenter can see the data and leave suggestions while the editor can change data, format cells, and share the file with others.

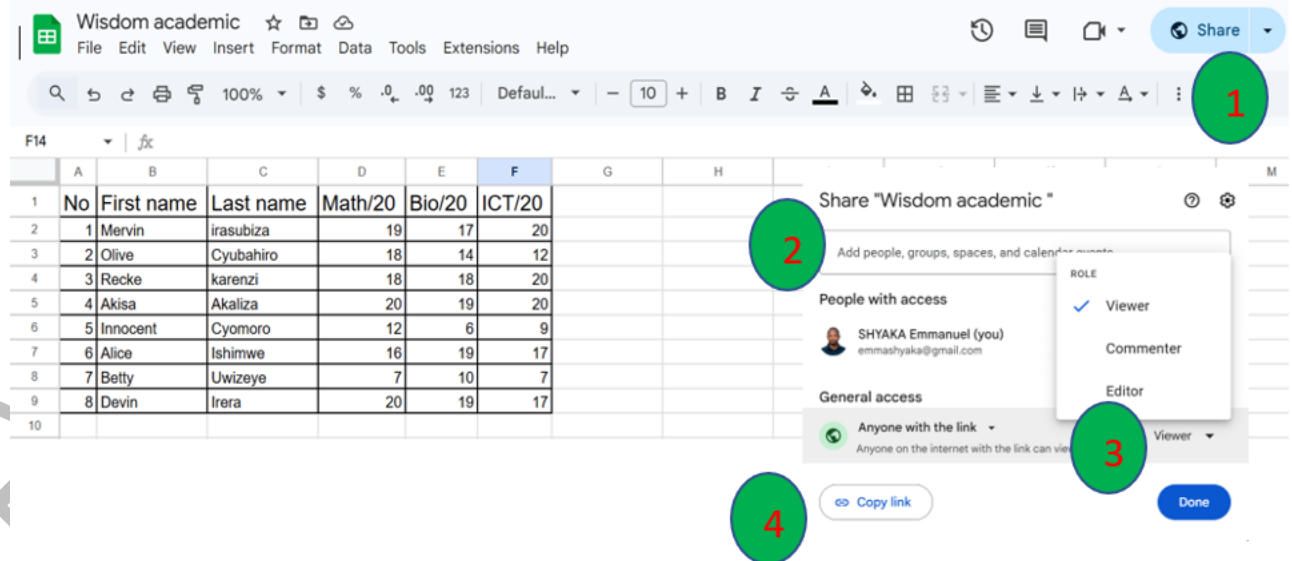


Figure 1. 5. Gmail inbox interface showing the Google apps

Step 5. Copy link: Alternatively, you can change "General access" from Restricted to Anyone with the link to provide broader access.

Application activity 1.1

Section A: Choose the correct answer.

- 1) The first step when logging into Google Workspace is:
a) Enter password b) Open a web browser c) Click Share Button d) Rename a sheet
- 2) The icon is used to access Google Apps in Google Workspace is ...
a) Star icon b) 9-dot grid icon c) Folder icon d) Email icon
- 3) How do you rename a sheet in Google Sheets?
a) Right-click and delete c) Restart the computer
b) Double-click the sheet name and type a new name d) Download the file again
- 4) What does the “Viewer” permission allow in Google Sheets?
a) Edit data b) Only view data c) Delete data d) Share the file
- 5) Which permission level allows a user to edit and share a Google Sheet?
a) Viewer b) Commenter c) Editor d) Guest

Section B: Answer the following questions.

- 1) Explain how Google Sheets supports collaboration among users working from different locations.
- 2) Outline the steps for creating a new Google Sheet
- 3) Differentiate between the four permission levels in Google Sheets: ownership, edit, comment, and view.
- 4) A student shares a spreadsheet with the Edit permission instead of View permission for everyone in the class.
a) Evaluate the possible risks or problems that may occur.
b) Identify the methods you should use to prevent errors.

1.2 Text formatting

Learning activity 1.2

Your class recently conducted a survey to find out students’ favorite subjects. The results were entered into a Google Sheet. However, when the worksheet was displayed in class by the teacher, the results appeared plain and difficult to read. Some titles were not clear, and the text was not well organized.

The teacher asks the class to improve the appearance of the report so that it can be clearly understood and easily presented to everyone.

- 1) Discuss how you could use formatting tools in Google Sheets to make the data clear, well organized, and visually appealing.
- 2) List 2 ways you could use to change color of text.

1.2.1 Bold, italics, and underline

In Google Sheets, formatting tools such as Bold, Italic, and Underline are used to improve the appearance and readability of data. These tools help users emphasize important information such as titles, headings, totals, or key values in a worksheet. The bold option makes text darker and thicker, italic slants the text slightly to the right, and underline draws a line beneath the text. Proper formatting helps make spreadsheets clear, organized, and professional. These formatting tools are found on the toolbar at the top of the Google Sheets window and can also be applied using keyboard shortcuts. Follow the steps below to get started:

Step 1. Open Google Sheets

Step 2. Select the cell or range of cells containing the text you want to format.

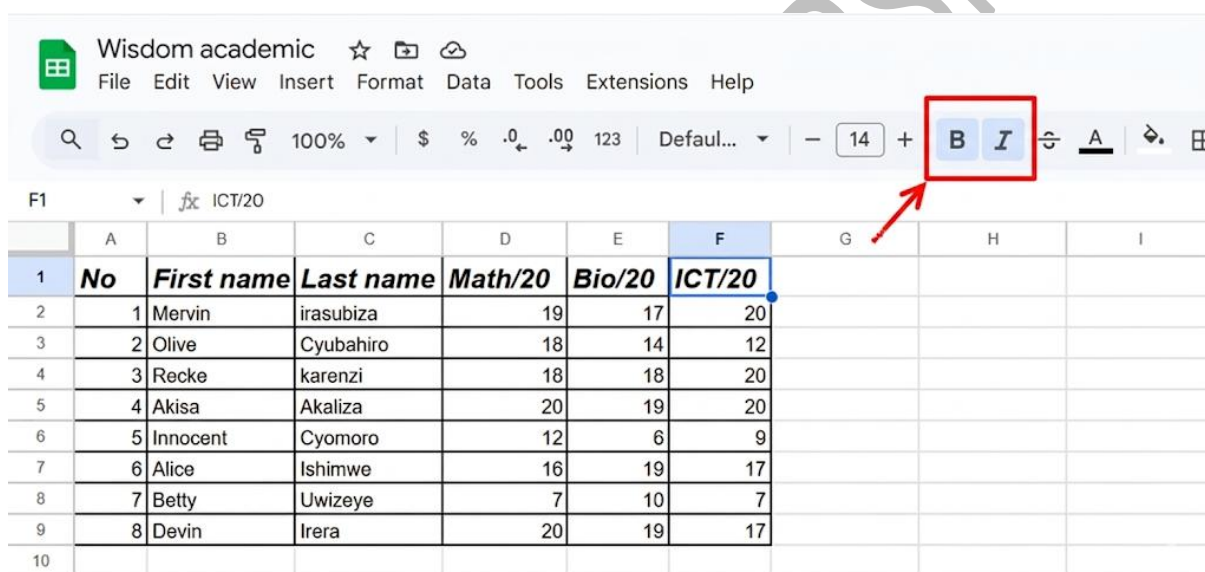


Figure 1. 6. Application of Bold and Italic

Step 3: Click the “**B**” (Bold) icon on the toolbar or press Ctrl + B.

Step 4: Click the “**I**” (Italic) icon on the toolbar or press Ctrl + I.

Step 5: Click the “**U**” (Underline) icon on the toolbar or press Ctrl + U.

1.2.2 Font size, color, alignment

Formatting tools like **Font Size**, **Font Color**, and **Alignment** help improve how a worksheet looks and organized. Font Size makes text bigger or smaller, font color changes the text color to highlight important information while the alignment places text to the left, center, right, top, middle, or bottom of a cell. Using these tools correctly helps create arranged, clear, and easy to read spreadsheets.

a) Font size

To change the font size select the cell or range of cells you want to modify then on the toolbar, click the Font Size dropdown which usually shows a number like 10 or 12

b) Font color

To change the font color select the cell or range of cells you want to modify then click the **A** tool located on the toolbar.

c) Alignment

Follow these steps to align cell content to the left, right or the center:

Step 1. Select the cell or range of cells whose content alignment you want to change

Step 2. On the toolbar, click the **Horizontal Align** button 

Step 3. Choose one of the alignment options namely left, center and right.

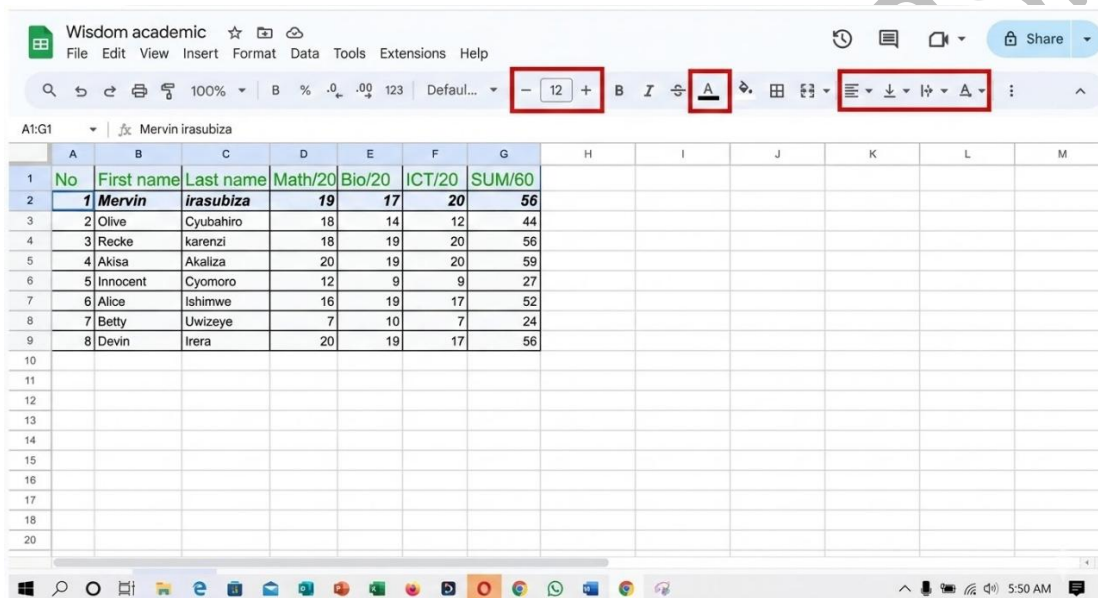


Figure 1. 7. Google Sheets toolbar highlighting font size, text color, and formatting options

1.2.3 Cell formatting feature

Fill color, cell merging, and text wrapping helps improve the appearance and organization of a worksheet. Fill Color is used to change the background color of a cell to highlight important information. Cell Merging combines two or more cells into one larger cell, often used for titles or headings. Text Wrapping allows long text to fit inside a cell by displaying it on multiple lines instead of overflowing into other cells.

a) Fill Color

Step 1: Open your worksheet in Google Sheets.

Step 2: Click the **Fill Color icon (paint bucket)** on the toolbar.

Step 3: Choose a color from the palette.

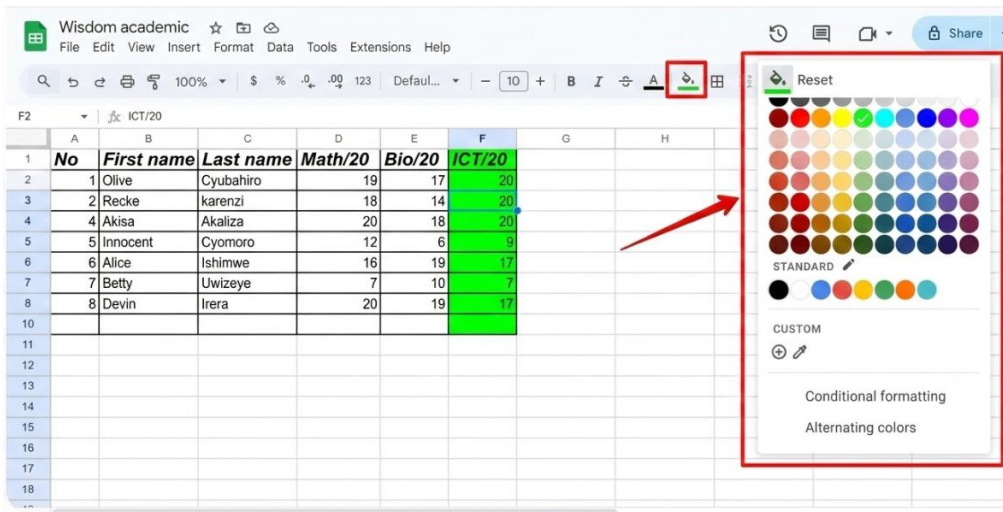


Figure 1. 8. Fill color palette

Step 4: The selected cell(s) will change to the chosen background color.

b) Merge Cells

Step 1: Select two or more adjacent cells you want to merge with.

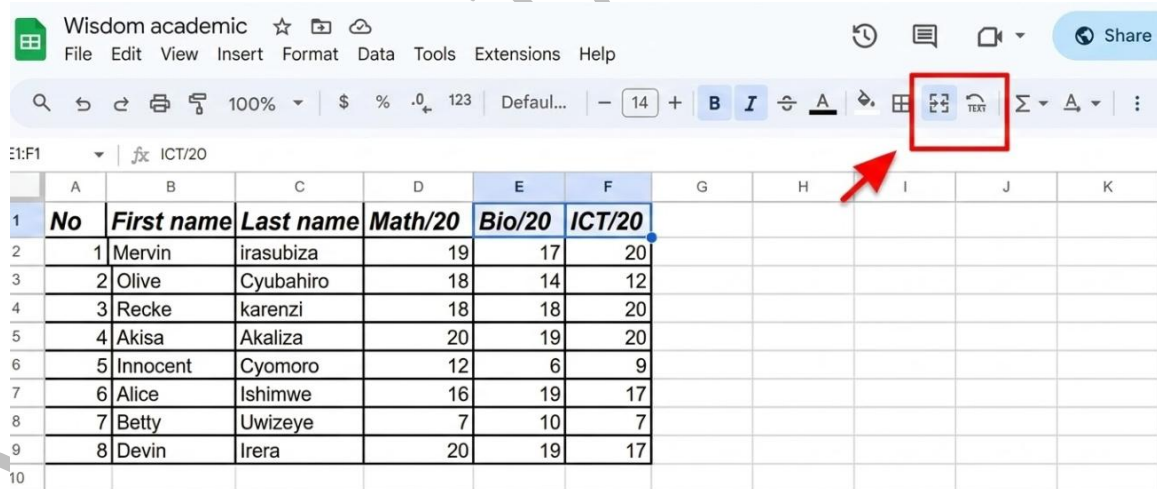


Figure 1. 9. Merging and wrapping text

Step 2: Click the **Merge Cells** icon on the toolbar.

c) Text wrapping

Step 1. Click the **Text Wrapping** icon on the toolbar after selecting the cells.

Step 2. Choose **Overflow -Wrap -Clip**.

The text will adjust based on your selection. You will realize that now all the text is visible in the cell.

Application Activity 1.2

Section A: Choose the correct answer.

1) What is the main purpose of using bold, italic, and underline tools in Google Sheets?

- a) To delete data
- b) To improve appearance and emphasize important information
- c) To hide cells
- d) To create charts

2) Which keyboard shortcut key is used for bolding in Google Sheets?

- a) Ctrl + I
- b) Ctrl + U
- c) Ctrl + B
- d) Ctrl + P

3) What does font color do in Google Sheets?

- a) Changes the background of the sheet
- b) Changes the text color
- c) Deletes text
- d) Merges cells

4) What is the purpose of merging cells?

- a) To split data into many rows
- b) To combine two or more cells into one larger cell
- c) To delete empty cells
- d) To change font style

5) What does text wrapping do in a cell?

- a) Deletes text automatically
- b) Changes font size
- c) Displays long text on multiple lines inside a cell
- d) Converts text to numbers

Section B: Answer the following questions

6) Design a plan to format a student report card with a title "End of Year Academic Remarks"

- a) Wrap the text in cell A1 only.
- b) Merge cells A1 to G1 for the title and use Wrap to ensure the long title stays within the merged area.
- c) Fill the entire sheet with Light blue color.

1.3 Calculations

Learning activity 1.3

At the end of the year, a teacher was overwhelmed by the task of preparing end-of-year reports for learners. The marks had been submitted as Word document soft copies, requiring manual calculation of totals, subject averages, and overall averages for each learner. As a result, the teacher considered buying a calculator to use.

- 1) How can Google Sheets help teachers do these calculations automatically?
- 2) Compare the accuracy of doing the calculations using Word files and using a Google Sheets

Arithmetic operations include addition, subtraction, multiplication, and division. They make calculations faster, easier, and more accurate. Functions like SUM, AVERAGE, and IF are used to add numbers, find the average, and make logical decisions.

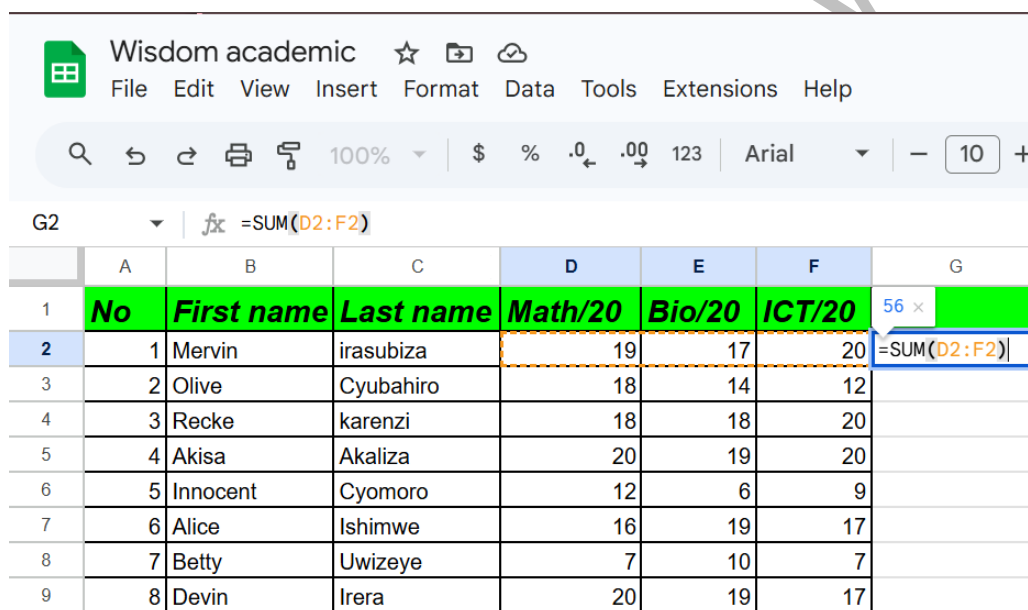
Operator	Name	Description	Example	Result
$x + y$	Addition	Sum of x and y	$2 + 2$	4
$x - y$	Subtraction	Difference of x and y	$5 - 2$	3
$x * y$	Multiplication	Product of x and y	$5 * 2$	10
x / y	Division	Quotient of x and y	$15 / 5$	3
$x \% y$	Modulus	Remainder of x divided by y	$5 \% 2$	1

Table 1.1 Arithmetic operations

1.3.1 SUM function

SUM function is commonly used in spreadsheet applications such as Microsoft Excel, Google Sheets, and LibreOffice Calc to quickly add numbers in a range of cells. Instead of adding numbers manually one by one, the SUM function automatically performs the addition. This makes calculations faster, more accurate, and easier to manage, especially when working with large datasets.

To use the SUM function in a Google Sheet select the cells then type the function such as =SUM(D2:F2) then press Enter.



The screenshot shows a Google Sheet interface with a menu bar (File, Edit, View, Insert, Format, Data, Tools, Extensions, Help) and a toolbar with various icons and settings. The spreadsheet has columns labeled A through G. Row 1 contains headers: A1 is 'No', B1 is 'First name', C1 is 'Last name', D1 is 'Math/20', E1 is 'Bio/20', F1 is 'ICT/20', and G1 is empty. Row 2 contains data: A2 is '1', B2 is 'Mervin', C2 is 'irasubiza', D2 is '19', E2 is '17', F2 is '20', and G2 is the formula '=SUM(D2:F2)'. The formula bar at the top shows the formula being entered in G2. The result of the formula, 56, is displayed in cell G2.

	A	B	C	D	E	F	G
1	No	First name	Last name	Math/20	Bio/20	ICT/20	
2	1	Mervin	irasubiza	19	17	20	=SUM(D2:F2)
3	2	Olive	Cyubahiro	18	14	12	
4	3	Recke	karenzi	18	18	20	
5	4	Akisa	Akaliza	20	19	20	
6	5	Innocent	Cyomoro	12	6	9	
7	6	Alice	Ishimwe	16	19	17	
8	7	Betty	Uwizeye	7	10	7	
9	8	Devin	Ilera	20	19	17	

Figure 1. 10. Using the SUM function for calculations in a Google Sheet

The total number of cells A1 to A5 will be displayed. Note that you can also click the Functions (Σ) icon and choose SUM.

To apply the function to other cells select the cells containing the sum result then position the cursor in the corner until it becomes a cross then press and hold down the left button and scroll down.

1.3.2 AVERAGE function

To use it select the cell then write the average function such as =AVERAGE (A1:A5) then press Enter. Immediately the result of the average value will appear.

If function is a logical function used to test a condition and return one value if the condition is true and another value if the condition is false.

This function is very useful when analyzing data such as determining whether a student has passed or failed, checking if a value meets a required limit, or categorizing data automatically.

Section A: Choose the correct answer.

- Section B: Answer the following questions.**

A	B	C	D	E	F	G	H
No	First name	Last name	Math/20	Bio/20	ICT/20		
1	Mervin	irasubiza	19	17	20		
2	Olive	Cyubahiro	18	14	12		
3	Recke	karenzi	18	18	20		
4	Akisa	Akaliza	20	19	20		
5	Innocent	Cyomoro	12	6	9		
6	Alice	Ishimwe	16	19	17		
7	Betty	Uwizeye	7	10	7		
8	Devin	Irera	20	19	17		

- 1) Write the exact formula you would enter into cell G9 to calculate the total score for Devin Irera.
- 2) Use IF to determine Innocent Cyomoro's status (pass mark: 30/60)

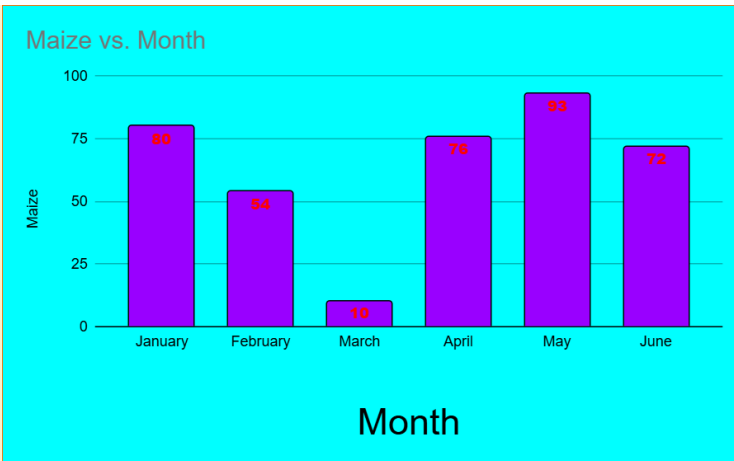
1.4 Creating charts

Learning activity 1.4

- 1) Open Spreadsheet program, enter data in the table below and save it as "Production"

Production			
Month	Soja	Beans	Maize
January	8	150	80
February	54	77	54
March	93	32	10
April	116	11	76
May	137	60	93
June	187	10	72

- a) Using production data make the graph below



b) Why are charts important in data presentation?

c) What type of chart is used to compare values?

Charts are visual tools used to represent data in a graphical form. In Google Sheets, charts help users understand information easily by showing comparisons, patterns, and trends. Instead of reading many numbers in a table, charts present data in a clear and attractive way. They are useful in education, business, and research because they make data simple to analyze and interpret.

1.4.1 Formatting charts

Formatting charts refer to the process of modifying the visual appearance of a chart or graph to make data easier to interpret and more professional. It involves adjusting elements such as titles, colors, fonts, and axes to highlight key information and ensure the chart communicates its message clearly.

To access these options, double-click your chart to open the Chart editor panel on the right side of the screen.

a) Chart style

This allows you to change the background color, font, and border color of the entire chart.

Follow these steps to access the different chart styles:

- Select your chart by clicking on it.
- **Click the three dots (:)** on the chart and choose **Edit chart**, or simply double-click the chart.
- The Chart editor panel will appear on the right side.

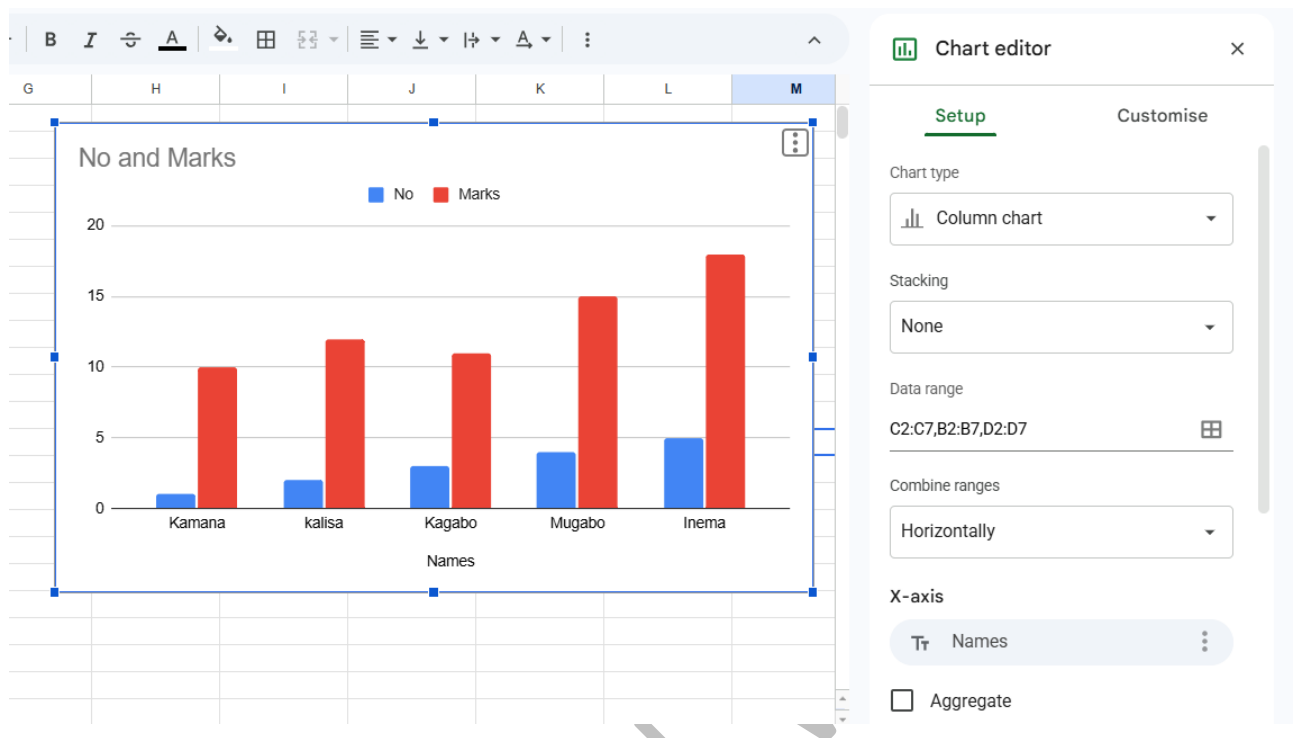


Figure 1. 11. Chart editor with Setup and customize tabs and style options

- Under the **Setup** tab, you can change the **Chart type** into Column chart, Bar chart, Line chart, Pie chart, Area chart, etc.
- Under the **Customize** tab, you can modify the chart's appearance by changing:

b) Chart and axis titles

A chart axis title is a label that describes what the values on an axis represent in a chart or graph. Use this to create a clear title for the chart and label the horizontal and vertical axes to know what the data represents.

Follow these steps to access the chart and axis titles:

- Select the chart then click on the three dots located at the top right corner then click **Edit Chart**
- In the Chart editor pane located to the left, click on **Customize**
- Now on that same Chart editor pane find different options including the chart and axis titles
- Click on **chart and axis titles** and change any of the available options depending on your choice.

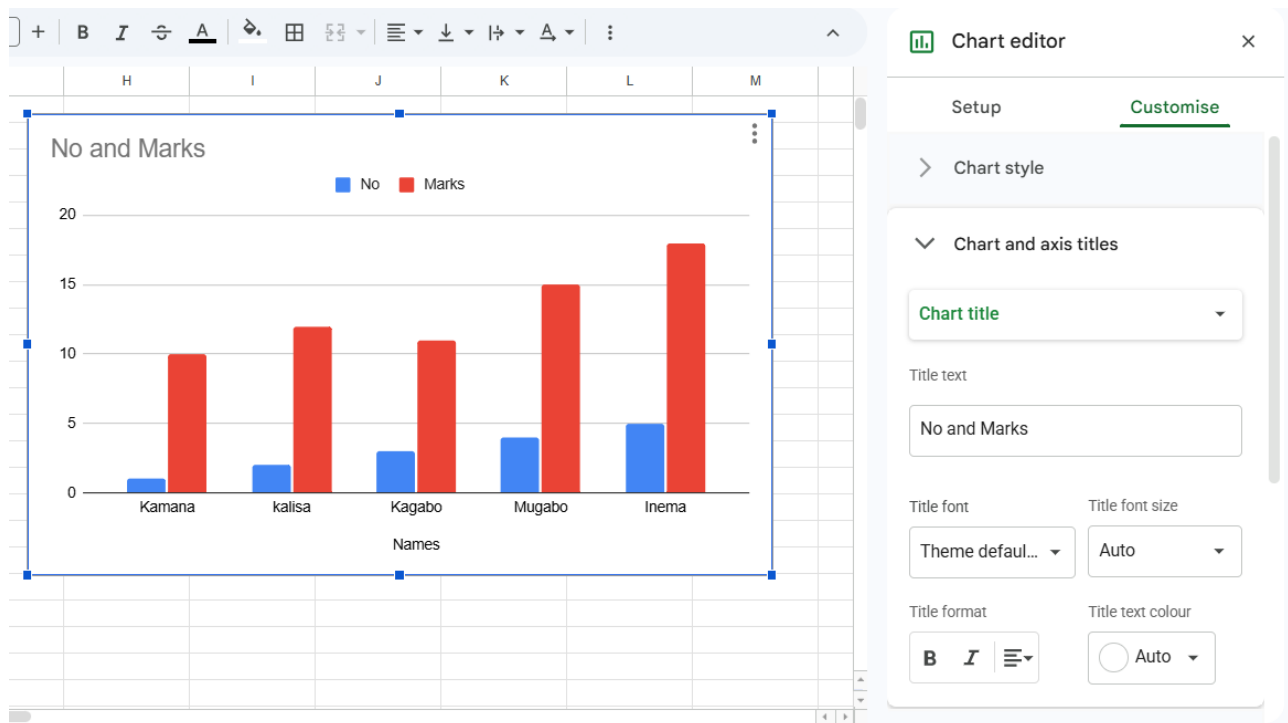


Figure 1. 12. Chart editor – Chart title settings

To access directly the chart and axis titles you can immediately click on the current axis title. In the chart above you will click on **Names** which is the current axis name.

c) Series

In Google Sheets, a series is a set of related data values that are displayed on a chart. Here you can change the color of the bars, lines, or pie slices. This is also where you can add data labels to show exact values on the chart. Data labels are pieces of text displayed on a chart that show the actual value of a data point, bar, slice, or marker.

d) Legend

A legend is a part of a chart that explains what the different colors or symbols represent. This tool allows you to change the position to different positions such as top, bottom, left or right and font style of the legend.

To customize the legend, click on the current legend () then click on **Legend** and change the available fields such as Position, Legend font, Legend font size, Legend format, etc.

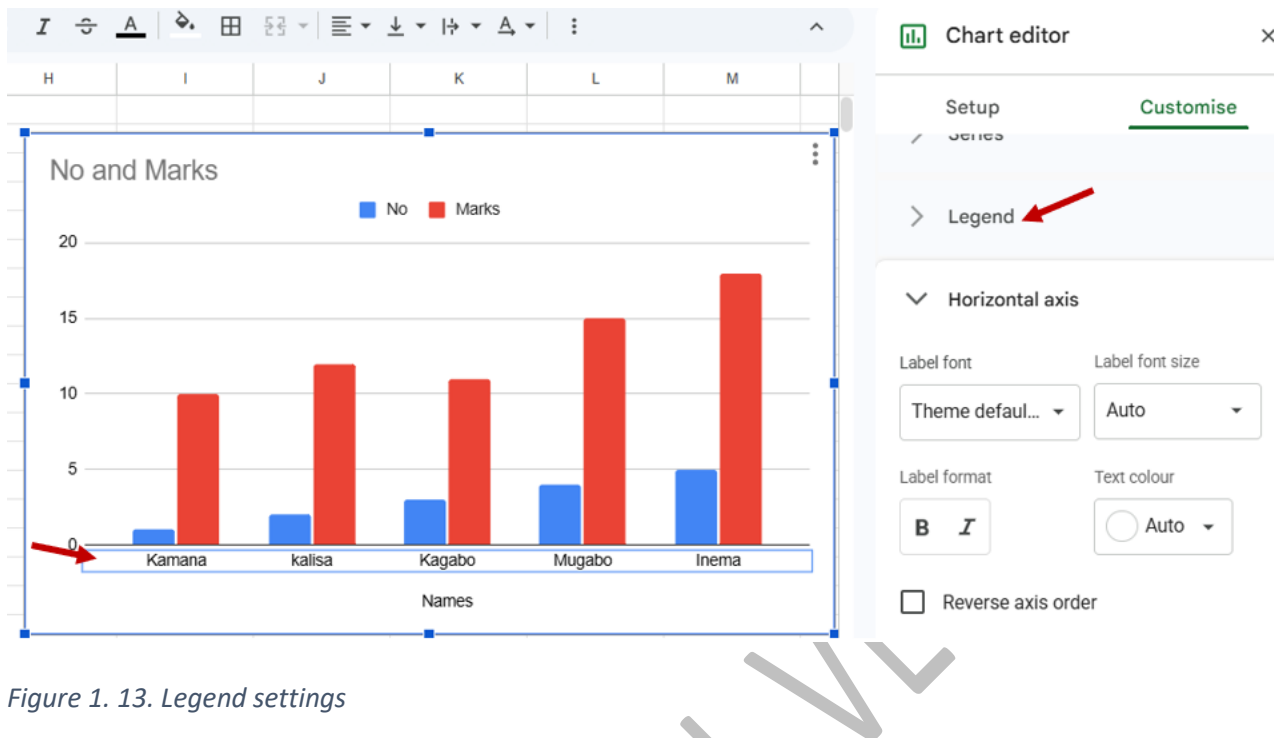


Figure 1. 13. Legend settings

Application activity 1.4

Section A: Choose the correct answer.

- What is the main purpose of using charts in Google Sheets?
 - To delete data
 - To make data harder to read
 - To represent data visually for easy understanding
 - To format text only
- Which chart type is best used to show parts of a whole in percentages?
 - Line Chart
 - Bar Chart
 - Pie Chart
 - Column Chart
- Which chart type is commonly used to show trends over time?
 - Pie Chart
 - Line Chart
 - Column Chart
 - Area Chart
- What does formatting charts in Google Sheets mainly involve?
 - Printing charts
 - Changing appearance such as colors, titles, and labels
 - Deleting charts
 - Copying spreadsheets
- Where can you access chart formatting options in Google Sheets?
 - File menu
 - Chart editor panel
 - Google Drive
 - Toolbar only

Section B: Answer the following questions.

- Make the following table in Google Sheet.

	A	B	C
1	NO	PRODUCT	PRICE
2	1	Laptop	390000
3	2	Desktop	350000
4	3	Classroom Projector:	400000
5	4	Epson	320000
6	5	External Monitors	360000

- Create a Column chart to compare the products' price.
- Use the Customize tab to change the color of the bars to green and add a title called "COMPANY PRODUCTS."

End unit assessment

Section A: Choose the correct answer.

1) Which of the following is the correct way to share a Google Sheet so that another user can only view but not edit it?

- Set permission to 'Editor'
- Set permission to 'Viewer'
- Set permission to 'Commenter'
- Share via email attachment

2) A student wants to combine cells A1 and B1 into one cell in Google Sheets. Which feature should they use?

- Text Wrapping
- Fill Color
- Cell Merging
- Freeze Rows

3) You want to add all values in cells B2 to B10 in Google Sheets. Which formula is correct?

- =ADD (B2:B10)
- =TOTAL(B2, B10)
- =SUM(B2:B10)
- =COUNT(B2:B10)

4) A student has a long text in cell C3 and wants it to display fully within the cell without overflowing into adjacent cells. Which text wrapping option should they choose?

- Overflow
- Clip
- Wrap
- Merge

Section B: Answer the following questions.

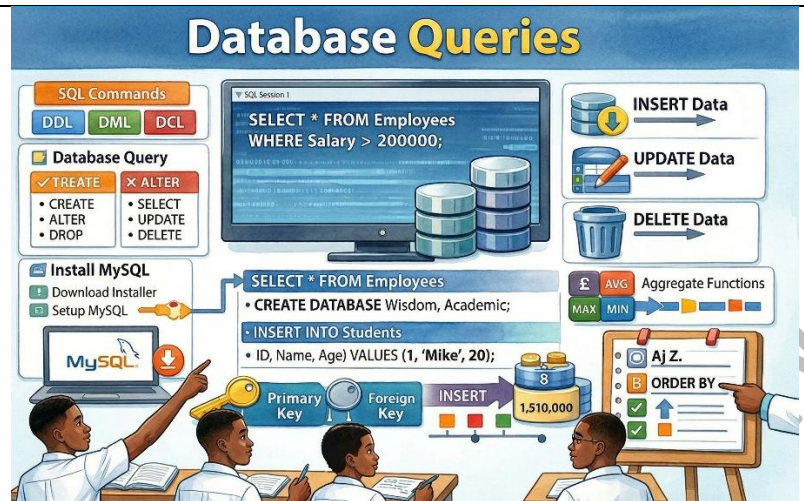
- List the three primary levels of permission you can set when sharing a Google Sheet with other users.
- What is the specific keyboard shortcut used to apply bold formatting to a selected cell in Google Sheets?
- Explain the difference between cell merging and text wrapping.
- How does the cloud-based nature of Google Sheets benefit users compared to traditional spreadsheet software?

- 5) A teacher has a list of student names in Column A and their Math marks in Column B. Write the exact formula she should enter to find the Average mark for students in cells B1 through B10.
- 6) Compare a Pie Chart and a Line Chart. Under what specific circumstances would you choose one over the other?
- 7) A student shares a spreadsheet with view permission for the whole class instead of editor permission. What are the risks this poses to the integrity of the data.
- 8) As an entrepreneur managing your shop, explain how Google sheet will help you in your daily work.
- 9) Would you like to use Google Sheets to calculate your home budget? Explain your answer.

EXPERIMENTAL VERSION

EXPERIMENTAL VERSION

UNIT 2: DATABASE QUERIES



Key unit competence:

Execute SQL commands for defining structure and manipulating data in a database

Learning objectives

Knowledge and understanding

- Describe common sql commands and the importance of constraints in maintaining data integrity within databases.
- Explain how sql clauses and aggregate functions are used to retrieve data for effective data analysis.

Skills

- Install a Database Management System such as MySQL.
- Manipulate and manage database by using DDL and DML statements
- Analyze and organize data by using aggregate functions and SQL clauses to filter and sort data.

Attitudes and values

- Value the use of Database Management Systems for organizing and managing data effectively.
- Appreciate the importance of data accuracy, integrity, and proper documentation in database management.
- Demonstrate curiosity in learning SQL and collaborate effectively with others during database-related tasks and projects.

2.0 Introduction

SQL or Structured Query Language is a programming language used to interact with databases. It allows users to request data from a database, insert new data, update existing data, delete data, and manage the structure of the database.

In this unit, you will create a database and its tables and manage it using Data Definition Language (DDL) and Data Manipulation Language (DML) commands.

Introductory activity

A secondary school has many employees working in different departments such as teaching, administration, and security. The school administration records information about employees including Employee ID, Name, Department, City, and Salary. Currently, this information is stored in notebooks and spreadsheets, which makes it difficult to search, update, organize, or analyze employee data quickly.

You are given the task to do that job of creating that database and manage it.

- 1) Before storing employee information, what should you create first: a database or a table? Explain.
- 2) If the school decides to add a new field such as email address to one of the tables. How can the table structure be modified?
- 3) Explain how to add new employee records to the database?
- 4) How can you display all employees stored in the database?
- 5) How can you find employees who live in Nyagatare?
- 6) How can you update an employee's salary in the database?

2.1 Introduction to SQL

Learning activity 2.1

A school wants to store student and teacher records using a computerized database instead of keeping information in notebooks. The ICT teacher installs MySQL using XAMPP so that students can create databases, store records, and search information easily using SQL queries.

- 1) Why is a database important in a school system?
- 2) What is the purpose of a database query?
- 3) Why is XAMPP installed on a computer?
- 4) Which service in XAMPP is used for managing databases?

5) Name the three main categories of SQL languages.

SQL or Structured Query Language is the standardized programming language used to communicate with relational databases such as MySQL, Oracle, SQL Server. It looks like English and is used to ask for or manipulate information in a database.

In most cases, SQL queries are not case sensitive meaning that queries written in capital letters are the same as those written in small letters.

2.1.1 Database queries

A database query is a request used to retrieve specific information from a database. Instead of searching through all the stored data manually, a query helps the user find, filter, and display the required data quickly.

Database queries are used to:

- Retrieve data from one or more tables in a database.
- Filter information using specific conditions.
- Sort and organize data for easier analysis.
- Update or modify records in a database.
- Delete unnecessary data from tables.

2.1.2 MySQL installation

The installation of MySQL typically involves downloading it, configuring the MySQL server, setting up user accounts including the root user, and verifying that the system is running correctly. In addition to installing MySQL directly, there are also software packages that simplify the process, such as XAMPP.

Instead of separately installing MySQL, Apache, PHP, and other tools, XAMPP installs them all at once and comes with server settings already set up, so you don't need to manually configure the MySQL server. It also already includes default settings, so beginners don't need to create everything from scratch immediately.

By installing XAMPP, you can develop and test websites on your own computer without needing an internet connection or a remote server. This is ideal for learning PHP, building database-driven web applications, and running content management systems like WordPress locally.

Steps for installing XAMPP

- **Download the XAMPP Installer:**

Open your web browser and go to the official XAMPP website

<https://www.apachefriends.org/download.html> then click on the appropriate software for Windows.

- **Run the XAMPP Installer**

Locate the downloaded installer file in your Downloads folder. Right-click on it and select 'Run as Administrator' to launch the installer with the required permissions.

If a User Account Control (UAC) warning appears asking 'Do you want to allow this app to make changes to your device?', click Yes to proceed.

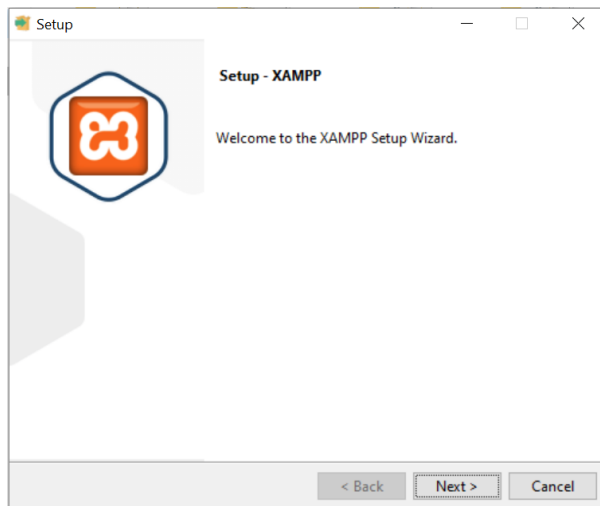


Figure 2. 1. The XAMPP Setup Wizard welcome Screen

- **Choose the components to install**

The XAMPP Setup Wizard will open. Click Next on the welcome screen and choose from the list the components available for installation such as MySQL, FileZilla FTP Server, Mercury Mail Server, Tomcat, etc.

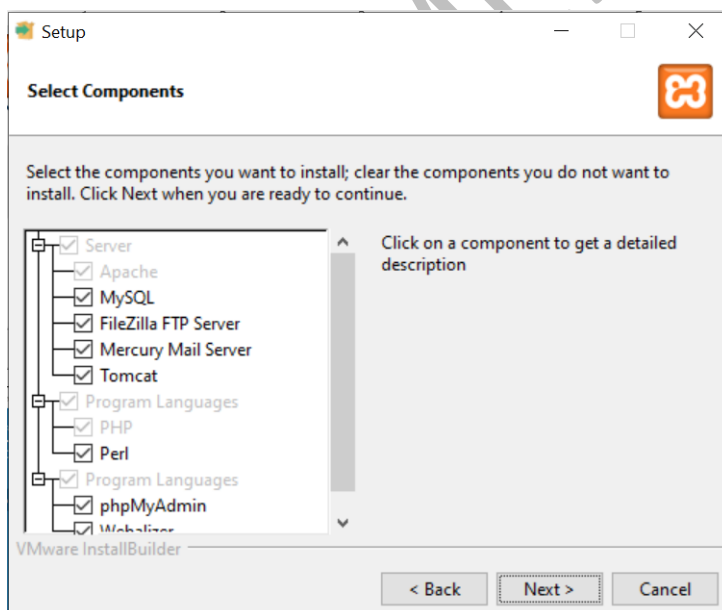


Figure 2. 2. Selection the components to be installed

- **Select the installation folder**

The installer will ask you to choose an installation directory and after choosing it click **Next** to continue. Later choose the installation language which in this case should be English.

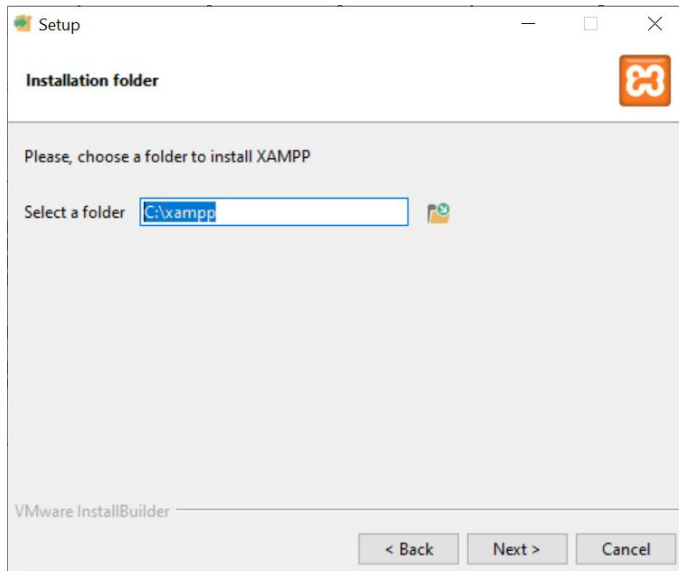


Figure 2. 3. The XAMPP installation folder

Note that the default path is C:\xampp. It is recommended to keep this default location, as some XAMPP scripts and configurations use this path. Avoid installing XAMPP in the Program Files folder, as Windows permission restrictions in that folder can cause issues when XAMPP tries to create or modify files.

- **Start the installation process**

After confirming your settings, click **Next** to begin the installation. The installer will copy all XAMPP files to your chosen directory. This may take a few minutes depending on your computer speed.

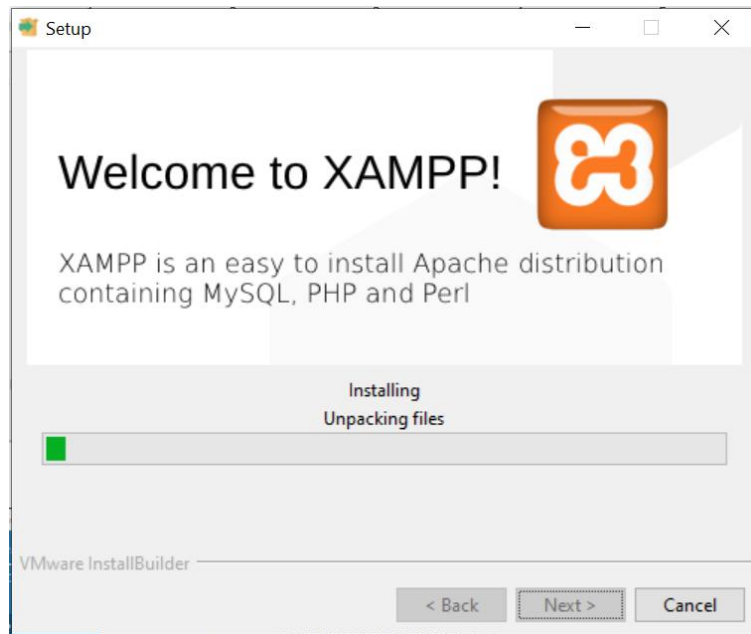


Figure 2. 4. Window for XAMPP installation

- **Complete the installation**

Once the installation is finished, you will see a completion screen. Make sure the option 'Do you want to start the XAMPP Control Panel now?' is checked.

Click **Finish**. The XAMPP Control Panel will launch automatically. This is the main hub from which you will start and stop the Apache web server, MySQL database and other services.

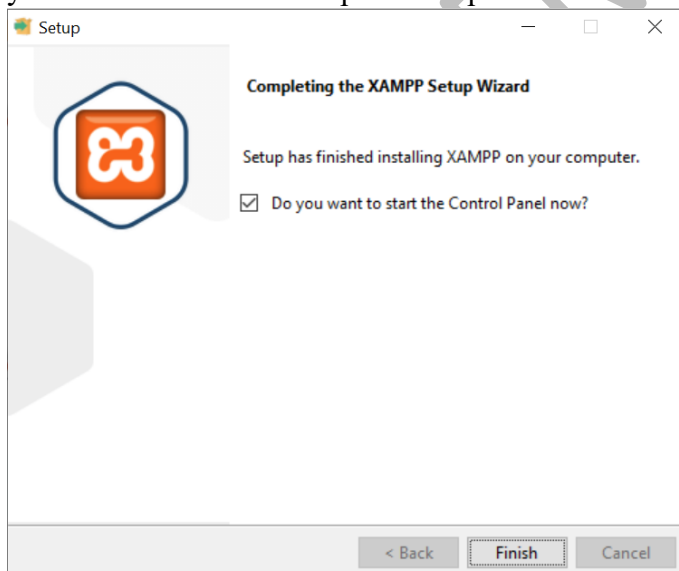


Figure 2. 5. XAMPP Setup Wizard completion screen

- **Start Apache and MySQL services**

After opening the XAMPP Control Panel, click the **Start** button next to Apache. After a moment, the Apache row should turn green, and the status will display the port numbers (80, 443), indicating that the web server is running.

Next, click the Start button next to MySQL. Similarly, the MySQL row will turn green and show port 3306. Both services must be running for your local web application to function correctly.

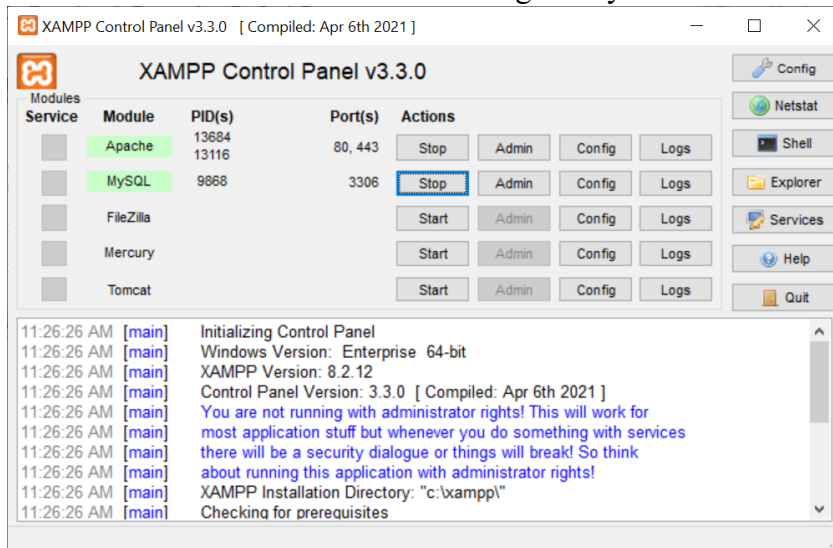


Figure 2. 6. Apache and MySQL running successfully on ports 80, 443 and 3306 respectively

- **Test the installation in your browser**

Open your web browser such as Chrome and type in the address bar “http://localhost” and press Enter. If XAMPP is running correctly, you will see the XAMPP dashboard. This confirms that Apache is working.

To test MySQL and access the database manager, click on phpMyAdmin in the XAMPP dashboard or navigate to: <http://localhost/phpmyadmin>

2.1.3 SQL languages

Structured Query Language (SQL) is used to manage and manipulate data in a database. SQL is divided into different categories based on the type of operations performed.

The three main categories are Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL).

Application activity 2.1

Section A: Choose the correct answer.

1) What is SQL mainly used for?

- a) Designing graphics
- b) Managing and manipulating data in a database
- 2) What is a database query?
 - a) A computer virus
 - b) A request to retrieve or manipulate data in a database
 - 3) What is the main purpose of XAMPP?
 - a) To edit videos
 - b) To provide a local web development environment
 - 4) Which components are included in XAMPP?
 - a) Word, Excel, PowerPoint
 - b) Apache, MySQL and PHP
 - 5) What are the three main categories of SQL?
 - a) HTML, CSS, JavaScript
 - b) Input, Process, Output
 - c) Creating presentations
 - d) Browsing the internet
 - c) A hardware device
 - d) A spreadsheet format
 - c) To create spreadsheets
 - d) To send emails
 - c) Chrome and Firefox
 - d) Python and Java
 - c) DDL, DML, DCL
 - d) Save, Edit, Delete

Section B: Answer the following questions.

- 1) Explain the purpose of installing MySQL on a computer.
- 2) Explain how database queries make data retrieval faster than manual searching.
- 3) Suggest the tables to be found in a school database
- 4) Explain how you would use DDL commands to create those tables in a database

2.2 Data Definition Language

Learning activity 2.2

- 1) What do you think we mean by *drop database command*?
 - a) Adding a column to the table
 - b) Adding a row to the database
 - b) Creating a second database
 - d) Deleting an existing database
 - 2) Which is not the structure of a table:
 - a. Table name
 - b) Column names
 - c) Data types
 - d) Database in which the table is
 - 3) What do you think is the use of these SQL commands?
 - a. Create database School;
 - b. Alter Table Students ADD City Varchar (50);

Data Definition Language (DDL) is a category of SQL commands used to define, create, modify, and delete the structure of a database and its objects. These database objects include tables, indexes, schemas, and views. DDL is used to design and organize how data will be stored in a database.

2.2.1 Create

The CREATE command in SQL is used to create a new database or database objects where data will be stored. It helps in building the structure of the database such as databases, tables, indexes, or views before any data is entered.

a) Creating a database in MySQL

Creating a database in MySQL is the process of making a new database where data can be stored, organized, and managed. A database acts like a container that holds tables.

The syntax of the query to use in creating a database is:

Create Database Database_Name;

Example: Create database STUDENTS;

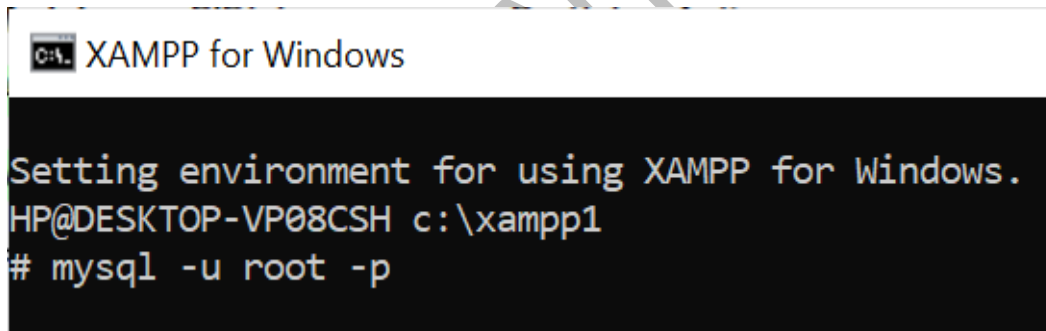
- **Opening MySQL**

It is necessary to open MySQL before creating a database or running any other query because MySQL is the database management system that receives and executes SQL commands. Before a database can be created, the user must access and connect to MySQL so that SQL commands can be entered and processed.

Open XAMPP Control Panel then start MySQL service by clicking Start next to MySQL. Open the command prompt and connect to MySQL

- **Writing the commands**

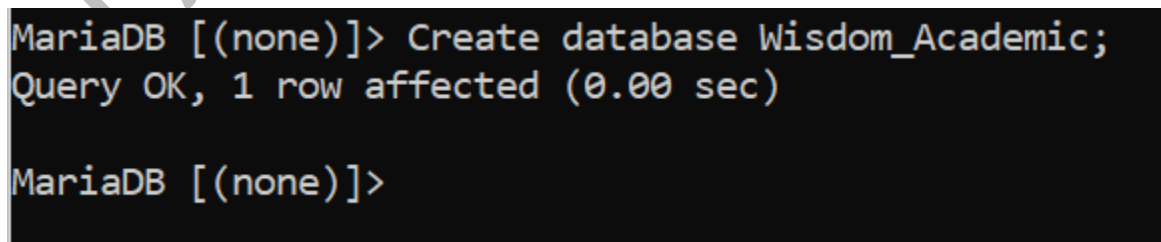
Write `mysql -u root -p` and Press ENTER then write the password there is any.

A screenshot of the XAMPP for Windows control panel window. The title bar says "XAMPP for Windows". The main content area is a black terminal window with white text. It shows the command prompt setting the environment for using XAMPP for Windows. The prompt is "HP@DESKTOP-VP08CSH c:\xampp1" and the command entered is "# mysql -u root -p".

```
C:\xampp1> Setting environment for using XAMPP for Windows.  
HP@DESKTOP-VP08CSH c:\xampp1  
# mysql -u root -p
```

Figure 2. 7. connecting to MySQL using the command

After having written the password type the command for creating the database such as **Create database Wisdom_Academic;** and press enter.

A screenshot of the MySQL command prompt. The prompt is "MariaDB [(none)]>". The command entered is "Create database Wisdom_Academic;". The output is "Query OK, 1 row affected (0.00 sec)". The prompt is now "MariaDB [(none)]>".

```
MariaDB [(none)]> Create database Wisdom_Academic;  
Query OK, 1 row affected (0.00 sec)  
  
MariaDB [(none)]>
```

Figure 2. 8. MySQL screen after successfully creating a new database

- **Displaying the databases created**

Displaying the created databases is a necessary step to verify that your command executed successfully. In MySQL, a server can host many different databases, so viewing the full list ensures you select the correct database and avoid running queries in the wrong place.

The syntax for showing the databases already created is: **Show databases;**

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| database26 |
| hct       |
| huye      |
| information_schema |
| my_database |
| mysql     |
| performance_schema |
| phpmyadmin |
| test      |
| wisdom_academic |
+-----+
```

Figure 2. 9. displaying all existing databases

As shown in the window above, there are different databases including the one that has just been created (wisdom_academic).

- b) Creating tables**

A database is simply like an empty container. To store and organize your data, a database must contain tables. A database can be compared to a folder in which files comparable to tables can be created. You cannot store data directly into a database without first creating at least one table in which to store data.

The syntax for the creating a table is:

```
CREATE TABLE table_name (column_name1 data_type, column_name2 data_type,
column_name3 data_type, .... )
```

Where table_name is the name that the table will have while column_name1 is the name of the first column in the table and data_type is the kind of data to be stored in that column such as integer, characters, etc.

Note that before creating a table you must select the database by using the command **Use Database Database_Name**

```
MariaDB [(none)]> use wisdom_academic;  
Database changed  
MariaDB [wisdom_academic]>
```

Figure 2. 10. Changing the database to wisdom_academic before creating tables

Now create a table like in the screen below:

```
MariaDB [wisdom_academic]> create table People(P_Id int, LastName varchar(50), FirstName varchar(50), Address varchar(40), City varchar(40));  
Query OK, 0 rows affected (0.21 sec)  
  
MariaDB [wisdom_academic]>
```

Figure 2. 11. Creation of Table

- **Describe the table created**

After creating a table, using the DESCRIBE command is a necessary practice to see if the created table has the desired structure. This command displays the table's columns with its data types.

For example in the image below the command “describe people” has been used so that the structure of the people table be shown as below:

```
MariaDB [wisdom_academic]> describe people;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| P_Id       | int(11)       | YES  |     | NULL    |       |  
| LastName   | varchar(50)   | YES  |     | NULL    |       |  
| FirstName  | varchar(50)   | YES  |     | NULL    |       |  
| Address    | varchar(40)   | YES  |     | NULL    |       |  
| City       | varchar(40)   | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
5 rows in set (0.05 sec)  
  
MariaDB [wisdom_academic]>
```

Figure 2. 12. Displaying the structure of the people table in the wisdom_academic database

2.2.2 Alter

The Alter command is used to modify the structure of an existing database table by allowing users to add, delete, or modify columns in that table.

The syntax is:

ALTER TABLE table_name ADD column_name datatype;

Whereby ALTER TABLE is a keyword, table_name is the exact name of the table to be modified, and ADD is the keyword used to insert a new column. column_name is the name of the new field, and datatype specifies the type of data it will store, such as INT for whole numbers or VARCHAR for text.

```
MariaDB [wisdom_academic]> ALTER TABLE People ADD Age INT;
Query OK, 0 rows affected (0.87 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [wisdom_academic]>
```

Figure 2. 13. Adding a new integer column AGE to the People table

Apart from the option to add a new column, you can modify an existing column by using this syntax:

ALTER TABLE table_name MODIFY column_name datatype;

```
MariaDB [wisdom_academic]> ALTER TABLE People MODIFY LastName VARCHAR(100);
Query OK, 0 rows affected (0.61 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [wisdom_academic]>
```

Figure 2. 14. Modifying the LastName column in the People table

When the DESCRIBE command is used like in the image below, you will find that the LastName column has been changed, and the Age column has been added.

```
MariaDB [wisdom_academic]> describe people;
```

Field	Type	Null	Key	Default	Extra
P_Id	int(11)	YES		NULL	
LastName	varchar(100)	YES		NULL	
FirstName	varchar(50)	YES		NULL	
Address	varchar(40)	YES		NULL	
City	varchar(40)	YES		NULL	
Age	int(11)	YES		NULL	

```
6 rows in set (0.02 sec)
```

Figure 2. 15. Viewing the updated schema of the people table

2.2.3 Drop

The DROP command is used to permanently delete a database object such as a table, database, index, or view from the database.

The DROP command can be confused by the DELETE command which only delete the content of a table while the Drop command delete a table (or database) structure together with its content.

Syntax: ALTER TABLE table_name DROP COLUMN column_name;

Whereby ALTER TABLE and DROP COLUMN are keywords that have to be used and written like that while table_name is the name of the table to be altered and column_name is the exact column name to be changed.

Example: ALTER TABLE People DROP COLUMN Address;

```
MariaDB [wisdom_academic]> ALTER TABLE People
-> DROP COLUMN Address;
Query OK, 0 rows affected (0.39 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [wisdom_academic]>
```

Figure 2. 16. Removing the Address column from the People table

Application activity 2.2

Section A: Choose the correct answer.

1) What is the main purpose of DDL commands?

- a) To display charts
 - b) To define and manage the structure of a database
 - c) To send emails
 - d) To calculate values
- 2) Which SQL command is used to create a new database?
- a) Select
 - b) Insert
 - c) Create
 - d) Delete
- 3) Which command is used to add a new column to an existing table?
- a) Drop
 - b) Alter table
 - c) Create table
 - d) Select
- 4) What does the Alter command do in SQL?
- a) Deletes the whole database
 - b) Modifies the structure of an existing table
 - c) Displays data only
 - d) Sorts records
- 5) What is the function of the Drop command?
- a) To add new data
 - b) To update records
 - c) To permanently delete a database object
 - d) To rename a table

Section B: Answer the following questions

- 1) Describe the difference between Alter and Drop commands
- 2) Explain why it is important to describe a table after creating or modifying it.
- 3) Write an SQL command to display all databases in MySQL.
- 4) Write an SQL statement to create a table called Students with the following columns:
Student_ID (INT), FirstName (VARCHAR(50)), Last Name (VARCHAR(50)), Age (INT)
 - a) Write an SQL command to add a column called City to the Students table.
 - b) Analyze how the CREATE, ALTER, and DROP commands work together when designing a database.
 - c) Compare the process of creating a table and modifying a table.
 - d) Explain the possible risks of using the DROP command without caution.

2.3 Data Manipulation Language

Learning Activity 2.3

A training center wants to store information about people registered in a program using a database system. The ICT teacher has already created a table called People in the database to store the participants' details.

The teacher asks students to use Data Manipulation Language (DML) commands to add new records, display stored information, and manage the data in the table using SQL commands such as INSERT and SELECT.

- 1) What SQL command used to add a new record into the People table?

- 2) How can students display all records stored in the People table?
- 3) How can students insert multiple records at the same time into the People table?
- 4) Why are DML commands such as INSERT and SELECT important when managing data in a database table?

DML is a language of SQL used to manage and manipulate data stored in database tables. allows users to insert, update, delete, and retrieve data from tables in a database.

2.3.1. Operation

In database systems, data must be stored, retrieved, modified, and removed when necessary. These actions are performed using basic SQL operations.

The most common operations are SELECT, INSERT, UPDATE, and DELETE. These operations are part of Data Manipulation Language (DML) and are essential for managing data in a database.

2.3.2. Inserting data into table

The INSERT statement is a SQL query used to insert data into the row of a table. Data insertion is accepted by the table if inly it meets the constraints of the database tables and its data types. For example if you attempt to insert integers in a char column, those data will not be inserted. In the same way if you insert null values (empty data) in a primary key column, that will not accepted by the table.

The syntax is:

- *INSERT INTO table_name (column1, column2, column3, ...) VALUES (value1, value2, value3);*

Whereby INSERT INTO and VALUES are keywords while column1, column2, column3 are the names of the three columns names and value1, value2, value3 are the three values that will be inserted respectively in the three columns.

- *INSERT INTO table_name VALUES (value1, value2, value3, ...);*

This second syntax above is used when data is to be inserted in all columns.

Example: *INSERT INTO People (P_Id, LastName, FirstName, Address, City, Salary) VALUES (7, 'Claire', 'MUKAMBUGA', 'ETP Kigali', 'Kigali', 250000);*

```
MariaDB [wisdom_academic]> INSERT INTO people (P_Id, LastName, FirstName, City, Age)VALUES (1, 'KAREGESHA', 'EMILE', 'Ruhango', 25);
Query OK, 1 row affected (0.14 sec)

MariaDB [wisdom_academic]>
```

Figure 2. 17. Entering a new record in the table

To insert multiple records simultaneously, use a single INSERT INTO statement followed by the table name and column list, then provide each set of data as a comma-separated list of values within parentheses.

Example:

INSERT INTO People (P_Id, LastName, FirstName, City, Age)

VALUES (1, 'MUGABO', 'Jean', 'Kigali', 30),

(2, 'UWIMANA', 'Alice', 'Huye', 22),

(3, 'GAKURU', 'Claude', 'Rubavu', 28),

(4, 'MUTONI', 'Grace', 'Musanze', 24),

(5, 'KAMANA', 'Patrick', 'Ruhango', 35);

```
MariaDB [wisdom_academic]> INSERT INTO People (P_Id, LastName, FirstName, City, Age)
-> VALUES (1, 'MUGABO', 'Jean', 'Kigali', 30),
-> (2, 'UWIMANA', 'Alice', 'Huye', 22),
-> (3, 'GAKURU', 'Claude', 'Rubavu', 28),
-> (4, 'MUTONI', 'Grace', 'Musanze', 24),
-> (5, 'KAMANA', 'Patrick', 'Ruhango', 35);
Query OK, 5 rows affected (0.19 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

Figure 2. 18. Entering multiple records at the same time

2.3.3. Selecting data from the table

Selecting data from a table is the process of retrieving records stored in a database table using SQL commands. This allows users to display all the data in a table or only the records that meet certain conditions. Without selecting data, it would be difficult to view and use the information stored in a database effectively.

To select data from your table, you use the SELECT statement, which is the main command for retrieving records.

The syntax for selecting data from a table is ***SELECT column_name(s) FROM table_name,***

However when the data to be displayed is from all the table columns, the column_name is replaced by * and the syntax becomes ***SELECT * FROM table_name***

Whereby SELECT and FROM are keywords while table_name is the table whose data is to be displayed and column_name is the name of the specific column to be shown.

Example 1: *SELECT * FROM People;*

```
MariaDB [wisdom_academic]> select * from people;
```

P_Id	LastName	FirstName	City	Age
1	KAREGESA	EMILE	Ruhango	25
2	MUGABO	Jean	Kigali	30
3	UWIMANA	Alice	Huye	22
4	GAKURU	Claude	Rubavu	28
5	MUTONI	Grace	Musanze	24
6	KAMANA	Patrick	Ruhango	35

```
6 rows in set (0.00 sec)
```

Figure 2. 19. Displaying all records using the Select statement

Example 2. Selecting specific columns: *SELECT FirstName, LastName, City FROM People;*

```
MariaDB [wisdom_academic]> SELECT FirstName, LastName, City FROM People;
```

FirstName	LastName	City
EMILE	KAREGESA	Ruhango
Jean	MUGABO	Kigali
Alice	UWIMANA	Huye
Claude	GAKURU	Rubavu
Grace	MUTONI	Musanze
Patrick	KAMANA	Ruhango

```
6 rows in set (0.00 sec)
```

Figure 2. 20. Retrieving FirstName, LastName, and City from the People table

From the example above, it is clear that when a few column are specified in the select command, only those specified columns' data are displayed.

Application activity 2.3

Section A: Choose the correct answer.

- Which SQL command is used to retrieve data from a table?
a) Insert b) Delete c) Select d) Update
- What does the INSERT command do?

- a) Deletes data from a table
- b) Updates existing records
- 3) Which statement is used to display all records from a table named “People”?
- a) SELECT People; b) SELECT * FROM People; c) GET ALL People; d) SHOW People;
- 4) What is the purpose of the UPDATE command?
- a) To remove records
- b) To modify existing data in a table
- c) Adds new records into a table
- d) Displays data
- c) To create a new table
- d) To sort data

Section B: Answer the following questions.

- 1) List four common DML operations used in SQL.
- 2) Explain why DML commands are important in managing data stored in database tables.
- 3) Describe the difference between SELECT * and selecting specific columns from a table.
- 4) Write an SQL query to display all records from the People table.
- 5) Write an SQL query to display only FirstName, LastName, and City from the People table.
- 6) Explain why it is important to retrieve and verify data after inserting records into a database.

2.4 Updating and deleting data in a table

Learning activity 2.4

A secondary school keeps employee information in a database table called People which contains details such as Employee ID, First Name, Last Name, City, and Salary. During a review of the records, the school administration noticed that some information needed correction. For example, the salary of one employee was increased, and the city of another employee changed after relocation, so the database administrator needed to update the existing records.

At the same time, one employee left the school, and the administration decided that the employee’s record should be removed from the table to keep only active employees.

- 1) If a person's address changes, should the old address remain in the database? Why?
- 2) What would you do if an employee's salary increases and the database still shows the old salary?
- 3) Why is it important to keep information in a database up to date?
- 4) Is it a good idea to delete a record of an employee who has left an organization? Explain your answer.
- 5) What problems might occur if outdated or incorrect information remains in a database?

2.4.1 Updating data with UPDATE

The UPDATE statement in SQL is a command used to modify existing data in one or more records of a table without changing the table structure. It allows users to change the values stored in specific columns of a table. However, it can also allow updating many records at the same time if there is no condition set in the “where” clause.

Syntax:

UPDATE table_name SET column1 = value1, column2 = value2 WHERE condition;

Whereby UPDATE, SET and WHERE are keywords while in “column1=value1” column1 is the column name and value1 is the value to be stored in the column whose name is column1. The “WHERE condition” specifies the column on which the command will be applied.

Example: *UPDATE people SET age = 40 WHERE P_Id = 3;*

```
MariaDB [wisdom_academic]> UPDATE people SET age = 40 WHERE P_Id = 3;
Query OK, 1 row affected (0.10 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Figure 2. 21. Updating the age to 40 for the record with P_Id = 3 in the people table

In the query above, the record whose P_Id is 3 was updated so that the age becomes 40 as displayed in the table below:

```
MariaDB [wisdom_academic]> select * from people;
+-----+-----+-----+-----+-----+
| P_Id | LastName | FirstName | City | Age |
+-----+-----+-----+-----+-----+
| 1 | KAREGESA | EMILE | Ruhango | 25 |
| 2 | MUGABO | Jean | Kigali | 30 |
| 3 | UWIMANA | Alice | Huye | 40 |
| 4 | GAKURU | Claude | Rubavu | 28 |
| 5 | MUTONI | Grace | Musanze | 24 |
| 6 | KAMANA | Patrick | Ruhango | 35 |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Figure 2. 22. After update: Alice's age changed to 40 (P_Id = 3)

2.4.2 Deleting data with DELETE

In a database, information is stored in tables in the form of records. Over time, some of these records may become unnecessary, incorrect, or outdated. The **DELETE command** in SQL is used to remove one or more records from a table.

The **DELETE** statement allows you to **remove one or more records** from a table. Its syntax is:

DELETE FROM table_name WHERE condition;

Whereby DELETE FROM and WHERE are keywords while table_name is the table in which the data to delete is stored and “condition” is the criteria for the column’s cell in which data must be deleted.

Example: *DELETE FROM people WHERE P_Id = 5;*

```
MariaDB [wisdom_academic]>
MariaDB [wisdom_academic]> DELETE FROM people WHERE P_Id = 5;
Query OK, 1 row affected (0.11 sec)
```

Figure 2. 23. Deleting the record with P_Id = 5 from the people table

The command above deletes the record of the people whose P_Id is 5. After deleting it you can see in the table below that the row with P_Id=5 is no longer there because it is already deleted.

```
MariaDB [wisdom_academic]> select * from people;
+-----+-----+-----+-----+-----+
| P_Id | LastName | FirstName | City   | Age |
+-----+-----+-----+-----+-----+
| 1    | KAREGESA | EMILE    | Ruhango | 25 |
| 2    | MUGABO  | Jean    | Kigali  | 30 |
| 3    | UWIMANA | Alice    | Huye    | 40 |
| 4    | GAKURU  | Claude   | Rubavu  | 28 |
| 6    | KAMANA  | Patrick  | Ruhango | 35 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Figure 2. 24. The People table after deleting the row having P_Id=5

Application activity 2.4

Section B: Choose the correct answer.

- 1) What is the purpose of the Update statement in SQL?
 - a) To remove records
 - b) To modify existing data in a table
 - c) To create a new table
 - d) To sort data
- 2) Which SQL statement correctly updates the age of a person with P_Id = 3?
 - a) Update people SET age = 40 where P_Id = 3;
 - b) Delete from people where P_Id = 3;
 - c) Select age from people where P_Id = 3;
 - d) Insert into people age = 40;
- 3) What is the function of the Delete statement?
 - a) To update records
 - b) To display data
 - c) To remove records from a table
 - d) To create a database
- 4) What happens if you use Delete without a Where clause?

- a) Only one record is deleted
- b) All records in the table are deleted
- 5) Which SQL statement correctly deletes a record with P_Id = 5?
 - a) Remove from people where P_Id = 5;
 - b) Delete people where P_Id = 5;
 - c) Delete from people where P_Id = 5;
 - d) Drop people where P_Id = 5;

Section B: Answer the following questions.

- 1) What SQL command is used to remove records from a table?
- 2) What might happen if the WHERE clause is omitted in DELETE command?
- 3) Write an SQL statement to update the age of a person to 35 where P_Id = 2 in the people table.
- 4) Analyze the following query and explain its result:
UPDATE people SET age = 50;
- 5) Create an SQL command that updates the city of a person to Kigali where P_Id = 6 and deletes the record of P_Id = 7.

2.5 SQL constraints

Learning activity 2.5

A secondary school is developing a database system to manage information about students, employees, and departments. The database administrator wants to ensure that the data entered into the tables is accurate and reliable. To achieve this, the administrator decides to apply specific rules so that each employee has a unique identification number, some fields such as names must always be provided, and employees must belong only to existing departments.

- 1) Why must each employee have a unique ID in the database?
- 2) What happens if a record is inserted without a name?
- 3) Why should employees belong only to existing departments?
- 4) Which rules help ensure data accuracy and table relationships?

Constraints are rules applied to columns in a table to control the type of data that can be stored. Any attempt to change the table content in a way that violates those constraints is rejected. Constraints are commonly used when creating or modifying tables, and they prevent invalid data from being inserted into the table.

Constraints help make sure data in a database present the following qualities:

- They ensure data integrity by making sure data is complete, correct, and not changed in a wrong way,
- They ensure accuracy making sure data is correct and free from errors,
- They ensure reliability by making sure data can be trusted and used without doubt.

2.5.1 Primary key

A Primary Key is a constraint used to uniquely identify each record in a table. Each table can have only one primary key and the values in this column must be unique and cannot be NULL.

Characteristics of a Primary Key:

- Uniquely identifies each row in a table.
- Cannot contain NULL values.
- Must contain unique values.
- A table can have only one primary key.

How to create a primary key

To create a primary key, include in your SQL query a statement to specify that a given column is a primary key like in the query below.

```
CREATE TABLE Employees ( Emp_ID INT PRIMARY KEY, FirstName VARCHAR(50), LastName VARCHAR(50));
```

In the query above Emp_ID is the primary key and ensures that each employee has a unique identification number.

2.5.2 NOT NULL

The NOT NULL constraint ensures that a column cannot contain empty or null values. This means that a value must always be provided when inserting a record into that column.

A NOT NULL constraint is added to a column by using that keyword in a query to create a table.

Example:

```
CREATE TABLE Employees ( Student_ID INT, Name VARCHAR(50) NOT NULL, Course VARCHAR(50));
```

In this query above, using the statement “Name VARCHAR(50) NOT NULL” mean that the Name column will never accept null values. Any attempt to insert null values will fail.

2.5.3 Foreign key

A foreign key is a column in one database table that refers to the primary key of another table. It creates a relationship between the two tables and helps maintain referential integrity to ensure that related data stays consistent.

This constraint is used to create a relationship between two tables thereby ensuring that the value in one table matches a value in the primary key of another table.

Example:

Consider two tables “Departments” and “Employees”. The Department table has a primary key (Dept_ID) while the Employees table has a Dept_ID column which is a foreign key and must contain only data which is in the Department Dept_ID column.

Creating those tables, use these SQL queries:

```
CREATE TABLE Departments (Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(50));
```

```
CREATE TABLE Employees ( Emp_ID INT PRIMARY KEY, Name VARCHAR(50), Dept_ID INT,  
FOREIGN KEY (Dept_ID) REFERENCES Departments(Dept_ID));
```

Application activity 2.5

Section A: Choose the correct answer.

1) What is the main purpose of constraints in a database?

- | | |
|--|--------------------|
| a) To delete tables | c) To display data |
| b) To ensure data integrity and accuracy | d) To sort records |

2) What is a primary key?

- | | |
|---|-------------------------------------|
| a) A column that stores duplicate values | c) A column that allows NULL values |
| b) A constraint that uniquely identifies each record in a table | d) A type of chart |

3) What does the NOT NULL constraint ensure?

- | | |
|--------------------------------------|--------------------------------------|
| a) A column can be empty | c) A column is deleted automatically |
| b) A column must always have a value | d) A column becomes a primary key |

4) What is the role of a foreign key?

- | | |
|--|----------------------|
| a) To delete records | c) To sort data |
| b) To create a relationship between two tables | d) To rename columns |

Section B: Answer the following questions.

- 1) Why must each employee have a unique ID?
- 2) Write an SQL command to create a table with a primary key and Name as NOT NULL.
- 3) Evaluate what would happen if the NOT NULL constraint was applied.

2.6 Aggregate Functions

Learning activity 2.6

A university wants to analyze employee salaries, student scores, and department sizes using aggregate functions like SUM, AVG, MAX, MIN, and COUNT.

- 1) Name one aggregate function used to calculate the total of a numeric column.
- 2) Which function finds the highest value in a column?
- 3) Write an SQL query to count the number of students in each department.
- 4) what will this query display as a result?

```
SELECT MAX(Salary) FROM Employees;?
```

Aggregate functions are special functions that perform a calculation on a group of rows and return a single result. They are commonly used to summarize data in a table, such as counting records, calculating totals, averages, and finding the highest or lowest values. They are often used with the SELECT statement to analyze data stored in a database table.

Consider the following table:

```
MariaDB [wisdom_academic]> select * from employees;
```

Emp_ID	FirstName	City	Salary
1	Emile	Ruhango	230000
2	Alice	Nyagatare	280000
3	Ruth	Muhanga	120000
4	John	Nyagatare	150000
5	David	Kigali	310000
6	Diane	Nyagatare	220000

6 rows in set (0.00 sec)

Figure 2.28: All employees: ID, first name, city, salary.

In the table above, some people are from the same city. An aggregate function can be used to calculate the total salary of people from that same city.

2.6.1 COUNT function

COUNT () function is an aggregate function used to count the number of rows (records) in a table or the number of values in a column.

Its syntax is: *SELECT COUNT(column_name) FROM table_name;*

Example: *SELECT COUNT(Emp_ID) FROM Employees;*

```
MariaDB [wisdom_academic]> SELECT COUNT(Emp_ID) FROM Employees;
+-----+
| COUNT(Emp_ID) |
+-----+
|          6 |
+-----+
1 row in set (0.00 sec)
```

Figure 2. 25. Counting total employees using the count function

This query counts the total number of employees in the Employees table.

2.6.2 SUM function

SUM () function is an aggregate function used to calculate the total value of a numeric column in a table. It adds all the numbers in the specified column and returns the overall total.

This function is commonly used when working with numerical data such as salaries, prices, quantities, marks, or sales amounts where it is necessary to know the total value of all records in a dataset.

Its syntax is: *SELECT SUM (column_name) FROM table_name;*

Here below is an example of a query that calculates the total salary of all employees:

SELECT SUM(Salary) FROM Employees;

```
MariaDB [wisdom_academic]> SELECT SUM(Salary) FROM Employees;
+-----+
| SUM(Salary) |
+-----+
|    1310000 |
+-----+
1 row in set (0.01 sec)
```

Figure 2. 26. Total salary of all employees calculated using the sum function

2.6.3 AVG function

AVG () function is an aggregate function used to calculate the average (mean) value of a numeric column in a table. It adds all the values in the specified column and then divides the total by the number of records to determine the average.

The AVG () function is commonly used when analyzing numerical data such as marks, salaries, prices, or scores, where it is necessary to determine the average value of the dataset.

Its syntax is *SELECT AVG(column_name) FROM table_name;*

Here below is the result of a query that calculates the average salary of employees by using the query *SELECT AVG(Salary) FROM Employees;*

```
MariaDB [wisdom_academic]> SELECT AVG(Salary) FROM Employees;
+-----+
| AVG(Salary) |
+-----+
| 218333.3333 |
+-----+
1 row in set (0.04 sec)
```

Figure 2. 27. Average employee salary

2.6.4 MAX function

The MAX() function in SQL is an aggregate function that returns the largest value in a column.

Its syntax is *SELECT MAX (column_name) FROM table_name;*

The query below shows the highest salary among employees:

SELECT MAX(Salary) FROM Employees;

```
MariaDB [wisdom_academic]> SELECT MAX(Salary) FROM Employees;
+-----+
| MAX(Salary) |
+-----+
|      310000 |
+-----+
1 row in set (0.01 sec)
```

Figure 2. 28. Calculation of the highest employee salary using the MAX () function

2.6.5 MIN function

The MIN () function is an aggregate function used to find the smallest (minimum) value in a specified column of a table. This function is used when working with numerical data when it is important to identify the smallest value among a group of records.

Its syntax is: *SELECT MIN(column_name) FROM table_name;*

Example: *SELECT MIN(Salary) FROM Employees;*

```

MariaDB [wisdom_academic]> SELECT MIN(Salary) FROM Employees;
+-----+
| MIN(Salary) |
+-----+
|      120000 |
+-----+
1 row in set (0.00 sec)

```

Figure 2.33: Lowest employee salary

The query above uses the MIN() aggregate function to find the lowest salary in the Employees table. When the query is executed, SQL examines all the values in the Salary column and compares them to determine the smallest value. Looking at the Employees table's data, the query will identify that 120000 is the lowest salary.

Application activity 2.6

Section A: Choose the correct answer.

- 1) What is the purpose of aggregate functions in SQL?
 - a) To create tables
 - b) To perform calculations on a set of values and return a single result
 - c) To delete records
 - d) To format data
- 2) Which function is used to count the number of records in a table?
 - a) SUM()
 - b) AVG()
 - c) COUNT()
 - d) MAX()
- 3) What does the SUM() function do?
 - a) Finds the average value
 - b) Counts rows
 - c) Adds all values in a numeric column
 - d) Finds the smallest value
- 4) Which function is used to find the highest value in a column?
 - a) MIN()
 - b) MAX()
 - c) AVG()
 - d) COUNT()
- 5) What does the AVG() function calculate?
 - a) Total value
 - b) Highest value
 - c) Average (mean) value
 - d) Number of records

Section B: Answer the following questions.

- 1) Which SQL function is used to count the number of employees in the table?
- 2) Explain how the AVG function works when calculating average salary.
- 3) Write an SQL query to count the total number of employees in the Employees table.
- 4) Create an SQL query that lists the total salary, average salary, highest salary, and lowest salary of employees in a single result.

2.7 Filtering and sorting data

Learning activity 2.7

A company has stored information about its employees in a database table called Employees. The table includes details such as Employee ID, First Name, Last Name, Address, City, and Salary.

One day, the company manager urgently needs specific information to prepare a report for a meeting. The manager wants to know which employees live in Nyagatare, which employees earn more than 200000, and how employees can be arranged from the highest salary to the lowest salary.

- 1) Explain how you would know staff who live in Nyagatare?
- 2) How can you know employees who earns more than 200,000?
- 3) How to list employees from the highest salary to the lowest?

When working with databases, users often need to retrieve specific information from a table. In SQL, filtering and sorting data help users select the exact records they need and arrange them in a meaningful order.

2.7.1 Filtering data

a) Using WHERE

The WHERE clause in SQL is used to filter records and retrieve only the data that meets a specified condition. Instead of displaying all records in a table, the WHERE clause allows users to select specific rows based on criteria such as values, ranges, dates, or text.

Example: *SELECT* FROM People WHERE City = 'Ruhango';*

This query displays only employees who live in Ruhango and this is specified in the clause “WHERE City=’Ruhango’”

```
MariaDB [wisdom_academic]> SELECT* FROM People WHERE City = 'ruhango';
+-----+-----+-----+-----+-----+
| P_Id | LastName | FirstName | City   | Age |
+-----+-----+-----+-----+-----+
| 1    | KAREGESA | EMILE    | Ruhango | 25 |
| 6    | KAMANA   | Patrick  | Ruhango | 35 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Figure 2. 29. Filtering the People table for records where City is Ruhango

b) Using AND condition

The AND condition is used to combine two or more conditions in a query. It ensures that all specified conditions must be true for a record to be included in the results. If even one condition is not met, the record will not be displayed.

Syntax: *SELECT column_name FROM table_name WHERE condition1 AND condition2;*

Example: *SELECT * FROM Employees WHERE City = 'Ruhango' AND age= 35;*

This query retrieves all columns (*) from the Employees table for employees who satisfy both conditions: they must be from Ruhango and their age must be 35 years.

```
MariaDB [wisdom_academic]> SELECT * FROM people WHERE City = 'Ruhango' AND age= 35;
+-----+-----+-----+-----+-----+
| P_Id | LastName | FirstName | City   | Age |
+-----+-----+-----+-----+-----+
| 6    | KAMANA   | Patrick   | Ruhango | 35  |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Figure 2.35 Filtering the People table for records where City is Ruhango and AGE IS 35

c) Using OR condition

When the OR condition is used in a query, a record is included in the result if at least one of the specified conditions is true.

Syntax: *SELECT column_name FROM table_name WHERE condition1 OR condition2;*

Example: *SELECT * FROM people WHERE City = 'Huye' OR City = 'Musanze';*

This query will retrieve all columns (*) from the People table for records where the City is either Huye or Musanze. The OR operator means that a record will be included in the results if it satisfies at least one of the specified conditions.

```
MariaDB [wisdom_academic]> SELECT * FROM people WHERE City = 'Huye' OR City = 'Musanze';
+-----+-----+-----+-----+-----+
| P_Id | LastName | FirstName | City   | Age |
+-----+-----+-----+-----+-----+
| 3    | UWIMANA  | Alice    | Huye   | 22  |
| 5    | MUTONI   | Grace    | Musanze | 24  |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Figure 2. 30. Filtering the People table for records where City is Huye or Musanze

2.7.2 Sorting data

ORDER BY is a clause used to sort the result of a query based on one or more columns in either ascending or descending order.

a) Ascending order

Ascending order (ASC) is the instruction used to arrange data from the smallest value to the largest value. It is the most common way to organize information, such as listing names alphabetically or sorting prices from cheapest to most expensive.

Example: *SELECT * FROM people ORDER BY age ASC;*

```
MariaDB [wisdom_academic]> SELECT * FROM people ORDER BY age ASC;
+-----+-----+-----+-----+-----+
| P_Id | LastName | FirstName | City   | Age |
+-----+-----+-----+-----+-----+
| 3    | UWIMANA  | Alice     | Huye   | 22  |
| 5    | MUTONI   | Grace     | Musanze| 24  |
| 1    | KAREGESA | EMILE     | Ruhango| 25  |
| 4    | GAKURU   | Claude    | Rubavu | 28  |
| 2    | MUGABO   | Jean      | Kigali | 30  |
| 6    | KAMANA   | Patrick   | Ruhango| 35  |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Figure 2. 31. All people record sorted by Age ascending

b) Descending order (DESC)

The descending order (DESC) is used to sort data from the highest value to the lowest value. It is essentially the reverse of the standard alphabetical or numerical order.

Example: *SELECT * FROM People ORDER BY age DESC;*

```
MariaDB [wisdom_academic]> SELECT * FROM people ORDER BY age DESC;
+-----+-----+-----+-----+-----+
| P_Id | LastName | FirstName | City   | Age |
+-----+-----+-----+-----+-----+
| 6    | KAMANA   | Patrick   | Ruhango| 35  |
| 2    | MUGABO   | Jean      | Kigali | 30  |
| 4    | GAKURU   | Claude    | Rubavu | 28  |
| 1    | KAREGESA | EMILE     | Ruhango| 25  |
| 5    | MUTONI   | Grace     | Musanze| 24  |
| 3    | UWIMANA  | Alice     | Huye   | 22  |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Figure 2. 32. All people record sorted in descending order by Age

c) Combining filtering and sorting

Combining filtering and sorting is using the WHERE clause to select specific records and the **ORDER BY** clause to arrange the selected records in a particular order. This allows users to retrieve only the required data and display it in a well-organized manner.

Example: *SELECT * FROM people WHERE age > 24 ORDER BY FirstName DESC;*

```
MariaDB [wisdom_academic]> SELECT * FROM people WHERE age > 24 ORDER BY FirstName DESC;
```

P_Id	LastName	FirstName	City	Age
6	KAMANA	Patrick	Ruhango	35
2	MUGABO	Jean	Kigali	30
1	KAREGESA	EMILE	Ruhango	25
4	GAKURU	Claude	Rubavu	28

4 rows in set (0.00 sec)

Figure 2. 33. Filtering people with age greater than 24 sorted by first name descending

The query above selects people whose salary is greater than 150,000 and displays only those records. The selected records are then sorted by first name in descending order, meaning the names are displayed from the highest alphabetical value to the lowest.

Application activity 2.7

Section A: Choose the correct answer.

- What is the purpose of the WHERE clause in SQL?
 - To delete table
 - To filter records based on a condition
 - To sort data
 - To create a database
- Which operator is used when all conditions must be true?
 - OR
 - NOT
 - AND
 - BY
- Which SQL statement correctly selects people who live in Huye or Musanze?
 - Select * from People where City = 'Huye' And City = 'Musanze';
 - Select * from People where City = 'Huye' Or City = 'Musanze';
 - Select * from People where City = 'Huye' Not 'Musanze';
 - Select * from People where City In All
- What does ORDER BY ASC do?
 - Sorts data from highest to lowest
 - Deletes records
 - Sorts data from lowest to highest
 - Filters records
- Which SQL statement correctly combines filtering and sorting?
 - Select * from People ORDER BY age > 24;
 - Select * from People WHERE age > 24 order by firstname desc;
 - Select * from People WHERE ORDER BY age desc;
 - Select * from People SORT age DESC;

Section B: Answer the questions

Employees Table

Emp_ID	FirstName	LastName	Address	City	Salary
1	Emile	Karegesa	Esc Byimana	Ruhango	230000
2	Alice	Niyomukiza	ETP Nyarurema	Nyagatare	280000
3	Ruth	Murebwayire	St Joseph	Muhanga	120000
4	John	Karamaga	ETP Nyarurema	Nyagatare	150000
5	David	Ndayishimiye	GS Remera	Kigali	310000
6	Diane	Mukamana	GS Rukomo	Nyagatare	220000
7	Eric	Habimana	GS Kabuga	Ruhango	180000
8	Grace	Uwimana	GS Nyamirama	Gatsibo	200000

- 1) Display employees who live in Nyagatare.
- 2) Show employees whose salary is greater than 200000.
- 3) Display employees who live in Nyagatare AND earn more than 200000.
- 4) Display employees who live in Nyagatare OR Ruhango.
- 5) Sort employees by salary in ascending order.
- 6) Sort employees by salary in descending order.

2.8 End unit assessment

2.8.1 Project activit

Project Title: School Employee Database Management System

- **Guidance:**

Using MySQL, You will create a computerized system to store and manage information about about each employee in the school, including their employee ID number, full name, gender, job position, department, salary, and contact information. It should also allow users to enter new employee data into the system, view all stored records or search for a specific employee, update existing information when changes occur, and delete records when an employee leaves the school.

- **Questions on the Project**

- 1) Explain the steps you followed to create the School database.
- 2) Describe the importance of having primary keys for the tables in the database.
- 3) Explain the importance of using aggregate functions in your new database.

2.8.2 Assessment questions

Section A: Choose the correct answer.

- 1) Which SQL command is used to remove an entire table and all its data from a database?
a) Delete b) Remove c) Truncate d) Drop
- 2) A school database has a Student table and an Enrollments table. Which SQL constraint links these two tables together?

- a) NOT NULL constraint
- b) Foreign Key constraint
- 3) Which SQL statement correctly retrieves only students whose marks are greater than 70 from the students table?

- a) Select *from Students Having marks > 70
- b) Select * from Students Where marks > 70
- c) Select * from Students If marks > 70
- d) Get * From Students Where marks > 70

4) Which aggregate function would a teacher use to find the highest mark scored in an exam from the Results table?

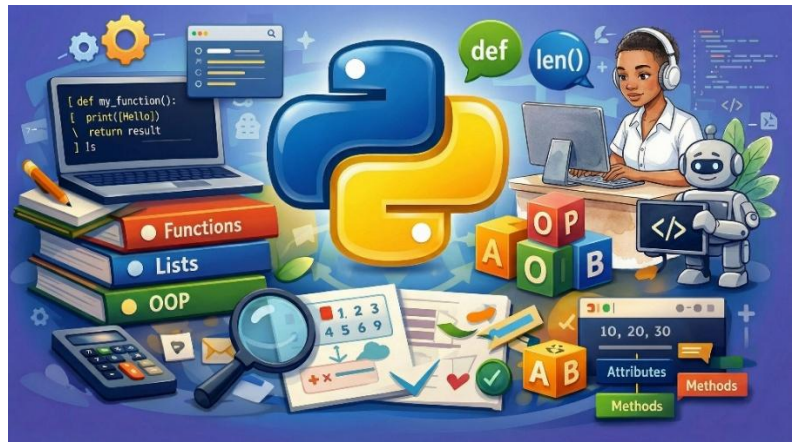
- a) SUM (marks)
- b) AVG (marks)
- c) MAX (marks)
- c) COUNT (marks)

Section B: Answer the following Questions.

- 1) What is the difference between a Primary key and a foreign key in a database table?
- 2) Describe the difference between Create, Alter, and Drop commands in SQL.
- 3) Explain how aggregate functions like SUM() and AVG() help in analyzing data in a table.
- 4) Update the salary of an employee with Emp_ID = 3 to 180,000 using an SQL command.

EXPERIMENTAL

UNIT 3: PYTHON PROGRAMMING II



Key unit competence:

Develop Python programs using functions, lists, and Object-Oriented Programming

Knowledge and understanding

- Explain functions in Python, including their purpose, creation, parameters, return values, and variable scope.
- Explain the use of lists for data storage and manipulation, including their common operations and methods.
- Explain basic Object-Oriented Programming concepts such as classes, objects, attributes, and methods.

Skills

- Use Python functions with parameters, return values, and proper variable scope.
- Create, access, and manipulate lists using loops and common list methods.
- Create a Python program that applies Object-Oriented Programming including classes, objects, attributes, and methods.

Attitudes and values

- Appreciate the importance of modular and reusable code.
- Demonstrate logical thinking and problem-solving through structured programming.
- Value collaboration and teamwork in coding projects.

3.0 Introduction

In this unit, you will learn advanced Python programming concepts that help you write better programs. You will learn how to use functions to break programs into small reusable parts, how to work with lists to store and

manage multiple pieces of data, and how to apply Object-Oriented Programming (OOP) to represent real-world objects using classes and objects.

This will ultimately make you be more capable of writing Python programs using functions, lists, and OOP concepts to solve different problems in a logical and efficient way.

Introductory activity

- 1) What do you think makes a computer program easy to read and understand?
- 2) Have you ever repeated the same task many times in a program? How could you avoid writing the same code again and again?
- 3) If you wanted to store the names of 20 students, would you prefer using 20 separate variables or one single structure? Why?
- 4) What do you understand by the word function and how can it be used in Python programming
- 5) A student and a class of students are different. Using this analogy explain how an element is different from an array.
- 6) a) Why is it important for programmers to write code that can be reused instead of writing everything from scratch each time?
b) Explain how this can be done in Python programming.

3.1. Functions in Python

Learning activity 3.1

- 1) Read the list of Python function features below and match each to the correct real-life example.

Features	Example in daily life
a) Code reusability	i) Dividing cooking into preparation, cooking, and serving
b) Modularity	ii) Using the same recipe card every day
c) Simple Syntax	iii) Adjusting ingredients for different portion sizes
d) Return Values	iv) Cooking a dish and serving it to customers
e) Flexible Arguments	v) Writing steps in short, easy instructions

2. Identify the features and benefits of Python functions.

A function is a named block of code that performs a specific task and can be reused whenever needed in a program. Functions are a core part of Python programming that allow you to write organized, reusable, and efficient. Instead of writing the same code multiple times, a function allows you to define it once and use it whenever needed.

Features of Python functions:

- **Code reusability:** Functions are defined once and can be used many times, reducing repetition in a program.
- **Modularity:** Functions divide a program into smaller, manageable parts, making it easier to understand and organize.
- **Dynamic typing:** There is no need to specify data types for parameters or return values.
- **Flexible arguments:** Functions support different types of arguments, including default, keyword, and variable-length arguments.
- **Return values:** Functions can return single or multiple values as results.
- **First-class objects:** Functions can be assigned to variables, passed as arguments, and returned from other functions, making them very flexible in use.
- **Improved program quality:** Functions make programs more readable, easier to maintain, less error-prone, and faster to develop by promoting structured and reusable code.

3.1.1. Creating and calling a function

Creating a function is defining a named block of code that can take inputs (parameters), perform operations, and optionally return results. Calling a function is executing that block of code by using its name in the program.

There are different types of functions namely Built in function and User Defined functions.

a) Build in function

A built-in function is the one that is already provided by Python and can be used directly without importing any module or writing the function yourself.

These functions are always available when you run a Python program and help perform tasks such as displaying output, calculating values, converting data types, etc.

Some of the most used built in functions in Python are `print ()`, `len ()`, `type ()`, `input ()`, `max ()`, `min ()`.

- **`print ()`:** This function is used to display output on the screen. It allows programmers to show text, numbers, variable values, and the results of calculations while a program is running.

Example:

Program	Output:
<pre>print("Hello, Python!") # Display text x = 10 print("The value of x is:", x) # Display variable value</pre>	<pre>Hello, Python! The value of x is: 10</pre>

- **`Len ()`:** This function is used to determine the number of items in an object, such as the number of characters in a string, elements in a list, etc.

Example:

```
text = "Python"
numbers = [1, 2, 3, 4, 5]
print(len(text))      # Output: 6
```

- **type():** It is used to identify the data type of a value, variable, or object. It helps programmers determine whether a piece of data is an integer, string, float, etc.

Example:

```
x = 10
y = "Hello"
print(type(x)) # Output: <class 'int'>
print(type(y)) # Output: <class 'str'>
```

- **input():** The input() function is used to receive data entered by the user through the keyboard while a program is running.

Example:

Program	Sample Input/Output:
<pre>name = input("Enter your name: ") print("Hello", name)</pre>	Enter your name: Alice Hello Alice

- **max():** It is used to find and return the largest value from a group of values or from a sequence such as a list or string.

Example:

```
numbers = [4, 8, 2, 10, 6]
print(max(numbers)) # Output: 10
```

b) User-defined functions

A user-defined function is the one created by the programmer to perform a specific task. Unlike built-in functions, these functions are not pre-defined in Python. They must be defined using the def keyword before they can be used.

Features of user-defined functions

- They are created using the def keyword
- They can accept parameters (inputs) and can return values using the return statement
- They can be called multiple times in a program

Syntax:

```
def function_name(parameters):  
    # code block  
    return value # optional
```

Example 1: Function Without Parameters

Program	Output
<pre>def greet(): print("Hello, welcome to Python!") greet() # Calling the function</pre>	Hello, welcome to Python!

Example 2: Function with parameters

Program	Output
<pre>def greet(): print("Hello, welcome to Python!") greet() # Calling the function</pre>	Hello Alice ! Hello John !

Example 3: Function with return value

Program	Output
<pre>def add(a, b): return a + b result = add(5, 3) print("Sum:", result)</pre>	Sum: 8

c) Calling a function

Calling a function in Python is telling the computer to execute a specific block of code that has been named and saved for later use. When you define a function using the `def` keyword, you are simply saving it in memory but it can do nothing on its own. Calling it is the act of using it.

To call a function as done in previous examples, you write the function's name followed immediately by a pair of parentheses `()`.

Example:

```
# Defining the function  
def greet():  
    print("Hello, welcome back!")  
# Calling the function  
greet()
```

3.1.2. Local and global variables

Python variables can either be local or global, depending on where they are defined and where they can be accessed.

Local variables are defined inside a function and can only be used within that function. They are created when the function is called and are destroyed when the function ends. Local variables are useful for storing temporary data or performing calculations that do not need to be accessed outside the function.

Example:

```
def greet():  
    message = "Hello, Python!" # Local variable  
    print(message)  
greet() # Output: Hello, Python!
```

Global variables are defined outside all functions and can be accessed anywhere in the program. They are useful when multiple functions need to share and modify the same data. A global variable can be read directly inside a function, but if you want to modify it, you must use the global keyword.

Example of a global variable:

```
message = "Hello, Python!" # Global variable  
  
def greet():  
    print(message) # Accessing global variable  
greet() # Output: Hello, Python!  
print(message) # Output: Hello, Python!
```

Application Activity: 3.1

Section A: Choose the correct answer

1) Which keyword is used to create a user-defined function in Python?

- a) func b) define c) def d) function

2) Which of the following is a built-in Python function?

- a) greet() b) calculate() c) print() d) myFunction()

3) What is the main benefit of using functions in programming?

- a) Makes programs longer c) Allows code reusability and better organization
b) Reduces code readability d) Deletes errors automatically

4) What is a local variable in Python?

- a) A variable used everywhere in the program
b) A variable defined outside all functions

- c) A variable defined inside a function and used only there
- d) A constant value

Section B: Answer the following questions

- 1) Why are local variables important in functions?
- 2) Write a Python function that uses a local variable to calculate the total cost of items
- 3) Write a simple Python function as follow.
Making a cup of tea – inputs could be water and tea leaves, output could be cup of tea.
- 4) Look at the code below and predict its output:

```
def multiply_by_two(number):  
    return number * 2  
  
result = multiply_by_two(7)  
print(result)
```

3.2. Arrays in Python

Learning Activity:3.2

The school librarian, Ms. Amina, has received a donation of books for the library and needs to organize and manage them efficiently using Python programming. The books have different titles, authors, and publication years, and she wants to store all this information in a single structure for easy access. As new books arrive, she needs to add them to the collection, and she also needs to remove books that are damaged or lost. Additionally, Ms. Amina may need to update information about the books, such as correcting titles or authors, and she would like to list all books in alphabetical or reverse order. She also wants to perform repetitive tasks efficiently, such as printing all book titles or counting the total number of books in the library.

- 1) How can Python lists help Ms. Amina manage the library more efficiently?
- 2) Which methods will allow her to add and remove books quickly?

In Python, an array is a data structure that can hold a collection of items, allowing you to store multiple values in a single variable. These items can be of the same type or different types, and they are stored in an ordered sequence.

In Python, an array is commonly implemented using a list. Lists are one of the most commonly used data structures because they are flexible and easy to use when working with multiple pieces of data.

In Python, a list is created by placing elements inside square brackets [], with each item separated by a comma. Lists allow you to store, access, and manipulate data efficiently.

3.2.1 Importance of list

Here is the importance of lists:

- **Efficient data storage:** Lists allow storing many values in one variable, reducing the need for multiple variables. For example: storing marks of many students in one list.
- **Easy data access:** Elements in a list can be accessed using an index, making it simple to retrieve specific data.
- **Flexibility:** They can have a dynamic size whereby they can grow or shrink as needed by adding or removing elements.
- **Supports different data types:** A list can store integers, strings, and other data types in the same structure.
- **Simplifies repetitive tasks:** Lists can be used with loops to perform operations on multiple items efficiently.
- **Easy modification of data:** You can update, delete, or insert elements at any time.

3.2.2 Creating and accessing lists

A list is created by placing elements inside square brackets [], separated by commas. Lists are used to store multiple values in a single variable, making programs simpler and more organized.

- **Creating lists**

The most common way to create a list is to place your items inside square brackets [], separating each item with a comma.

Example:

```
# Creating a list of numbers
numbers = [10, 20, 30, 40]

# Creating a list of strings
names = ["Alice", "John", "Mary"]

# Creating a mixed list
mixed_list = [1, "Hello", 3.5, True]

# Creating an empty list
empty_list = []
```

- **Accessing list elements**

In Python, each element of a list has a position called an index meaning the very first element in the list is at index 0, the second is at index 1, and so on. To access an item, place the index number inside square brackets [] immediately after the list variable.

Example:

```
fruits = ["apple", "banana", "orange", "grape"]  
# Accessing the first element  
print(fruits[0]) # Output: apple  
# Accessing the second element  
print(fruits[1]) # Output: banana  
# Accessing the last element using negative index  
print(fruits[-1]) # Output: grape
```

- **Accessing a range of elements**

Accessing a range of elements in Python is called **slicing**. It allows you to extract a group from a list instead of just a single item. To slice a list, you use square brackets with a colon : separating your boundaries as **list[start:stop:step]**. The “step” part is optional and when it is not specified, Python assumes that the step is 1.

Example:

```
numbers = [10, 20, 30, 40, 50]  
# Access elements from index 1 to 3  
print(numbers[1:4])  
# Output: [20, 30, 40]
```

3.2.3. Modifying list elements

In Python programming language, lists elements can be changed after creation. This allows programmers to easily add, remove, and update items in a list depending on the needs.

a) Adding elements to a list

You can add new elements to a list using different methods:

- **append():** Adds an element at the end of the list
- **insert():** Adds an element at a specific position. While inserting an element, you must specify the index at which it will be inserted.
- **extend()** Adds multiple elements from another list

Example:

Program	Output
<pre>numbers = [10, 20, 30] # Add element at the end numbers.append(40)</pre>	<pre>[10, 15, 20, 30, 40, 50, 60]</pre>

<pre># Insert element at index 1 numbers.insert(1, 15) # Add multiple elements numbers.extend([50, 60]) print(numbers)</pre>	
--	--

b) Removing elements from a List

Elements can be removed using one of these methods:

- **remove()**: Removes a specific value.
- **pop()**: Removes an element located at a specified index or the last element if no index is given.
- **Del**: Deletes an element or entire list. It is not a function meaning that you have to specify the list name and the index in parenthesis.

Example:

Program	Output
<pre>fruits = ["apple", "banana", "orange", "grape"] # Remove specific item fruits.remove("banana") # Remove last item fruits.pop() # Remove item at index 0 del fruits[0] print(fruits)</pre>	['orange']

c) Updating list elements

You can change the value of an element by assigning a new value using its index.

Example:

Program	Output
<pre>marks = [70, 80, 90] # Update second element marks[1] = 85 print(marks)</pre>	[70, 85, 90]

3.2.4 Iterating through lists using loops

Iterating through a list is going through each element one by one to perform a specific action, such as displaying values, calculating totals, or modifying elements. This is commonly done using loops.

a) Using a For loop

Using For loop in accessing list elements is a process by which, instead of reaching items one by one using manual indices (like list[0], list[1]), a for loop lets Python automatically handle this from start to finish. Two methods that can be used is accessing list elements directly or using indices.

- **Accessing list elements directly**

In the example in the example below Python will loop search through the list from the beginning to the end by the use of the for loop. If you forget num, in or : the program will give an error. Num is a variable representing each item in the list during each iteration.

Program example	Output
<code>numbers = [10, 20, 30, 40]</code> <code>for num in numbers:</code> <code>print(num)</code>	10 20 30 40

- **Accessing list elements using a for loop with index:**

This means that instead of accessing items directly, the loop moves through the different indices of the list (0, 1, 2, 3...)

Once you have the index number, you look inside the list at that exact position to retrieve your data.

Program example	Output
<code>fruits = ["apple", "banana", "orange"]</code> <code>for i in range(3):</code> <code>print("Index:", i, "Value:", fruits[i])</code>	Index: 0 Value: apple Index: 1 Value: banana Index: 2 Value: orange

Note that “for i in range(3):” can be replaced by “for i in range(len(fruits)):” whereby the function len() calculates the length of the fruits list.

b) Accessing list elements using a While loop

Accessing list elements using a while loop means you have a counter variable to move through the list's index numbers one by one. Contrary to the for loop which handles the progression automatically, a while loop runs as long as a specific condition is true.

Program example	Output
<code>marks = [70, 80, 90, 85]</code> <code>i = 0</code> <code>while i < len(marks):</code> <code>print(marks[i])</code> <code>i += 1</code>	70 80 90 85

3.2.5. Common list methods

Python lists come with built-in methods that make it easy to manipulate and manage data. These methods allow you to add, remove, organize, and analyze list elements efficiently. The common list methods are `append()`, `remove()`, `pop()`, `sort()`, `reverse()`, `insert()`.

- **append():** Adds an element to the end of the list

Program example	Output
<pre>numbers = [10, 20, 30] numbers.append(40) print(numbers)</pre>	[10, 20, 30, 40]

- **remove():** Removes a specific element from the list.

Program	Output
<pre>fruits = ["apple", "banana", "orange"] fruits.remove("banana") print(fruits)</pre>	["apple", "orange"]

- **pop():** Removes and returns an element at a given index (default is the last element).

Example:

```
marks = [70, 80, 90]
marks.pop()
print(marks) # Output: [70, 80]
```

- **sort():** Sorts the list in ascending order

```
numbers = [30, 10, 20]
numbers.sort()
print(numbers) # Output: [10, 20, 30]
```

- **Reverse():** This reverses the order of elements in the list.

Example:

```
numbers = [1, 2, 3, 4]
numbers.reverse()
print(numbers) # Output: [4, 3, 2, 1]
```

- **insert():** This inserts an element at a specific position.

Example:

```
names = ["Alice", "John"]
names.insert(1, "Mary")
print(names) # Output: ["Alice", "Mary", "John"]
```

Application activity:3.2

Section A: Choose the correct answer

1) How are lists created in Python?

- a) Using parentheses ()
- b) Using curly braces {}
- c) Using square brackets []
- d) Using angle brackets <>

2) Which method is used to add an element at the end of a list?

- a) remove()
- b) insert()
- c) append()
- d) pop()

3) Which loop is commonly used to iterate through a list?

- a) if statement
- b) while statement only
- c) for loop
- d) switch statement

Section B: Answer the following questions

1) Create a list of items sold in the canteen: ["Bread", "Juice", "Soda", "Chips"] and answer the questions.

- a) Use a loop to display all items sold.
- b) Add "Cake" to the list using append().
- c) Remove "Soda" from the list using remove().
- d) Sort the list alphabetically using sort() and display it.
- e) Write a for loop that prints each item with a message like "Bread is available"
- f) Write a while loop that prints items until "Chips" is reached.
- g) Update "Juice" to "Fruit Juice" in the list.

2) Explain why lists are better than using multiple separate variables.

3.3 Introduction to Object-Oriented Programming (OOP)

Learning activity 3.3

Your school library wants to manage its books and members efficiently. Every book has a title, author, and number of copies, and every member has a name, grade, and borrowed books.

Instead of keeping separate lists for each book or member, you can create classes in Python keep that information.

A Book class can store the properties of a book such as title, author, copies and have methods like display_info() or borrow_book() while a member class can store information about students and methods like borrow_book() or return_book().

1) In Object-Oriented Programming (OOP), a class is

- a) A specific book in the library
- b) A template for creating objects
- c) A type of loop
- d) A Python operator

- 2) Which of the following could be a class in a library management system?
 a) Book b) Mathematics for Rwanda c) John d) Grade 5
- 3) What is an object in OOP?
 a) A programming language b) A model for creating data
 c) A specific instance of a class d) A type of variable
- 4) If Book is a class, which of the following is an object?
 a) Title b) Author c) Mathematics for Rwanda d) Number of copies
- 6) In a library system, borrow_book() is an example of:
 a) An attribute b) A class c) A method d) An object
- 7) What is the main purpose of a method in a class?
 a) To store data about an object b) To define actions an object can perform
 c) To create variables d) To display error messages
- 8) What is the purpose of the __init__() method in a class?
 a) To initialize the attributes of an object b) To delete an object
 c) To print information d) To stop the program
- 9) Why is OOP useful in a library management system provided in the scenario above?
 a) It organizes related data and actions together b) It removes the need for data storage
 c) It replaces Python functions completely d) It only works with books and libraries
- 10) In your own words explain OOP.

3.3.1 Concept of object and class

Object-Oriented Programming (OOP) in Python is a way of writing programs by organizing them as real-world things called objects. In this approach, a class acts as a template or model that defines what an object should look like and what it can do.

For example, in a school library system, a Book class can describe common features of all books such as title, author, and number of copies, as well as actions like displaying book information or borrowing a book.

An object is a specific example created from a class, such as a particular book like “Mathematics for Rwanda.” Each object has its own data but follows the structure defined by the class. The information stored in an object is called attributes, while the actions it can perform are called methods.

Example of a car:

A class is the general design or template of the car. It defines what every car should have and what it can do. A Car class may describe that every car has a color, model, and speed, and it can move or stop.

An object is a real, specific car created from that class. A “Toyota Corolla” is an object. Each car object has its own values, such as red color or blue color.

Attributes are the characteristics or properties of the car object. These include things like color, model, engine size, or speed. For example, a car might have the attribute: color = red.

Methods are the actions or behaviors that the car can perform. These include actions like start(), stop(), accelerate(), or brake(). These methods describe what the car can do.

3.3.2 Creating a class in Python

A class is created using the keyword Class. It serves as a blueprint for creating objects by defining their attributes (variables) and methods (functions).

Syntax of a class:

```
class ClassName:
    def __init__(self, parameters):
        # attributes
        self.variable = parameters
    def method_name(self):
        # method body
```

With this syntax here are more explanations:

- Class is the keyword used to define a class
- ClassName is the name of the class such as Student, Car, Book, etc.
- def __init__(self, parameters) is a special method called the constructor. A constructor runs automatically when an object is created and is used to initialize attributes by assigning initial values to the object. The keyword self refers to the current object being created, while the parameters represent the values passed during object creation and are used to set up the object's attributes.
- The line "self.variable = parameters" creates and assigns attributes to the object. The "self.variable" means the attribute belongs to the object while "parameters" are the values given when the object is created.

Example:

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks
    def display(self):
        print("Name:", self.name)
        print("Marks:", self.marks)
```

3.3.3 Creating and using objects

An object is an instance of a class. After creating a class, objects are created to use the properties (attributes) and behaviors (methods) defined in that class.

- **Creating an object**

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks
# Creating objects
student1 = Student("Alice", 85)
student2 = Student("John", 90)
```

- **Using objects**

Using an object is interacting with the specific instance (student1) that you created from the Student template. Once an object is created, using it typically involves two actions namely accessing attributes and calling methods.

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks
    def display(self):
        print("Name:", self.name)
        print("Marks:", self.marks)
# Creating objects
student1 = Student("Alice", 85)
# Accessing attributes
print(student1.name)
print(student1.marks)
# Calling method
student1.display()
```

The output will be:

```
Alice
85
Name: Alice
Marks: 85
```

Note that student1 is an object of the Student class. Through this object, you can interact with the class components: student1.name directly accesses its specific data attribute, while student1.display() calls the behaviors and functions defined within that template.

Application activity 3.3

Section A: Choose the correct answer

1) Object-Oriented Programming is:

- a) A method of designing websites
- b) A programming approach that uses objects and classes
- c) A type of database
- d) A spreadsheet tool

2) What is a class in Python?

- a) A function that runs automatically
- b) A blueprint or template for creating objects
- c) A loop structure
- d) A variable type

3) What is an object in OOP?

- a) A database
- b) A programming error
- c) An instance of a class
- d) A loop

4) Which keyword is used to create a class in Python?

- a) def
- b) class
- c) object
- d) create

5) What is the role of the `__init__()` method in a class?

- a) To delete objects
- b) To display output
- c) To initialize object attributes
- d) To loop through data

Section B: Do the questions

Your school library wants to manage all its books and members efficiently. Every book has a title, author, and number of copies, and every member has a name, grade, and borrowed books.

- 1) Define a Book class with attributes title, author, and copies, and a method `display_info()` that prints all the book details.
- 2) Create two objects of the Book class and call `display_info()` for each.
- 3) Define a Member class with attributes name, grade, and `borrowed_books`, and a method `borrow_book()` to add a book to `borrowed_books`.
- 4) Create a member object and let the member borrow one of the books created.
- 5) Explain why using classes and objects makes managing books and members easier than using separate variables.

End unit assessment

1) Which keyword is used in Python to define a function?

- a) function
- b) define
- c) def
- d) func

2) What is the correct way to access the third element in a Python list named 'marks'?

- a) `marks[3]`
- b) `marks(2)`
- c) `marks[2]`
- d) `marks.get(3)`

3) A variable declared inside a function in Python is called a:

- a) Global variable
- b) `marks.get(3)`
- c) Static variable
- d) Public variable

4) Which Python method is used to add a new element at the end of an existing list?

- a) `list.add()`
- b) `list.insert()`
- c) `list.extend()`
- d) `list.append()`

5) In OOP, which term refers to a blueprint used to create objects with shared attributes and methods?

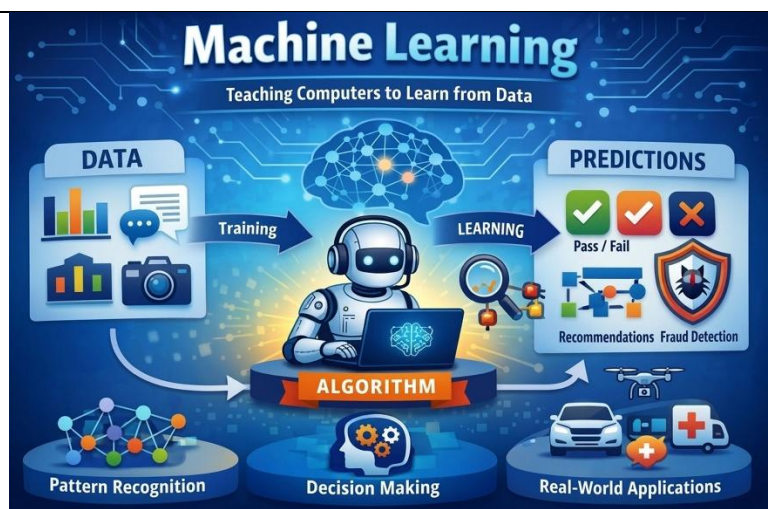
- a) Function
- b) Module
- c) Class
- d) Array

Section B:

- 1) What is a Python list? Explain two reasons why lists are useful in programming.
- 2) Explain the difference between a class and an object in OOP.
- 3) Write a Python function called `greet_user()` that prints "Hello, welcome to Python programming!"
- 4) Write a Python function that accepts two numbers as parameters and returns their sum. Call the function with sample values and display the result.
- 5) Create a class `Book` with attributes `title` and `author`, and a method `display_info()` that prints the book details. Instantiate an object of the class and call the method.
- 6) Explain why using functions makes a program easier to maintain.
- 7) Compare using a Python list to store data versus using separate variables for each item. Give one advantage of lists.
- 8) Examine the following code and predict the output:

```
def add_numbers(a, b=5):  
    return a + b  
  
result = add_numbers(3)  
print(result)
```


UNIT 4: MACHINE LEARNING



Key unit competence:

Learners should be able to develop machine learning models to solve real-world problems responsibly and ethically using collected and analyzed data.

Learning objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Describe key concepts of data, including data types, sources, formats, and characteristics of quality data.
- Explain different methods of data acquisition and data preprocessing techniques.
- Explain the main categories of Machine Learning and the core steps of ML pipeline.

Skills

- Prepare datasets by checking accuracy, completeness, consistency, and timeliness before analysis.
- Manage data using spreadsheets or databases, and use visualization tools to interpret graphs and charts for decision-making.
- Build and test simple ML models using practical tools

Attitudes and values

- Demonstrate a positive attitude toward quality data for effective analysis and informed decision-making.
- Show curiosity and openness toward Artificial Intelligence (AI) and Machine Learning technologies and their practical applications.
- Appreciate the importance of Machine Learning in solving real life problems.

4.0 Introduction

In this unit, you will learn the concepts of Machine Learning that help you train computers to solve real-world problems. You will learn how to collect and clean data to ensure it is accurate and complete, how to use spreadsheets to organize and visualize information for better decision-making, and how to build and test simple machine learning models using practical tools. You will also learn how to handle data responsibly and ethically, respecting privacy and consent.

This will ultimately make you more capable of developing machine learning models using high-quality data to solve real life challenges in an ethical way.

Introductory activity

One Friday in the school computer lab, Aline was working on a project about students' performance in Mathematics. She collected students' marks from the last three tests and entered them into an Excel spreadsheet. She noticed that some students who regularly completed homework scored higher marks, while others who missed classes often performed poorly.

Curious about this pattern, Aline asked the ICT teacher, if the computer could help predict which students might need extra support before the final exam. The teacher explained that by using Machine Learning, the computer could analyze past data such as attendance, homework completion, and previous marks to identify patterns and make predictions.

Choose the right answer

- 1) Why is it important to use accurate and complete data when training a Machine Learning model?
 - a) To ensure the model takes as long as possible to train.
 - b) Because a model's outputs becomes better.
 - c) It allows the model to work without needing any computer algorithms.
 - d) It guarantees that every student will receive a passing grade.
- 2) Should teachers use only a machine learning model to decide which students need extra support? Explain.
 - a) Yes, because these models are always 100% accurate and lack human bias.
 - b) Yes, because it saves teachers time from having to talk to their students.
 - c) No, because models can make mistakes and the lack human context that a teacher understands.
 - d) No, because using computer models in schools is strictly against educational laws.
- 3) What are the challenges that can arise when using Machine Learning in education?
 - a) Students might learn the material too quickly, causing them to get bored.
 - b) Computer screens might emit too much blue light during data entry.

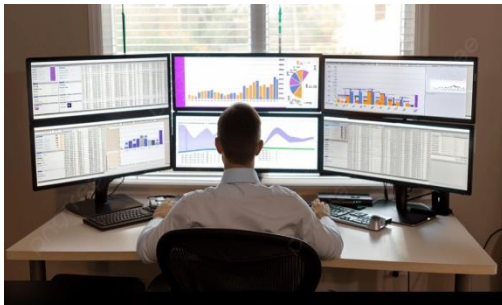
- c) There can be issues related to data privacy and relying too heavily on automated decisions.
- d) The model may accidentally delete the school's data.

Answer these questions

- 1) How could a computer help Aline analyze the students' data?
- 2) What types of data could be collected to help predict students' final exam performance?
- 3) How can a machine learning model learn from past student data?
- 4) How could the model identify students who may need extra academic support?
- 5) Why can the model's predictions not always be correct and how can it be improved?

4.1 Data fundamentals

Learning activity 4.1



The school headteacher wants to evaluate student performance in three key subjects: ICT, Physics and Mathematics in order to compare students' performance and identify. This would help in deciding about providing extra academic support, allocating teachers, and changing the teaching strategies.

To do this evaluation, the school has various records including students' exam marks stored in spreadsheets,

attendance records, teachers' written comments and practical project reports.

However, this information is in different formats and locations. Therefore, the headteacher requests that it gets organized in order to generate a performance report that will support effective decision-making.

- 1) What are data in computing?
 - a) Processed and meaningful facts
 - b) Raw facts and figures that are not organized
 - c) Final reports for decision-making
 - d) Conclusions drawn from analysis
- 2) Which of the following best describes knowledge?
 - a) Raw numbers and facts
 - b) Organized tables
 - c) Understanding used to make decisions
 - d) Unprocessed observations
- 3) A teacher revising a topic after seeing many students fail is an example of:
 - a) Data collection
 - b) Information storage
 - c) Knowledge application
 - d) Data deletion
- 4) Structured data is best described as:
 - a) Unorganized files
 - b) Data in rows and columns
 - c) Audio recordings
 - d) Handwritten notes only
- 5) Why is structured data important?
 - a) It is difficult to analyze
 - b) It cannot be stored
 - c) It is easy to search and analyze
 - d) It contains only text
- 6) Exam results collected directly from students are:
 - a) Secondary data
 - b) Primary data
 - c) External data
 - d) Digital data only

- 7) Using last year's data to make decisions for this year fails meet which data quality characteristic?
a) Timeliness b) Accuracy c) Uniqueness d) Structure
- 8) The main purpose of organizing school data is to:
a) Delete records b) Support decision-making
c) Reduce attendance d) Replace teachers

Machines learn from data in order to identify patterns, make predictions, and support decision-making. For better learning from it data should be correct.

4.1.1 Data, information, and knowledge

a) Data

Data are raw facts and figures that are collected from different sources. They can be numbers, words, symbols, or measurements that have not yet been organized. Data can be collected through observation, surveys, experiments, or counting.

Example: The numbers 45, 67, 89, 56, and 72 are students' Mathematics marks. These numbers alone are data because they are just raw figures.

b) Information

Information is data that has been processed and organized in a meaningful way. When data are arranged, summarized, or analyzed, they become information. Information helps people understand situations clearly and answer important questions. It is more useful than raw data because it gives meaning to the facts collected.

Example: If the teacher says that the average Mathematics mark is 66% and 70% of students passed, this is information because the raw marks were processed.

c) Knowledge

Knowledge is the understanding and skills gained from information and experience. It is the ability to use information correctly to solve problems or make decisions. Knowledge goes beyond knowing facts; it involves understanding how and why something happens. Knowledge helps people think critically and take appropriate action. It is the highest level compared to data and information because it involves deep understanding.

Example: After seeing that many students failed algebra, a teacher decides to revise that topic before the next test. This action shows knowledge because the teacher used information to improve learning.

4.1.2 Data types and structures

a) Data types

Data types refer to the different kinds of data that can be collected, stored, and processed in a computer system. Different types of data are used depending on the purpose of the task.

- **Numerical data** consists of numbers that can be counted or measured. Examples include marks obtained in exams, attendance percentages, heights, weights, or ages. Numerical data can be divided into two types namely discrete data and continuous data.
- **Discrete data** are whole numbers such as 80, 90, or 45, and continuous data are those that can take any value within a given range, including fractions (eg: 75.5) and decimal values.
- **Categorical data** consists of labels or names used to classify information into groups. This type of data does not involve numbers for calculation but is used for identification. For example, subject names like ICT, Mathematics and Physics are categorical data. Other examples include gender, class level, or department name. Categorical data helps in organizing and grouping information.
- **Text data** refers to written words, sentences, or paragraphs. It includes comments, descriptions, and reports. For example, a teacher's remark such as "Good performance" or "Needs improvement in programming" is text data. Text data is important for explaining information in detail.
- **Boolean data** has only two possible values: True or False. It is used to represent conditions or decisions. Examples include Passed/Failed, Yes/No, or Present/Absent. Boolean data is very useful in programming because it helps computers make logical decisions based on given conditions.

b) Data structures

Data structures refer to the way data is organized and stored so that it can be used efficiently. Proper organization of data makes it easier to access, manage, and analyze information. There are three main types of data structures namely structured, unstructured, and semi-structured data.

- **Structured data** is data that is organized in rows and columns. It is usually stored in tables, spreadsheets, or databases. This type of data follows a clear format, which makes it easy to search, sort, and analyze. Because it is well organized, structured data can easily be used for calculations like finding averages or totals.

Example: Students' marks recorded in Microsoft Excel with columns for names and subjects.

- **Unstructured data** is data that does not follow a predefined format or arrangement. It is not organized in rows and columns, which makes it more difficult to analyze using simple tools. Special tools or advanced software are often needed to analyze this type of data.

Examples of unstructured data include teacher comments, essays, project reports, images, audio files, and videos.

- **Semi-structured data** has some level of organization but is not fully arranged in tables like structured data. It may use tags, markers, or labels to separate information. Semi-structured data is easier to analyze than unstructured data but still requires special processing methods.

Examples include HTML web pages which contain tags like <title>, <p>, <h1> but also contain text

4.1.3 Data formats

Data format is the way data is stored, organized, and represented in a file or system, which determines how it can be read, processed, and shared by computer programs.

A data frame is created from data stored in a data format when the data is loaded into a program for analysis.

A data frame is a table-like structure used to store data in an organized way. It allows data to be arranged so that it is easy to read, analyze, and manipulate. A data frame contains rows, which represent individual records, and columns, which represent different variables or types of information.

Student Name	Web	Database	Networking
Ilera	85	78	70
Irasubiza	90	88	75
Karenzi	91	86	75

Table 4. 1. A Data Frame organizes data into rows (records) and columns (variables)

The benefits of data frames include easy organization of data, simple analysis, and efficient filtering and sorting. They are also useful for preparing datasets for machine learning projects, where structured and clean data is essential.

4.1.4 Data sources

Data comes from different places depending on the purpose of collection. Knowing where data comes from helps in understanding how reliable it is and how it can be used effectively.

- Primary data sources** are collected directly from the original source. This type of data is first-hand and often very accurate. Examples include exam results, surveys conducted by the school, and interviews with students. Since it is collected personally, it is considered more trustworthy for analysis.
- Secondary data sources** are data that have already been collected by someone else. These are useful for research or studying trends without the need to collect new data. Examples include government reports, published research papers, and statistical databases. Secondary data saves time but may need verification for accuracy.
- Internal sources** are data obtained from within an organization or institution. Schools often use internal data for management and planning purposes. Examples include student attendance logs, library records, and exam results.
- External sources** are data obtained from outside the organization. This type of data helps in comparison, benchmarking, and studying trends beyond the organization. Examples include online educational platforms, public datasets, and statistics from other organizations.
- Digital sources** refer to data that is collected and stored in digital form. These are often automated and easy to access. Examples include online databases, learning management systems (LMS), and data from sensors or IoT devices. Digital data makes it easier to analyze large amounts of information quickly.

4.1.5 Characteristics of quality data

The main characteristics of quality data are:

- ### Application activity 4.1

1) Which of the following is an example of structured data?

- 1) Which of the following is not a type of data?
- a) Teacher comments c) Audio recordings
b) Exam marks in Excel table d) Video files
- 2) Which data type includes labels such as Web, Database, and Networking?
- a) Numerical data b) Text data c) Categorical data d) Boolean data
- 3) Which of the following is an example of Boolean data?
- a) Student marks b) Attendance records c) Passed/Failed d) Student names

- 1) What does completeness mean, and why is it important?
- 2) Which quality characteristic ensures that data is up-to-date and useful for decision-making?
- 3) Why should each student record be unique in the school database?
- 4) Using last year's attendance record for this year's planning.
- 5) A Mathematics test that measures only Math skills, not reading comprehension.
- 6) Recording students' favorite colors when analyzing exam performance.
- 7) A student appears twice in the school database by mistake.
- 8) How can the school maintain timeliness when collecting attendance records?

9) Analyze why incomplete data can lead to wrong decisions.

4.2 Data collection, preparation, and cleaning

Learning activity:4.2

A school wants to use Machine Learning to predict students' exam performance. To do this, they need to collect data from surveys, attendance records, and past exam scores. Before the data can be used, it must be organized, cleaned, and prepared. Some records may be missing, duplicated, or contain errors. Teachers and data analysts must also ensure that student data is collected and stored ethically by respecting privacy and consent.

- 1) What is data collection in Machine Learning, and why is it important?
- 2) Give two examples of reliable sources for data collection in schools.
- 3) What does data preparation involve?
- 4) Name three tasks performed during data cleaning.
- 5) Why is data ethics important when collecting and storing information?

In Machine Learning, data collection, preparation, and cleaning are important to make sure models learn correctly.

Data collection is gathering information from reliable sources like surveys, experiments, or databases. Data preparation organizes the data into a clear format and selects the important parts while data cleaning fixes errors, removes duplicates, fills missing values, and removes irrelevant information. These steps make the data accurate, complete, and ready for machine learning, helping models give better predictions and decisions.

4.2.1. Methods of data acquisition

Data acquisition methods are the different ways used to **collect data from various sources** for analysis and machine learning. These methods help gather reliable and relevant data that can be used to train machine learning models and support decision-making. The common methods of data acquisition include the following:

- **Surveys and questionnaires:** Surveys collect data by asking people questions about a specific topic. They can be conducted online, on paper, or through mobile devices.
- **Observation:** Observation involves collecting data by carefully watching and recording events, behaviors, or conditions. For example, a teacher may observe student participation in class to collect learning behavior data.
- **Experiments:** Experiments collect data by performing controlled tests to study cause and effect relationships. For example, researchers may conduct experiments to study how changes in temperature affect machine performance.

- **Sensors and IoT devices:** Sensors automatically collect data from the physical environment, such as temperature, pressure, motion, or light. Also, IoT devices collect and send data to computer systems for storage and analysis.
- **Manual data entry:** Data can be collected manually by individuals entering information into forms, spreadsheets, or databases. Examples include recording student attendance or entering exam marks.
- **Web scraping and data mining:** Web scraping is the process of automatically extracting data from websites. It is often used to collect large amounts of information from online sources such as news sites, product listings, or social media.
- **Application Programming Interfaces (APIs):** They allow systems or applications to request and retrieve data from other online services or databases.
- **Published datasets and data repositories:** Researchers and organizations often share datasets through repositories such as GitHub or the UCI Machine Learning Repository. These datasets can be reused for machine learning research and development.
- **Simulations:** Simulations generate data using computer models that imitate real-world systems. For example, climate models can simulate environmental conditions to produce data for scientific studies.

4.2.2. Data preprocessing techniques

a) Data preprocessing

Data preprocessing is a crucial step in data analysis and machine learning, aimed at preparing raw data for accurate and efficient analysis. It involves several key processes, including data cleaning, integration, transformation, reduction, normalization, and feature selection.

b) Data cleaning

Data cleaning focuses on identifying and correcting errors in the dataset to improve data quality. This includes removing incorrect records, fixing inconsistent values, handling missing data, identifying and handling unusual data, and eliminating duplicate entries. Proper data cleaning ensures that the dataset is accurate, consistent, and reliable for further analysis.

- **Handling missing data:** Missing data can be handled by removing affected records, computing replacing values using statistical measures such as mean, median, or retrieving the missing information if possible.
- **Handling unusual data:** These are values that differ significantly from other observations. They may result from errors or rare events and can distort analysis. They can be handled by removing, correcting, or transforming them.
- **Handling duplicate data:** Duplicate records appear when the same data is recorded more than once, often due to data entry or merging issues. These duplicates should be identified and removed to avoid biased or misleading results.

c) Data integration

Data integration involves combining data from multiple sources into a unified dataset. For example, student attendance records and exam scores from different systems can be merged to provide a comprehensive view for analysis.

d) Data transformation

Data transformation converts data into a consistent and suitable format for analysis. This includes normalizing values, converting units, and encoding categorical variables into numerical formats required by machine learning algorithms.

4.2.3 Handling missing data

Missing data occurs when some values in a dataset are not recorded or are unavailable. This can happen due to data entry errors, system failures, or incomplete data collection. Missing data can be handled by removing records with missing values, replacing the missing values with estimated values such as the **mean, median, or mode**, or collecting the missing information again if possible.

- **Handling outliers**

Outliers are data values that are significantly different from the rest of the dataset. These unusual values may occur because of measurement errors, incorrect data entry, or rare events. Outliers can affect statistical analysis and machine learning models. They can be handled by identifying them using statistical methods or visualization techniques and then removing them, correcting them, or replacing them with more reasonable values.

- **Handling duplicate data**

Duplicate data occur when the same record appears more than once in a dataset. This may happen during data entry, data merging, or system errors. Duplicate records can lead to incorrect analysis and biased results. To handle duplicates, the dataset should be checked carefully and repeated entries should be removed to keep only the correct and unique records.

4.2.4 Data ethics in collection and storage

Data ethics refers to the responsible and respectful handling of data during its collection, processing, and storage. In machine learning and data analysis, it is important to ensure that data is collected and used in a way that respects people's rights, privacy, and security.

When collecting data, organizations should obtain permission or consent from individuals whose data is being collected. People should be informed about how their data will be used, stored, and protected. Only necessary and relevant data should be collected to avoid misuse.

Data must also be stored securely to prevent unauthorized access, loss, or misuse. This can be done using security methods such as passwords, encryption, and controlled access systems.

Another important aspect of data ethics is fairness and transparency. Data should not be used in ways that discriminate against individuals or groups.

Application Activity:4.2

Section A: Choose the correct answer

- 1) What is data collection in machine learning?
 - a) Deleting unnecessary data
 - b) Gathering information from different sources
 - c) Formatting data into tables
 - d) Sorting data
- 2) Which of the following is a method of data acquisition?
 - a) Drawing charts
 - b) Surveys and questionnaires
 - c) Formatting text
 - d) Printing documents
- 3) What is the main purpose of data cleaning?
 - a) To store data
 - b) To remove errors and improve data quality
 - c) To create databases
 - d) To display data
- 4) Which of the following is an example of handling missing data?
 - a) Ignoring the data
 - b) Replacing missing values with mean or median
 - c) Deleting the whole database
 - d) Changing data types
- 5) Why is data ethics important?
 - a) To increase file size
 - b) To protect privacy and ensure responsible data use
 - c) To delete old data
 - d) To speed up computers

Section B: Answer the following questions

- 1) Explain why data preparation is important in machine learning.
- 2) Why is it important to remove duplicate data?
- 3) A dataset contains some missing student marks. How would you handle this problem?
- 4) Explain the effects of using unclean data in machine learning.
- 5) Design a simple survey to collect data about student study habits.

4.3 Data analysis and representation

Learning activity 4.3

A secondary school wants to analyze the performance of students in Mathematics to improve teaching and learning. The school has collected raw data including student names, test scores, and attendance, but data is not well organized, and no analysis has been done yet.

- 1) Explain the school organize, visualize, and analyze the data so that teachers can make better decisions.
- 2) Explain Why is it important to organize data before analysis?

4.3.1 Data organization in spreadsheets and databases

Data organization is the process of arranging data in a structured and systematic way so that it can be easily stored, accessed, and analyzed. In computing and data management, data is commonly organized using spreadsheets such as Excel and databases such as Access.

In Microsoft Excel, you can:

- Use `=AVERAGE (B2:D2)` to calculate each student's average mark.
- Sort by Attendance % to see who is absent most often.
- Create a bar chart showing subject performance.

A (Student Name)	B (Math Mark)	C (Science Mark)	D (English Mark)	E (Attendance %)
Nyiraneza	85	78	92	95%
Ndagijimana	62	71	68	88%
Shyaka	94	89	96	100%
Keza	47	55	61	76%

Table 4. 2.Data organization in spreadsheets: rows represent records, columns represent

Databases are structured collections of data stored electronically in a computer system. Data in databases is usually organized into tables, where each table contains related information. Each table consists of rows (records) and columns (fields). They are designed to handle large amounts of data and allow multiple users to access and manage data efficiently. Examples of database management systems include MySQL, Microsoft Access, and Oracle Database.

Here is an example of how data can be organized in MySQL

Table Name: Students

Row (Record)	Student Name (Field/Column)	Age (Field/Column)	Grade (Field/Column)
Record 1	Alice Kamikazi	15	10
Record 2	Brian Shyaka	16	10
Record 3	Cynthia Karabo	14	9

Table4.3: Databases organize data into tables with rows.

This table shows a structured collection of data in which each row represents an individual record (a student) and each column represents a field, such as Student Name, Age, and Grade. All the data is organized using the same format, making it easy to store, manage, and analyze.

4.3.2 Data visualization techniques

Data visualization techniques are methods used to **represent data in visual forms** such as charts, graphs, and diagrams. These techniques help people understand large amounts of data more easily by showing patterns, trends, and relationships clearly. Instead of reading many numbers in tables, visual representations allow users to quickly interpret and analyze information.

Common data visualization techniques include:

- A bar chart: It is used to compare values between different categories.
- A line graph: It shows how data changes over time.

- A pie chart: It represents proportions or percentages of a whole, making it easy to see how different parts contribute to the total.
- A histogram: It is used to show the distribution of numerical data.
- A scatter plot: It helps display relationships between two variables.

Data visualization is widely used in spreadsheets, databases, and data analysis tools to present information clearly. By using these techniques, users can identify trends, and make comparisons. Proper data visualization facilitates understanding of complex datasets and supports better decision-making.

4.3.3 Descriptive and inferential statistics

In data analysis and machine learning, statistics help us to understand patterns in data and make conclusions. The two main types of statistics are descriptive statistics and inferential statistics.

a) Descriptive statistics

Descriptive statistics are methods used to summarize and describe the main features of a dataset. They help present data in a simple and meaningful way using measures such as mean (average), median, mode, range, and standard deviation, as well as tables, charts, and graphs. For example, a teacher may calculate the average score of students in an exam to understand the general performance of the class.

b) Inferential statistics

Inferential statistics are methods used to make predictions or generalizations about a larger population based on a sample of data. Instead of analyzing every member of a population, inferential statistics use a smaller group of data to estimate trends or relationships. Techniques such as hypothesis testing are commonly used in inferential statistics. Hypothesis testing is a statistical method used to determine whether there is enough evidence in a sample of data to support a claim about a population.

Application activity 4.3

Section A: Choose the correct answer

- 1) What is the main purpose of data in an organization?

a) To decorate reports	c) To increase storage usage
b) To support decision-making	d) To replace employees
- 2) Which of the following is an example of qualitative data?

a) 45 students	b) 78% score	c) Student feedback comments	d) 120 marks
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- 3) What does data analysis help to do?

a) Delete all records	c) Discover useful information and patterns
b) Hide important information	d) Increase errors in data
- 4) Which tool is commonly used to organize and analyze data?

a) Microsoft Paint	b) Google Sheets	c) Notepad	d) Calculator
--------------------	------------------	------------	---------------

5) What is a dataset?

- a) A single value
- b) A collection of related data
- c) A computer virus
- d) A type of hardware

Section B: Answer the following questions

- 1) What is the difference between descriptive and inferential statistics?
- 2) Why are predictions based on small samples unreliable?
- 3) Create a new spreadsheet using Google Sheets.
 - a) Create the following columns: Student Name, Subject, Marks, Attendance (%).
 - b) Fill in at least 6 student records with fictional or real data.
 - c) Sort the data by Marks in descending order.
 - d) Filter students with attendance above 85%.

4.4 Machine Learning foundation

Learning activity 4.4

A technology company wants to help schools improve student learning. They have collected a lot of information from different schools, such as how many hours students study, their exam scores, attendance, and how they use online learning platforms. The data is very large and complicated, making it hard for teachers to see patterns or understand it. Your team of data analysts has been hired to use Machine Learning (ML) to study data and give suggestions to help students do better in school.

- 1) What is the main challenge that the company is trying to solve?
- 2) Based on the scenario, which type of ML would you use to predict students' exam scores? Explain.

Machine Learning (ML) is a subfield of AI that focuses on building systems capable of learning from data and making decisions based on data. Unlike traditional programming, where a human writes explicit rules, ML uses algorithms to discover patterns within datasets.

4.4.2 AI, Machine Learning, and Deep Learning

AI, Machine Learning (ML), and Deep Learning (DL) are closely related fields in computer science that focus on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning from data, recognizing patterns, making decisions, and solving problems.

a) Artificial Intelligence

AI refers to the development of computer systems that can simulate human intelligence such as reasoning, problem-solving, and understanding language. AI includes many technologies such as robotics, expert systems, machine learning, and natural language processing.

b) Machine Learning

Machine Learning is a subset of AI that focuses on developing algorithms that allow computers to learn from data and improve their performance without being explicitly programmed. Instead of following fixed instructions, machine learning models analyze data, identify patterns, and make predictions or decisions based on the information they learn.

c) Deep learning

Deep learning is a specialized subset of machine learning. It uses artificial neural networks with multiple layers to analyze large amounts of data and learn complex patterns. Deep learning is widely used in applications such as image recognition, speech recognition, language translation, and autonomous vehicles.

4.4.3 Types of ML algorithms

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed. Depending on how the learning process is carried out and the type of data available, machine learning can be categorized into different types. Each type uses a different approach to identify patterns, make predictions, or support decision-making. The main types of machine learning are Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

a) Supervised learning

In supervised learning, a machine learning model is trained using a labeled dataset, which means that every input data point is paired with the correct output or answer. During the training process, the model analyzes many examples of input output pairs and learns the relationship between them. The main objective is for the model to understand how inputs correspond to outputs so that it can correctly predict results for new and unseen data in the future.

Consider the example below which illustrates the working mechanisms of supervised learning:

- The model is given training data as Tomato, Carrot and Banana. Because each image has a correct label attached to it is called **labeled data**.
- The labeled data is fed into the machine learning algorithm during the **training phase**. The model analyzes the examples and learns the characteristics of each category. The goal is to make the model learn the relationship between the images (inputs) and their labels (outputs).
- New data: After training, the model receives new images that it has not seen before. The model compares these new images with the patterns it learned during training.

- **Prediction (output):** The trained model predicts the correct label for each new image. It predicts the images are respectively carrot, banana and tomato.

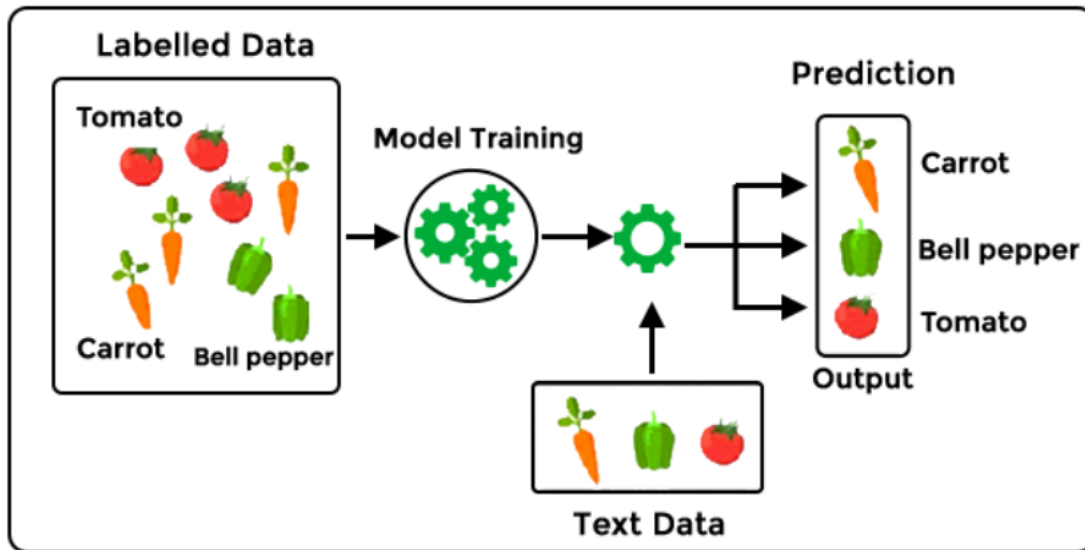


Figure 4. 1. Supervised learning working mechanism

b) Unsupervised learning

It is a type of machine learning that learns patterns and structures within the data without human supervision.

In unsupervised learning, a machine learning model is trained using data that does not have predefined labels or correct answers. Unlike supervised learning, where the system learns from examples with known outcomes, unsupervised learning allows the model to explore the data independently and discover hidden relationships.

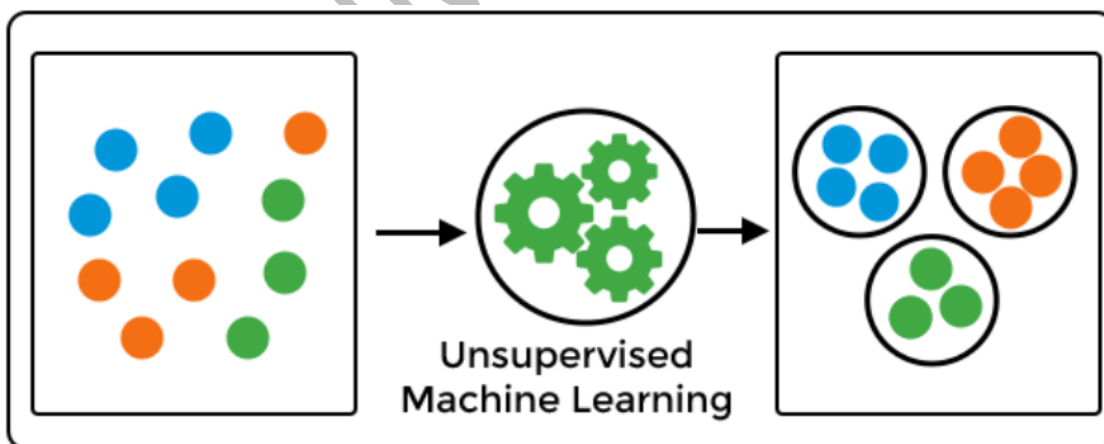


Figure 4. 2. The working mechanism of unsupervised learning

c) Semi supervised learning

It is a type of machine learning that partially implements both supervised and unsupervised. This kind of learning algorithms fall between supervised and unsupervised learning methods.

In semi-supervised learning, a machine learning model is trained using a combination of both labeled and unlabeled data, making it a bridge between supervised and unsupervised learning. In many real-world situations, obtaining fully labeled data is expensive, time-consuming, or requires expert knowledge, such as in medical imaging or large-scale text classification.

Semi-supervised learning addresses this challenge by using a small amount of labeled data to guide the learning process, while also taking advantage of the large amount of unlabeled data available. The algorithm first learns patterns, relationships, and structures from the labeled examples, then extends or generalizes this knowledge to the unlabeled data, improving its overall performance.

Consider the example below:

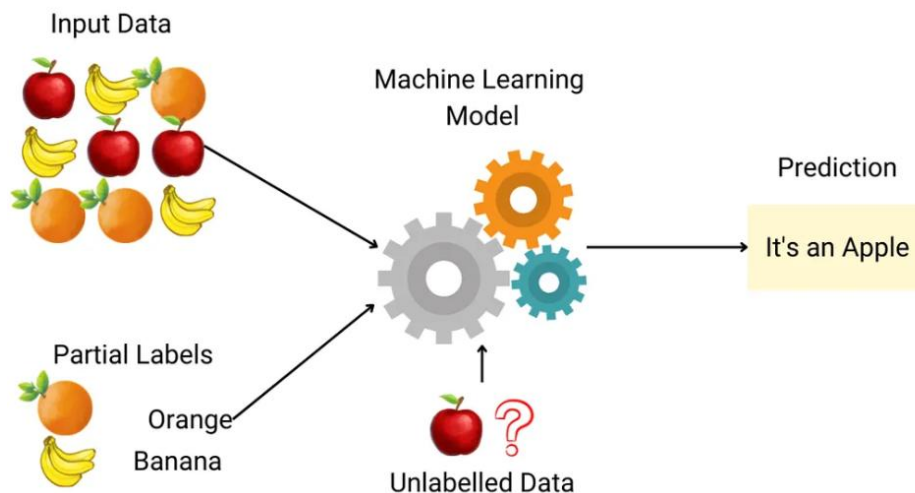


Figure 4. 3. Semi supervised learning working mechanism

In the example above the dataset for training is composed mixed fruits that are not labelled. However there partially labelled fruits namely the orange and the banana. After learning from the labelled data, the model sees a new fruit (unlabeled apple) and predicts that it is an apple.

d) Reinforcement machine learning

It is a type of machine learning model that is similar to supervised learning but does not use sample data to train the algorithm. This model learns by trial and error and gets related feedbacks. When it gets positive feedback the action that led to that feedback is retained.

This learning model knows feedback is positive or negative because a human programmer defines it using positive and negative numbers in the code.

4.4.4 Machine Learning applications

Machine Learning is applied in many real-world scenarios to improve efficiency. Some applications include:

- **Healthcare:** ML helps in diagnosing diseases, predicting patient outcomes, and personalizing treatment plans. Examples include analyzing medical images and predicting the risk of chronic diseases.
- **Finance:** ML is used for fraud detection. It helps banks and financial institutions make faster and more accurate decisions.
- **E-commerce:** ML analyses customer behaviors and create appropriate recommendations for each customer
- **Education:** ML personalizes learning experiences, predicts student performance, and identifies areas where students may need additional support.
- **Agriculture:** ML analyzes weather patterns, soil conditions, and crop health to improve farming productivity and reduce waste.
- **Entertainment and media:** ML recommends movies, music, and content based on user preferences.

Application activity 4.4

Section A: Choose the correct answer

- 1) What is the main purpose of Machine Learning?
a) To replace all human intelligence b) To allow computers to learn from data and make decisions
c) To store large amounts of data d) To design computer networks
- 2) Which statement correctly describes the relationship between AI, ML, and DL?
a) ML is broader than AI b) DL is not related to ML
c) DL is a subset of ML, which is a subset of AI d) AI is a subset of ML
- 3) Which Machine Learning category works with unlabeled data?
a) Supervised learning c) Semi supervised learning
b) Unsupervised learning d) Reinforcement learning

Section B: Answer the following questions

- 1) Why is Deep Learning considered a subset of Machine Learning?
- 2) Differentiate supervised learning and unsupervised learning?
- 3) What is semi-supervised learning, and how does it combine supervised and unsupervised learning?
- 4) Imagine a school has collected data on students' study hours, attendance, and online engagement. Describe a step-by-step approach using ML to help predict which students might need extra support.

4.5 ML workflow and model building

Learning activity 4.5

A school wants to develop a Machine Learning system that predicts whether a student will pass or fail based on their study time, attendance, and assignment scores.

Learners will act as data scientists to design the process of building this system using the ML workflow.

- 1) Where does the data come from?
- 2) How will you prepare the data?

ML workflow refers to the systematic process of developing, training, evaluating, and deploying a machine learning model to solve real-world problems. It provides a structured approach that helps ensure the model is accurate, reliable, and useful.

4.5.1 ML pipeline

A **Machine Learning pipeline** is a structured sequence of steps that guides the development and deployment of ML models. It ensures that data is efficiently processed, models are accurately trained, and predictions are reliable. The ML pipeline provides a systematic approach to solving real-world problems using machine learning.

Key stages of an ML Pipeline

- **Data collection:** Gathering raw data from various sources such as sensors, databases, surveys, or online platforms. High-quality and relevant data is crucial for effective ML models.
- **Data preparation and cleaning:** Organizing, formatting, and cleaning the collected data. This includes handling missing values, correcting errors, removing duplicates, and ensuring data consistency.
- **Data exploration and analysis:** Understanding the dataset by performing statistical analysis and visualizations to identify patterns, correlations, and important features.
- **Feature engineering:** Selecting, transforming, and creating relevant features from the raw data to improve model performance.
- **Model selection:** Choosing the appropriate ML algorithm (supervised, unsupervised, or reinforcement learning) based on the problem type and data characteristics.
- **Model training:** Feeding the prepared dataset into the chosen model to learn patterns and relationships. This involves splitting data into training and validation sets to evaluate performance.
- **Model evaluation:** Testing the model's accuracy, precision, recall, or other metrics on unseen data to ensure it generalizes well and performs reliably.
- **Hyperparameter tuning:** Adjusting model parameters to optimize performance and reduce errors.
- **Deployment:** Integrating the trained model into real-world applications for predictions or decision-making.
- **Monitoring and maintenance:** Continuously tracking model performance, retraining with new data, and updating the model to adapt to changes in the environment.

4.5.2 Datasets

A dataset is a collection of data used to train and evaluate machine learning models. It can include numbers, text, images, or audio depending on the problem being solved.

To ensure accurate learning and predictions, datasets are usually divided into parts namely **training set**, **validation set** and **test set**.

- **Training set:** The training set is the largest portion of the data. The model uses this data to learn patterns and relationships. Typically, it accounts for **60–80%** of the total dataset.
- **Validation set:** The validation set is a separate part of the dataset used during training to adjust the model's settings (hyperparameters) and check its performance. It helps to detect overfitting, which happens when the model works well on training data but poorly on new data. The validation set shows how well the model performs on data it hasn't seen before. It usually makes up about 10–20% of the dataset.
- **Test set:** The test set is used after the model has been trained. It helps to check how well the model performs on new, unseen data. The test set usually makes up 20–40% of the dataset.

4.5.3 Model evaluation metrics

In Machine Learning, splitting data into different sets typically **training** and **testing** set is essential for building reliable models. The training set is used to learn patterns, the validation set helps tune model parameters, and the test set evaluates final performance. This approach ensures that the model **generalizes well to new, unseen data** and avoids problems like overfitting.

There are several types of machine learning models, and each type is evaluated using specific performance metrics depending on the task. In the content below, examples of metrics are given depending on the task to be performed by the model specifically classification models, regression models, clustering models.

a) Classification models

These models predict categories or classes such as yes or no, disease or no disease, etc.

Examples of metrics include:

- Accuracy: Proportion of correctly predicted instances
- Precision: Correct positive predictions out of all predicted positives

b) Regression models

These models predict continuous values such as price and temperature.

Examples of metrics include:

- Mean Absolute Error (MAE): Average of absolute errors
- Mean Squared Error (MSE): Average of squared errors
- Root Mean Squared Error (RMSE): Square root of MSE

- R-squared (R^2): Proportion of variance explained by the model

c) Clustering models

This is a type of unsupervised machine learning model that groups data points into clusters based on similarity, without using predefined labels. The idea is that items in the same cluster are more alike than items in different clusters.

Examples of metrics include:

- Silhouette score: Measures how similar an object is to its cluster
- Davies Bouldin Index: Evaluates cluster separation and compactness
- Inertia: Measures cluster compactness

4.5.4 Concept of overfitting and underfitting

Machine Learning, building a model that generalizes well to new data is essential. **Overfitting** and **underfitting** are common issues that affect model performance.

a) Overfitting

Overfitting happens when a machine learning model learns the training data too well, including small errors, noise, or random patterns that don't represent the real-world situation. It is a kind of memorizing the training data instead of generalizing from it. As a result, the model performs very well on the training data but fails to make accurate predictions on new, unseen data.

Solutions:

- Reduce model complexity.
- Increase training data.

b) Underfitting

Underfitting happens when a machine learning model cannot learn the patterns in the training data. This means the model is too simple to capture the relationships between inputs and outputs. As a result, it performs poorly on both training and testing data.

Solutions:

- Increase model complexity.
- Add more relevant features.
- Train for longer or use a better algorithm.

Application activity 4.5

Section A: Choose the correct answer

1) What is the purpose of a Machine Learning workflow?

- To store data permanently
- To follow a structured process for building and deploying ML models

- c) To replace human decision-making completely
- d) Designing computer hardware
- 2) Which stage of the ML pipeline involves handling missing values and removing duplicates?
 - a) Model training
 - b) Data collection
 - c) Data preparation and cleaning
 - d) Deployment
- 3) What is the main role of the training dataset?
 - a) To test the final model
 - b) To deploy the model
 - c) To help the model learn patterns from data
 - d) To visualize results
- 4) Which metric is commonly used for evaluating regression models?
 - a) Accuracy
 - b) Precision
 - c) Mean Squared Error (MSE)
 - d) Recall
- 5) What is overfitting in Machine Learning?
 - a) When the model is too simple to learn patterns
 - b) When the model performs poorly on both training and test data
 - c) When the model learns training data too well including noise
 - d) When the model ignores the training data

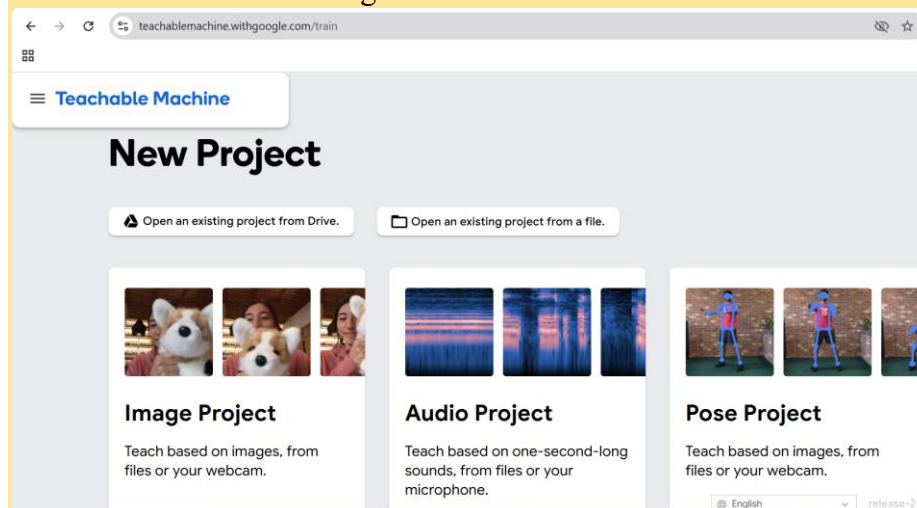
Section B: Answer the following questions

- 1) Why is ML pipeline important in developing ML models?
- 2) Explain the difference between training, validation, and testing datasets.
- 3) Why is it important to split a dataset into training, validation, and testing sets?

4.6 Tools for Machine Learning

Learning activity 4.6

Here is a screen of the Google Teachable Machine



- 1) Which URL address is used to access the Google Teachable Machine?
- 2) Which project type should a user select if they want to train a model using one-second-long sounds?
 - a) Image Project
 - b) Audio Project
 - c) Pose Project
 - d) Voice Project
- 3) What are the two input sources for an Image Project?

- | | | | |
|--|--|-----------------|------------------|
| a) Files or your webcam | b) Microphone or your webcam | | |
| c) Google Drive or your webcam | d) Files or your microphone | | |
| 4) If a user already has a saved project stored on their cloud storage, which button should they click? | | | |
| a) Teachable Machine Menu | b) Open an existing project from a file. | | |
| c) New Project | d) Open an existing project from Drive. | | |
| 6) If you wanted to create a model that recognizes the difference between a clap and a snap, which project type should you use | | | |
| a) Pose Project | b) Image Project | c) File Project | d) Audio Project |
| 7) Explain the importance of Google Teachable Machine | | | |

Tools for Machine Learning are software programs, libraries, platforms, or environments that help developers and data scientists create, train, test, and deploy machine learning models efficiently. These tools provide built-in functions and resources to handle tasks such as data preprocessing, model building, evaluation, and visualization, making it easier to apply machine learning to real-world problems without having to program everything from scratch.

4.6.1 Google Teachable Machine

Google teachable machine is a web-based tool that allows anyone to create machine learning models without needing to code. It provides a simple and interactive interface to train models for tasks like image recognition, audio recognition, and pose detection.

Google Teachable Machine has a user-friendly interface that allows users to easily drag-and-drop training examples to teach the model.

In Google Teachable Machine you can create these types of projects:

- **Image project:** Recognizes objects, gestures, or faces.
- **Audio project:** Recognizes sounds or voice commands.
- **Pose project:** Detects body movements and poses.

a) Steps to access and use Google teachable Machine

To open the Google Teachable Machine, follow these steps:

Step 1: Open the Teachable Machine by writing in the browser's address bar <https://teachablemachine.withgoogle.com/> then click on **Get Started**

Teachable Machine

Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

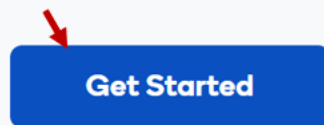


Figure 4. 4. No-code training for image, sound, and pose models

Step 2. In the new window that will appear choose your options such as opening an existing project from the drive or from a file. If you want to create a new project, click on anyone of the project types such as Image Project, Audio Project or Pose Project.

New Project

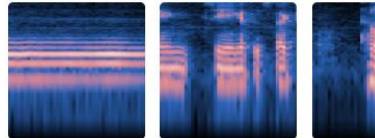
 Open an existing project from Drive.

 Open an existing project from a file.



Image Project

Teach based on images, from files or your webcam.



Audio Project

Teach based on one-second-long sounds, from files or your microphone.



Pose Project

Teach based on images, from files or your webcam.

Figure 4. 5. Teachable Machine: image, audio, or pose projects

Step 3. If you want an image project, click on its icons located to the left of the screen then click on any one of the two available options depending on your choice. The options are Standard image model and Embedded image model.

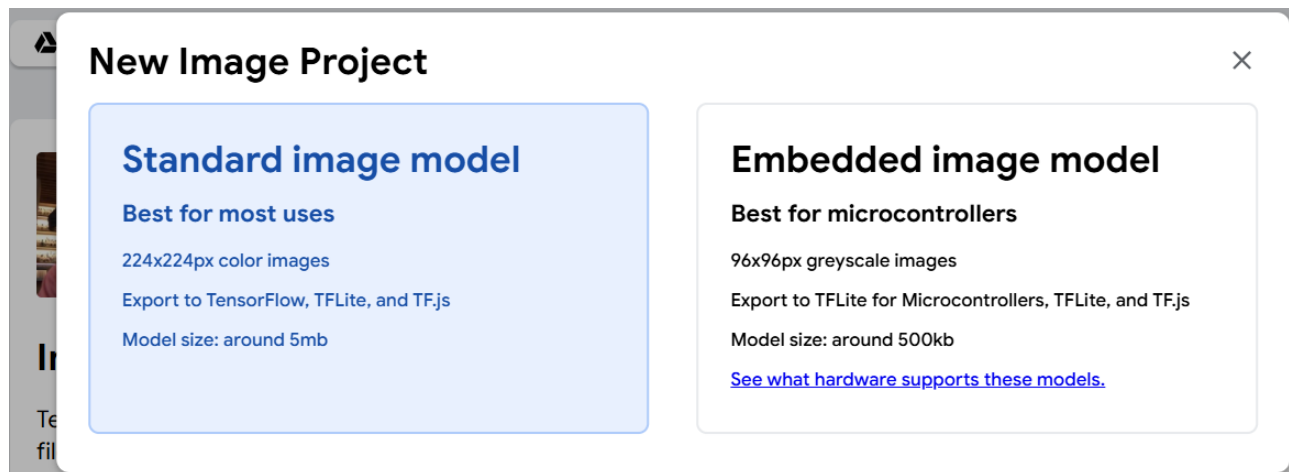


Figure 4. 6. Teachable Machine Image Project

Step 4. By default, there are two classes (Class 1 and Class 2). Rename them according to your project choice. If your project is to identify a tablet and an agenda, the first class can be renamed as “Tablet” and the second class can be “Agenda”. Note that you can add a new class by clicking on Add a class which is below Class 2

Step 5. Collect images: If your images are already in a folder on your computer, click on **Upload**. If they are to be taken using the computer camera, click on **Webcam**.

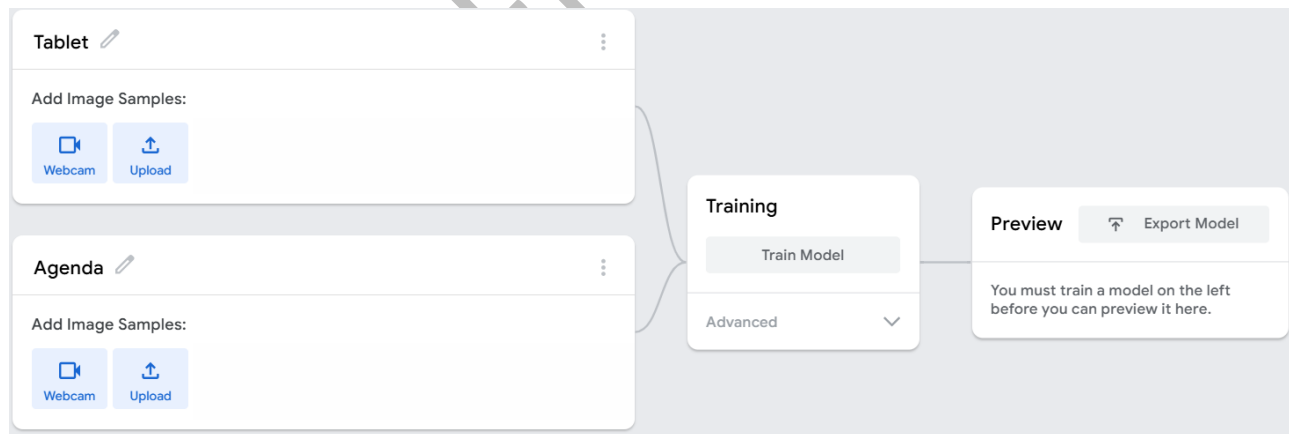


Figure 4. 7. Teachable Machine: add samples for two classes, then train & preview

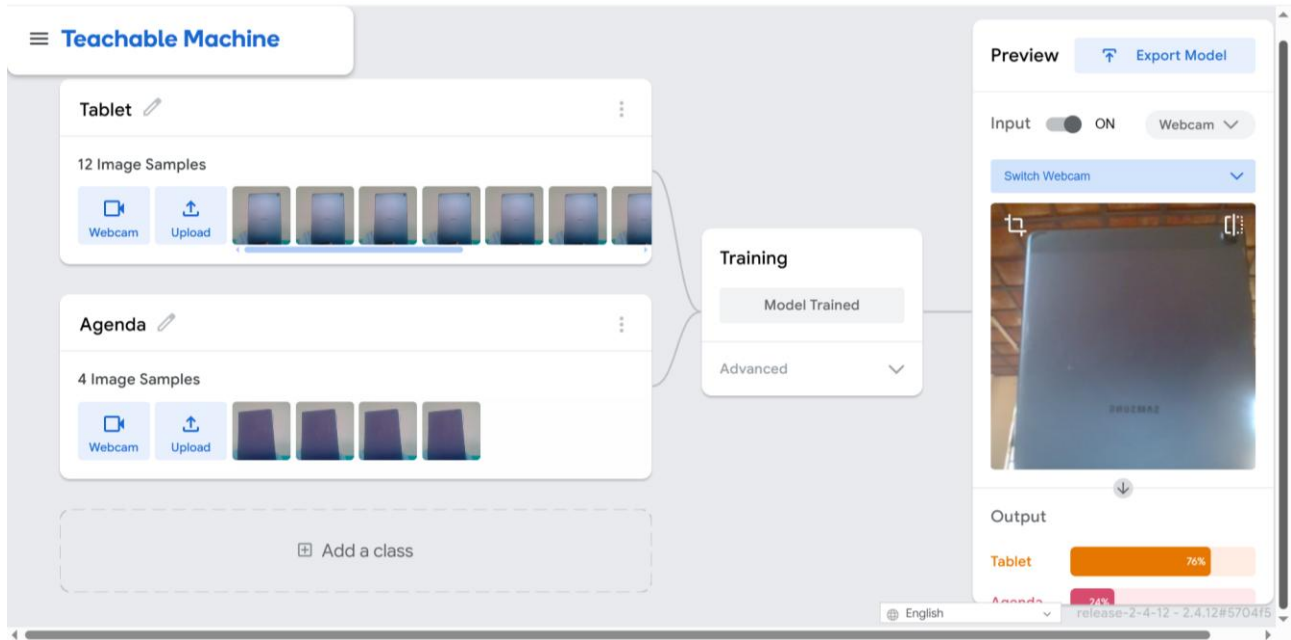


Figure 4. 8. Teachable Machine: Tablet vs. Agenda, 76% confidence for Tablet

Step 6. Train your model by clicking on **Train Model**

Step 7. Preview: The model created using Google Teachable Machine has successfully completed training (shown by “Model Trained”). It is designed to classify two categories: Tablet and Agenda, using 12 images for Tablet and 4 images for Agenda.

In the live webcam preview, the model is currently detecting an object that appears to be a tablet. The prediction results show:

- Tablet: 76% confidence
- Agenda: 24% confidence

Once the accuracy of the trained model is high, it can be exported for use in different projects.

b) Advantages of Google Teachable Machine

- **Real time training:** Models can be trained quickly using webcam, microphone, or uploaded files.
- **Export options:** Once trained, models can be exported for use in websites, applications, or other projects.
- **No coding required:** It makes machine learning accessible for beginners, students, and educators.

4.6.2. Scratch ML extensions

Scratch ML extensions are additional features in the Scratch programming environment that allow users to explore machine learning concepts in a simple and visual way. Scratch uses block-based

coding, which makes it easy for beginners and secondary school students to create programs without writing complex code. With ML extensions, users can build simple models that can recognize images, sounds, text, or body movements. These extensions help in understanding how machines learn from data by training models using examples and then testing them in real time.

4.6.3. Python based notebooks

Python based notebooks are interactive tools used for writing and running Python code in a step-by-step manner. They allow users to combine code, text, images, and results in one place, making them very useful for learning and practicing machine learning. The most common examples are Jupyter Notebook and Google Colab.

The Python code below uses a supervised learning algorithm to categorize data based on its features.

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Train model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Test model
predictions = model.predict(X_test)
accuracy = accuracy_score(y_test, predictions)

print("Model Accuracy:", accuracy)
```

Application activity 4.6

Section A: Choose the correct answer

- 1) What are Machine Learning tools?
 - a) Devices used to store data
 - b) Software and platforms used to create and deploy ML models
 - c) Only programming languages
 - d) Hardware components for computers
- 2) What is the main advantage of Google Teachable Machine?
 - a) Requires advanced programming skills
 - b) Only works with text data
 - c) Allows users to create ML models without coding
 - d) Cannot export models
- 3) Which type of project can be created using Google Teachable Machine?
 - a) Image recognition
 - c) Pose detection
 - b) Audio recognition
 - d) All of the above
- 4) What is the purpose of Scratch ML Extensions?
 - a) To develop complex AI systems only
 - b) To help beginners learn machine learning using block-based coding
 - c) To replace Python programming
 - d) To store large datasets
- 5) Which of the following is an example of a Python based notebook used in ML?
 - a) Microsoft Word
 - b) Jupyter Notebook
 - c) Paint
 - d) Calculator

Section B: Answer the following questions

- 1) How can you improve the accuracy of a Teachable Machine model if it is making incorrect predictions?
- 2) How does Scratch ML Extensions help beginners understand machine learning concepts without coding?
- 3) Explain why evaluating a model with a testing dataset is important.

4.7. End unit assessment

4.7.1. Project activity

Project Title: Requirements for developing a Machine Learning Tool to predict students who may need extra academic support

Guidance for the project

- In your groups identify the type of data needed and plan on the methodologies you will use to collect, clean, and prepare the dataset.
- Plan how you will split the dataset into training, validation, and testing sets and also how you will visualize the data (e.g., spreadsheets, charts).

- Choose an appropriate type of Machine Learning that you will use.
- Describe how you will evaluate the model and ensure it is accurate and reliable.
- Have a plan on what you will do to make sure ethics principles are applied properly

Questions on the project

- 1) Explain how your Machine Learning model will predict students that need extra support?
- 2) What are the problems that may hinder the accuracy of that model's predictions?
- 3) Explain the kind of web application in which you will include your developed model.

4.7.2. Assessment questions

Section A: Choose the correct answer

1) Which category of Machine Learning is used when the training data does not include labeled outputs?

- | | |
|---------------------------|-----------------------------|
| a) Supervised Learning | c) Unsupervised Learning |
| b) Reinforcement Learning | d) Semi-supervised Learning |

2) A dataset is split into 80% training and 20% testing. What is the main purpose of the testing portion?

- | | |
|---|--------------------------------------|
| a) To train the model with more data | c) To clean the data before training |
| b) To evaluate the model's performance on unseen data | d) To label the dataset |

3) Which of the following best describes overfitting in a Machine Learning model?

- The model performs poorly on both training and testing data
- The model performs well on training data but poorly on new and unseen data
- The model has too few parameters to learn patterns
- The model ignores the training data completely

4) Which of the following is a correct characteristic of quality data used in Machine Learning?

- | | |
|--|---|
| a) Contains many missing values | c) Accurate, complete, and relevant |
| b) Collected from a single source only | d) As large as possible regardless of quality |

5) A hospital wants to build an ML system that groups patients by similar symptoms without pre-existing categories. Which ML approach is most appropriate?

- | | |
|------------------------|-----------------------------|
| a) Supervised learning | b) Semi supervised learning |
|------------------------|-----------------------------|

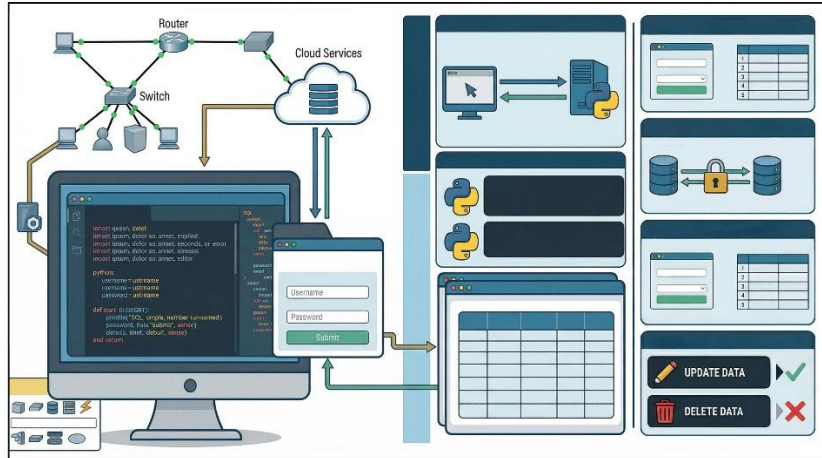
c) Unsupervised learning

d) Reinforcement learning

Section B: Answer the following questions

- 1) Differentiate data, information, and knowledge.
- 2) With examples, describe three types of data.
- 3) Explain four characteristics of quality data.
- 4) Describe the main steps of the Machine Learning pipeline.
- 5) Explain the difference between training, validation, and testing datasets.
- 6) Classify the following as structured, unstructured, or semi-structured data:
 - a) Excel spreadsheet
 - b) Student essay
 - c) JSON file
- 7) Given a dataset of student marks, explain how you would split it into training, validation, and testing sets.
- 8) Explain how poor-quality data can affect machine learning results.

UNIT 5: WEB DATABASE CONNECTIVITY



Key Unit Competence: Connect a web application to database using python successfully

Learning objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Describe the structure of a web application, including its components and the difference between front-end and back-end.
- Explain how web applications work, including frontend–backend connection and data collection using HTML forms.
- Explain data handling in web applications, including database design and CRUD operations using Python and SQL.

Skills

- Design and develop web interfaces and databases by creating HTML forms and SQL database structures.
- Develop functional web applications by connecting Python programs to databases and implementing CRUD operations.
- Test and debug web applications to ensure proper communication between the frontend and backend.

Attitude and values

- Appreciate the role of Python in modern web development and the importance of accuracy in programming and data management.
- Demonstrate creativity and innovation in designing user-friendly web interfaces.
- Collaborate effectively with peers in developing and testing web applications.

5.0 introduction

Python is a popular programming language for web development due to its simple syntax, rich libraries, and strong community support. It is mainly used on the server side to handle user requests, process data, interact with databases, and return responses.

Like other programming languages, Python uses frameworks which are collections of pre-written code, libraries, and tools that provides a skeleton for building applications. Common Python web frameworks include **Flask, Django, and FastAPI**.

Flask is lightweight and easy to use, Django is suited for large applications with built-in features, and FastAPI is designed for high-performance APIs. This unit focuses on Flask because it is simple and clearly shows how web applications connect to databases.

Introductory activity

A secondary school wants to replace its paper-based student registration process with an online system. To achieve this, they design a simple web application that will allow students to enter their personal information using a web form, submit the information online, and store it permanently in a database. The system should also enable users to view, update, and delete student records whenever necessary. The ICT Teacher explains that the application must include a frontend where students interact with forms and buttons, a backend developed using Python to process requests, and a database responsible for storing student information permanently.

- 1) What information should students enter in the registration form?
- 2) Which technologies can be used to design the frontend of the application?
- 3) What is the role of Python in the system?
- 4) Why is a database important in a web application?
- 5) Why might a school need to update or delete student records?

5.1 Python in web development

Learning activity: 5.1

A school wants to create a simple web application where students can enter their names and receive a welcome message. The system should have a frontend (form where students type their names) and a backend (Python using Flask to process and respond).

- 1) Identify which part of the system represents the frontend and which represents the backend.
- 2) Give two examples of technologies used in the frontend and two used in the backend.

5.1.1 Components of web application

Every web application is divided into two major parts that work together namely the **frontend** that is visible by users and the **backend** which is the logic behind the scenes.

Frontend (client side)	Backend (server side)
Everything users see and interact with in their browsers	The brain of the application invisible to the user.
Built with HTML, CSS, and JavaScript.	Built with Python (Flask/Django), SQL databases, and servers.
Runs inside the user's web browser.	Runs on a web server
Focus on user experience and visual design.	Focuses on data processing, security, and business logic.
Sends requests to the backend (e.g. form submissions).	Receives requests, processes them, queries the database, returns data.

The backend and front end can be illustrated like in the image below:

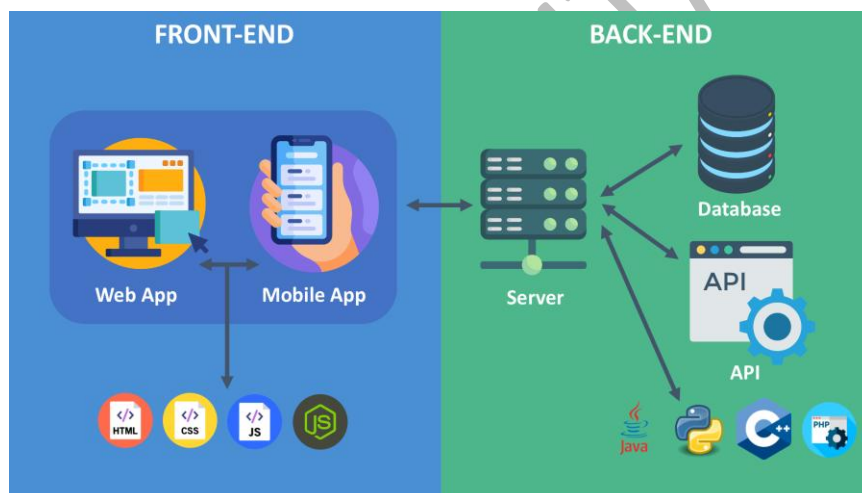


Figure 5. 1. Frontend and Backend the two sides of a web application

a) Frontend or the client side

The frontend is the presentation layer that includes everything the user interacts with. The front end has buttons, forms, text, images and animations.

A frontend is built using three core technologies:

- **HTML (Hypertext Markup Language):** Defines the structure and content of the web page, headings, paragraphs, input fields, buttons, and tables.
- **CSS (Cascading Style Sheets):** Controls the visual presentation, colours, fonts, layout, spacing, and responsive design for different screen sizes.
- **Javascript:** Adds dynamic behaviour responding to user clicks, validating forms before submission, fetching data from the server without reloading the page (AJAX), and controlling animations.

b) Backend or the server side

The backend handles everything the user cannot see. When a user submits a registration form, the backend receives that data, validates it, stores it in the database, and sends a confirmation back. The backend is responsible for:

- **Business Logic:** The rules that govern how data should be processed. For example the backend checks if the user is over 18 to register if this is one of the rules.
- **Database communication:** Reading from and writing to the database using SQL queries.
- **Authentication and security:** Verifying who the user is and protecting sensitive information.
- **APIs:** Also known as Application Programming Interfaces, it provides structured endpoints that the frontend can call to get or send data.

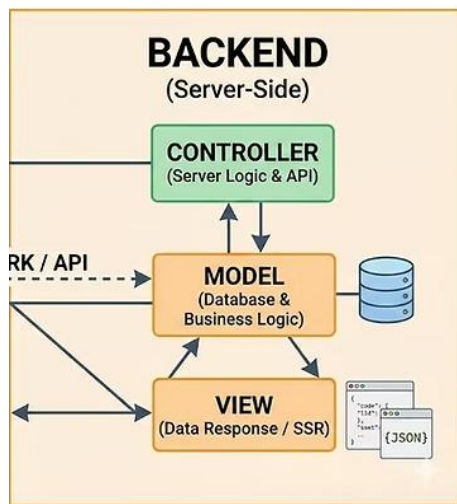


Figure 5. 2. Backend (Server-Side) components

5.1.2 Front-end to back-end connection

The frontend and backend do not run in the same place. The frontend runs in the user's browser while the backend runs on the server. They communicate over the internet using HTTP protocol. Whereby HTTP works sends requests from a client to a server and returns responses from the server to the client, enabling web pages and online applications to function.

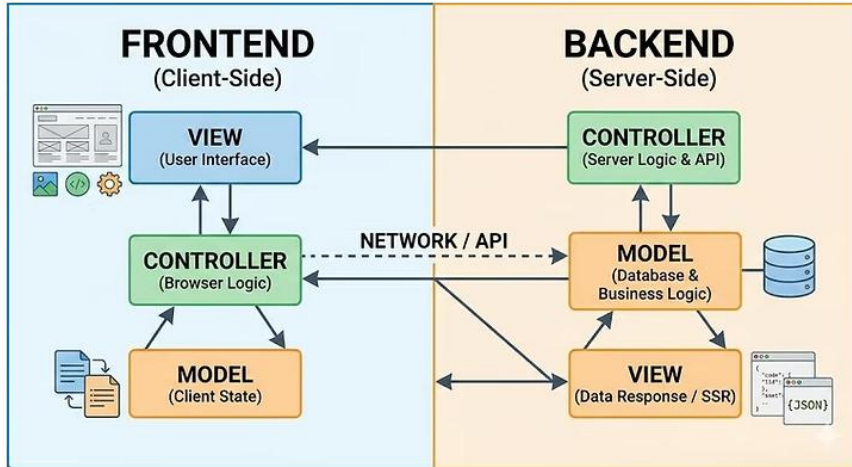


Figure 5. 3. HTTP communication between frontend and backend

a) HTTP request methods

HTTP defines several methods that describe what action the frontend wants the backend to perform:

Method	Use case	Example in a student application
GET	Retrieve/read data	View all students
POST	Submit/create new data	Register student
PUT/PATCH	Update existing data	Edit student
DELETE	Remove data	Delete student

When the user submits an HTML form, the browser sends an HTTP request to a specific URL on the Flask server. Flask reads that request, extracts the submitted data, and processes it.

b) Creating a Flask application

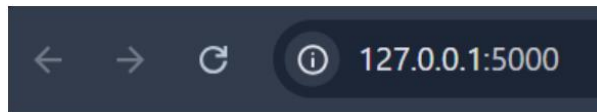
A Flask application is a Python program that uses the Flask framework to create a web application. It defines how the application responds to user requests and displays content in a web browser. In this section, we will create a simple Flask application that displays a message when a user visits the home page.

Step 1: Install Flask and create your first Flask Application. Create a new Python file named app.py which will contain the code for your first Flask web application.

```
from flask import Flask
app = Flask(__name__)
```

```
@app.route('/')  
def hello_world():  
return 'Hello World'  
  
if __name__ == '__main__':  
    app.run()  
  
Flask(__name__) creates the Flask application  
@app.route('/') connects the home URL (/) to a function  
The function returns text shown in the browser  
app.run() starts the local server
```

Step 3: Run the Flask Application to get the web page below



Hello World

Figure 5. 4. Web page created in Flask

Step 4: Handling a POST request using an HTML form

To demonstrate the process of **POST requests** within the Flask framework, we will build an interactive form. This setup allows a user to submit their name to the server, which then processes the data and reflects it back on the page.

Before writing any code, organize your project folder like in the image below:

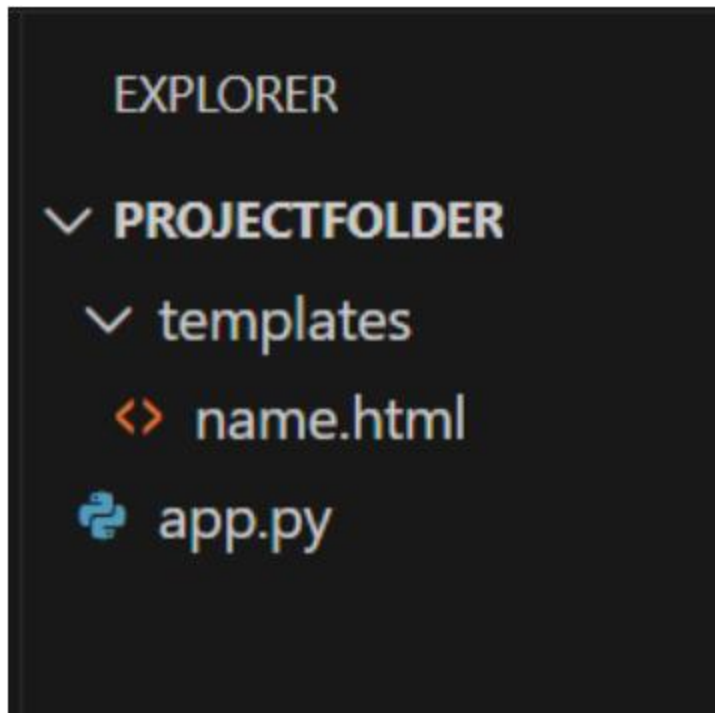


Figure 5. 5. Files organization

The files in the image above include:

app.py: Main Flask application file

templates: Flask automatically looks here for HTML files

name.html: HTML form used to send POST data

- **Create the HTML Form**

Inside the templates folder, create a file named name.html and add the following code:

```
<!DOCTYPE html>
<html>
<body>
  <h3>Enter Your Name</h3>
  <form method="post">
    <input type="text" name="username" required>
    <br><br>
    <input type="submit" value="Submit">
  </form>
</body>
```

</html>

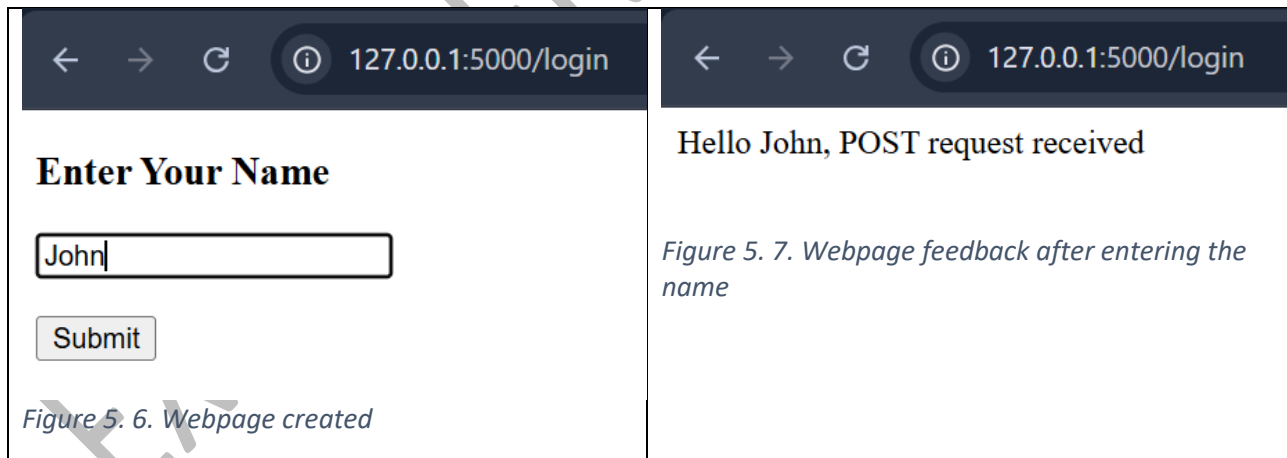
This form created uses the “post” method to send data securely. The input field sends the value using the name username and clicking **submit** sends the data to the flask server.

- **Updating app.py:**

```
from flask import Flask, request, render_template
app = Flask(__name__)
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        name = request.form['username']
        return f"Hello {name}, POST request received"
    return render_template('name.html')
if __name__ == '__main__':
    app.run(debug=True)
```

- **Result**

Open the browser and visit <http://127.0.0.1:5000/login> to get this web page. Then enter your name and hit the submit button.



Interpretation:

render_template() displays the HTML form

methods=['GET', 'POST'] allows both request types

GET request shows the form in the browser

POST request reads submitted form data using request.form

debug=True helps during development

Application Activity: 5.1

Section A: Choose the correct answer

- 1) Why is Python commonly used in web development?
 - a) It only works on the frontend
 - b) It has simple syntax, rich libraries and strong community support
 - c) It replaces HTML and CSS
 - d) It does not support databases
- 2) Which of the following is an example of a frontend technology?
 - a) Python
 - b) SQL
 - c) HTML
 - d) Flask
- 3) What is the main role of the backend in a web application?
 - a) Designing user interfaces
 - b) Displaying images
 - c) Processing data and handling logic
 - d) Styling web pages
- 4) Which HTTP method is used to submit data from a form?
 - a) GET
 - b) POST
 - c) DELETE
 - d) PUT
- 5) What does the Flask function render_template() do?
 - a) Deletes data from the database
 - b) Sends HTTP requests
 - c) Displays an HTML page to the user
 - d) Stops the server

Section B: Answer the following question

- 1) What is the role of Python in web development?
- 2) Name two Python frameworks used in web development and explain one difference between them.
- 3) What is the difference between **frontend** and **backend** in a web application?
- 4) Explain how a user's request moves from the browser to the server and back.
- 5) What is the role of HTTP in communication between frontend and backend?
- 6) Why is it important to use method="post" in an HTML form?
- 7) What happens when a user submits a form in a Flask application?
- 8) Identify two possible errors that can occur when connecting frontend and backend.
- 9) Explain why backend security is important in web applications.

5.2 Frontend and backend design

Learning activity 5.2

A school wants to build a Student Registration System.

1. Explain in detail how that system will work.
2. Suggest the tables names that will be in that system's database
3. That system as a submit button on one of its forms. Can this button be changed into "Send" button? If yes explain how it will be done.

4. On a paper design different pages which that registration system will have

Before writing any Python code, a good developer first plans the frontend and backend design of a web application. The frontend focuses on what the user will see and interact with, such as the forms used to enter information. The backend focuses on how the data will be stored and managed, including designing the database structure.

Before writing any Python code, a good developer plans both the frontend (what forms the user will fill in) and the backend database (what data needs to be stored).

This section walks through designing from scratch for a Student Registration system.

5.2.1 Design an HTML form

An HTML form is used to collect data from users. It is part of the front-end and allows users to input information such as names, emails, or marks. The data entered in the form is sent to the back-end for processing.

a) Key attributes of the <form> tag

Attribute	Explanation
Action	The URL the form data is sent to (e.g., /register sends to <code>http://localhost:5000/register</code>)
Method	The HTTP method to use: GET (data in URL) or POST (data in body, more secure for forms)
Enctype	How form data is encoded. For file uploads use <code>multipart/form-data</code> . Default is <code>application/x-www-form-urlencoded</code>

In web development, the <form> tag relies on two key attributes namely action and method. Action, defines the destination URL for the data while method specifies the transfer protocol usually GET for data retrieval or POST for data submission.

b) Steps to design an HTML form

To design a functional form, begin by establishing the container with the <form> tag. Within this, incorporate input fields like text boxes to capture user data, and pair them with labels to ensure the purpose of each field is clear. Finally, include a submit button to trigger the data transfer to the server.

- Example of an HTML form

```
<!DOCTYPE html>  
<html>
```



```
<body>

<h2>Student Registration Form</h2>

<form action="/submit" method="POST">
<label>Name:</label>
<input type="text" name="name"><br><br>

<label>Email:</label>
<input type="text" name="email"><br><br>

<label>Age:</label>
<input type="number" name="age"><br><br>

<input type="submit" value="Submit">
</form>

</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
<title>Student Registration Form</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
```

```
<h2>Student Registration Form</h2>

<form>
<label>Name:</label>
<input type="text" name="name">

<label>Email:</label>
<input type="text" name="email">

<label>Age:</label>
<input type="number" name="age">

<input type="submit" value="Submit">
</form>

</body>
</html>
```

Student Registration Form

Name:

Email:

Age:

Figure 5. 8. HTML Student Registration Form

- Examples of HTML input types

Input type	Example	Description
Text	<code><input type='text'></code>	Single-line plain text. Use for names, cities.
Email	<code><input type='email'></code>	Browser validates email format automatically.
Password	<code><input type='password'></code>	Characters are hidden
Number	<code><input type='number' min='1'></code>	Numeric input with optional min and max.
Tel	<code><input type='tel'></code>	Phone number. Shows numeric keypad on mobile.
Date	<code><input type='date'></code>	Calendar date picker in the browser.
Checkbox	<code><input type='checkbox'></code>	True/false selection. Can select multiple.
Radio	<code><input type='radio'></code>	Select exactly one from a group of options.
select	<code><select><option>...</select></code>	Dropdown list of options.

Input type	Example	Description
textarea	<textarea rows='4'>	Multi-line text for comments or descriptions.

c) Styling the form with CSS

A plain HTML form works but looks unprofessional. Cascading Style Sheets (CSS) is used to make forms visually appealing and user-friendly. Below is a style sheet that styles the registration form above.

```
body {
font-family: Arial, sans-serif;
background-color: #5fa65f; /* green background */
margin: 0;
padding: 0;
}
h2 {
text-align: center;
color: #2f5fb3; /* blue text */
margin-top: 20px;
}
form {
background-color: #f2f2f2;
width: 350px;
padding: 25px;
margin: 40px auto;
border-radius: 15px;
box-shadow: 0px 5px 15px rgba(0,0,0,0.2);
}
label {
display: block;
color: #2f5fb3; /* blue text */
font-weight: bold;
margin-top: 10px;
}
input[type="text"],
input[type="number"] {
width: 100%;
padding: 10px;
margin-top: 5px;
margin-bottom: 15px;
border-radius: 6px;
border: 1px solid #ccc;
}
input[type="submit"] {
width: 100%;
padding: 12px;
```

```

background-color: #ff4f87; /* pink button */
color: white;
border: none;
border-radius: 8px;
font-size: 16px;
cursor: pointer;
}
input[type="submit"]:hover {
background-color: #e04375;
}

```

Once styled, the form will be transformed like in the image below:

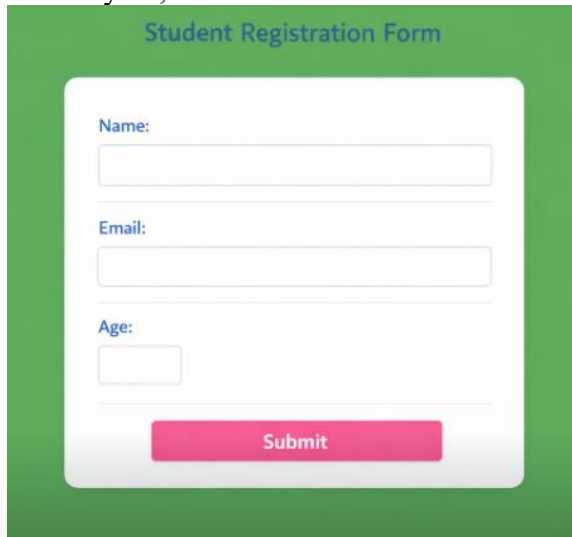


Figure 5. 9. Student Registration form created using HTML and CSS

Note: Always add type='submit' to the submit button and type='reset' to the clear button. Without type attributes, buttons in a form default to type='submit', which may cause unexpected behavior.

5.2. 2 Design databases

Designing a database is essential in web development as it defines how data is organized, stored, and managed. It involves identifying data, creating tables, and setting relationships to support CRUD operations efficiently.

- **Database schema for the students table**

A database schema is the structure of a database which defines how data is organized, including tables, columns, data types, relationships between tables and constraints.

Column Name	Data Type	Constraints	Description
Id	Integer	Primary key, Autoincrement	Unique identifier for each student record
Name	Varchar(100)	Not null	Full name of the student
Email	Varchar(150)	Unique, not null	Student email must be unique across all records
Phone	Varchar(20)	—	Optional phone number
Age	Integer	—	Student age in years
Course	Varchar(80)	Not null	Name of the course the student is enrolled in
Notes	Text	—	Optional additional information about the student
created_at	Datetime	Default Current_Timestamp	Automatically records when the student was registered

- SQL create table statement

Create database

CREATE DATABASE *student_db*;

Use the database

USE *student_db*;

Create table

CREATE TABLE *students* (

id **INT** **AUTO_INCREMENT** **PRIMARY KEY**,

name **VARCHAR**(100),

email **VARCHAR**(100),

age **INT**);

Application activity 5.2

- 1) What is the main purpose of an HTML form?
 - a) To store data permanently
 - b) To collect user input and send it to the backend
 - c) To style web pages
 - d) To create databases
- 2) Which attribute of the <form> tag specifies where the data is sent?
 - a) method
 - b) action
 - c) enctype
 - d) name
- 3) Which input type is best for validating email automatically?
 - a) text
 - b) number
 - c) email
 - d) password
- 4) What is the role of CSS in a web form?
 - a) To process data
 - b) To connect to the database
 - c) To improve appearance and layout
 - d) To store user inputs
- 5) In a database table, what is the purpose of a primary key?
 - a) To store duplicate values
 - b) To connect frontend and backend
 - c) To uniquely identify each record
 - d) To format text

- 1) What is the difference between the action and method attributes in a <form> tag?
- 2) Which HTTP method is more suitable for submitting sensitive data?
- 3) Give three examples of HTML input types and describe their uses.
- 4) Why is it important to use labels in an HTML form?
- 5) Explain the purpose of the following SQL constraints:
 - a) Primary key
 - b) Not null
 - c) Unique

Learning activity 5.3

The school hires you as a web developer to help solve this problem.

- 1) Why do you think the submitted data is not appearing in the records?
- 2) Where should the data go after submission?

Connecting to a database is the process of linking a web application to a database so that it can store, retrieve, and manage data. This connection is usually handled by the back-end using a programming language such as Python. Through this connection, the application can perform operations like inserting new data, reading existing data, updating records, and deleting information.

In a web application, connecting a form to a database allows user data to be collected, processed, and stored permanently. This connection involves three main components: the HTML form (front-

end), Python (Flask back-end), and the database (such as MySQL or SQLite). To successfully connect these components, a series of steps must be followed carefully.

Step 1: Ensure Python and Flask are installed and your virtual environment is active.

Step 2: Install the MySQL connector using pip.

Step 3: Create a Python file named app.py.

Step 4: Write code to connect Flask to MySQL.

Finally, write a route that receives form data and inserts it into the database.

```
import mysql.connector
from flask import Flask, request, render_template
app = Flask(__name__)
# Database connection configuration
db = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="student_db"
)
@app.route('/')
def home():
    return render_template('form.html')
@app.route('/submit', methods=['POST'])
def submit():
    name = request.form['name']
    email = request.form['email']
    age = request.form['age']
    cursor = db.cursor()
    # Insert data into database
    cursor.execute(
        "INSERT INTO students (name, email, age) VALUES (%s, %s, %s)",
        (name, email, age)
    )
    db.commit()
    return "Data inserted successfully!"
```

To connect the form to the database, follow these steps:

Step 1: Prepare the working environment. This involves installing Python on the computer and setting up a virtual environment. The virtual environment helps to manage project dependencies separately. After that, Flask must be installed because it is the framework that will handle communication between the form and the database. If MySQL is used, the MySQL connector must also be installed so that Python can communicate with the database.

Step 2: Create the database and its table. Before any connection can happen, a database must exist to store the data. Using SQL, a database such as student_db is created, followed by a table such as students with fields like name, email, and age.

Step 3: Design the HTML form. The form is created using HTML and includes input fields that match the database columns. For example, the form collects name, email, and age. The form must include the action attribute, which specifies where the data will be sent, and the method attribute, which is usually set to POST for sending data securely.

Step 4: Create the Python Flask application. In this step, a Python file (commonly called app.py) is created. Flask is initialized, and routes are defined. A route is a function that responds to a specific URL. One route displays the form, while another route receives the submitted data.

Step 5: Establish a connection between Python and the database. This is done using a database connector such as mysql.connector for MySQL or sqlite3 for SQLite. The connection requires details such as host, username, password, and database name. Once connected, Python can send SQL commands to the database.

Step 6: Receive form data in Flask. When the user submits the form, Flask captures the data using request.form. Each field in the form is accessed using its name attribute. This allows Python to retrieve the values entered by the user.

Step 7: Execute an SQL query to insert the data into the database. A cursor is created from the database connection, and an INSERT statement is executed. It is important to use placeholders (such as %s for MySQL or ? for SQLite) instead of directly inserting values. This improves security and prevents SQL injection which is a cyberattack technique where an attacker inserts malicious SQL code into an application's input fields.

Step 8: Save the data permanently. After executing the INSERT query, the commit() function must be called to ensure that data is written to the database.

Step 9: Close the database connection. After completing the operation, the connection should be closed to free system resources and avoid errors.

The final step is to run and test the application. The Flask server is started, and the form is opened in a web browser. When the user fills and submits the form, the data should be successfully stored in the database. This confirms that the connection between the form, Python, and the database is working correctly.

Application Activity: 5.3

Section A: Choose the correct answer

- 1) What is the purpose of connecting a web application to a database?
 - a) To design web pages
 - b) To store, retrieve, and manage data
 - c) To create animations
 - d) To improve internet speed
- 2) Which of the following is used in Python to connect to a MySQL database?

a) html.connector b) mysql.connector c) flask database d) python.sql

3) In a Flask application, how is form data accessed after submission?

a) request.data b) request.input c) request.form d) request.get

4) Why are placeholders such as %s used in SQL queries?

a) To increase speed c) To prevent SQL injection and improve security

b) To format output d) To delete data

5) What is the purpose of the commit() function in database operations?

a) To delete records c) To permanently save changes to the database

b) To display data d) To close the connection

Section B: Answer the following questions

- 1) What is meant by connecting a web application to a database?
- 2) Mention the three main components involved in connecting a form to a database.
- 3) Why is it important to use placeholders (e.g., %s or ?) in SQL queries?
- 4) What is the role of the commit () function in database operations?
- 5) List any three steps involved in connecting a form to a database.

5.4 Inserting and retrieving data

- 1) Consider the Student Registration System. What is the difference between registering a new student and looking up a student's existing information?
- 2) When a user clicks a "Submit" button on a web form, which HTTP method is typically used to send that data safely to the server.
- 3) If you want to add a new row of information into a database table, which SQL keyword do you use?
- 4) If you want to view existing information from a database table, which SQL keyword do you use?
- 5) What does the asterisk symbol mean when you write a database query like SELECT * FROM students?

5.4.1 Inserting inputs into a database

To insert input into a database, several steps must be followed.

- The user enters data into the form fields such as name, email, and age.
- The form sends this data to the Flask application.
- Flask receives the data using request.form.
- A connection to the database is established.
- A cursor is created to execute SQL commands.
- SQL INSERT statement is written to store the data.
- The data is saved using the commit() function.
- Finally, the connection is closed.

The steps described above can be illustrated in the program below:

```

import mysql.connector
from flask import Flask, request, render_template
app = Flask(__name__)
# Connect to database
db = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="student_db"
)
@app.route('/')
def home():
    return render_template('form.html')
@app.route('/submit', methods=['POST'])
def submit():
    # Step 1: Get form data
    name = request.form['name']
    email = request.form['email']
    age = request.form['age']
    # Step 2: Create cursor
    cursor = db.cursor()
    # Step 3: Insert data into table
    cursor.execute(
        "INSERT INTO students (name, email, age) VALUES (%s, %s, %s)",
        (name, email, age)
    )
    # Step 4: Save data
    db.commit()
    return "Data inserted successfully!"
if __name__ == '__main__':
    app.run(debug=True)

```

When the user submits the form, Flask receives the input values using **request.form**. A cursor is then created to allow execution of SQL queries and the INSERT statement adds the data into the students table. To ensure safe insertion of data in the database, the %s placeholders are used. After executing the query, the commit() function saves the data permanently and a message is returned to confirm that the data has been successfully inserted.

5.4.2 Displaying stored data from a database

When data is stored in a database, it can be accessed using SQL queries. The most common SQL command used to retrieve data is the SELECT statement. In a web application, Python (Flask) sends a SELECT query to the database, receives the results, and displays them on a web page using HTML.

To display stored data from a database, these steps are followed:

- A connection to the database is established using Python.
- A cursor is created to execute SQL commands.
- A SELECT query is written to retrieve data from the table.
- The results are fetched using methods such as fetchall().

- The data is passed to an HTML template.
- The HTML page displays the data in a structured format such as a table.

These steps can be illustrated in the program below:

```
import mysql.connector
from flask import Flask, render_template

app = Flask(__name__)
# Database connection
db = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="student_db"
)
@app.route('/students')
def display_students():
    cursor = db.cursor(dictionary=True)
    # Retrieve data
    cursor.execute("SELECT * FROM students")
    students = cursor.fetchall()
    return render_template('students.html', students=students)
if __name__ == '__main__':
    app.run(debug=True)
```

The program connects to the database and creates a cursor then the SELECT query is executed to retrieve all student records and the fetchall() function collects all rows from the table. Finally data is passed to an HTML template called students.html, where it will be displayed.

Application activity:5.4

Section A: Choose the correct answer

- 1) What is the first step when inserting data into a database through a web form?
 - a) Execute SQL query
 - b) Create a cursor
 - c) User enters data into the form
 - d) Commit the transaction
- 2) Which Flask object is used to retrieve form input values?
 - a) request.data
 - b) request.form
 - c) request.input
 - d) request.values
- 3) Which SQL statement is used to insert data into a database?
 - a) Select
 - b) Update
 - c) Insert
 - d) Delete
- 4) Which function is used to retrieve all records from a query result?
 - a) fetchone()
 - b) getall()
 - c) fetchall()
 - d) selectall()
- 5) What is the role of render_template() in displaying database data?
 - a) It deletes data
 - b) It executes SQL queries
 - c) It sends data to an HTML page for display
 - d) It connects to the database

Section B: Answer the following question

- 1) What is the purpose of inserting data into a database in a web application?
- 2) List the steps involved in inserting form data into a database using Flask.

- 3) What is the role of request.form in handling user input?
- 4) Write a simple SQL INSERT statement to add a student record into a table.
- 5) What is the function of commit() in database operations?
- 6) What is the purpose of the SQL SELECT statement?
- 7) Write a simple SQL query to retrieve all records from a table named students.
- 8) Explain how database retrieval can be useful in a real-life school system.
- 9) What is the difference between inserting data and retrieving data?
- 10) Why is it important to close a database connection after operations?

5.5 Updating and deleting data

Learning activity 5.5

- 1) A student accidentally registers at a school with a misspelled name and a wrong date of birth. What should be done to that wrong record?
- 2) If the student transfers to a new school, what action should be done about the current information in the database?
- 3) Suppose I have a list of all students in the classroom, and I say: "Change the age to 16." What is wrong with that instruction?
- 4) If you tell a database to "DELETE FROM students" but forget to mention which student, what do you think will happen to the rest of the records in that table?
- 5) Why do we use a unique value like a StudentID instead of a student's FirstName when we want to modify or remove a specific person's record?
- 6) Why is it necessary to run db.commit() after updating or deleting a row?
- 7) When you submit a form to insert data, you stay on a success page or go somewhere else. Once a user successfully edits a profile, where is the most logical place to send them next so they can verify the changes?

5.5.1 Update data into database

Updating data is changing the values of one or more fields in an existing record. This is done using the SQL **UPDATE** statement. Unlike inserting new data, updating modifies data that is already stored in the database. It is important to specify which record to update using a condition, preferably the **ID**.

To update data using Python (Flask), follow these steps:

- The existing data is retrieved from the database and displayed in a form and the user edits the information in the form.
 - The updated data is sent to the Flask application using the POST method then Flask receives the new data using request.form.
 - A connection to the database is established.
 - A cursor is created to execute the SQL command.
 - An UPDATE query is executed with placeholders.
- The changes are saved using commit(). Finally, the database connection is closed.

Example of a program to illustrate updating data:

```
import mysql.connector
from flask import Flask, request, render_template, redirect, url_for
app = Flask(__name__)
db = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="student_db"
)
@app.route('/edit/<int:id>')
def edit(id):
    cursor = db.cursor(dictionary=True)
    cursor.execute("SELECT * FROM students WHERE id = %s", (id,))
    student = cursor.fetchone()
    return render_template('edit.html', student=student)
@app.route('/update/<int:id>', methods=['POST'])
def update(id):
    name = request.form['name']
    email = request.form['email']
    age = request.form['age']
    cursor = db.cursor()
    cursor.execute(
        "UPDATE students SET name=%s, email=%s, age=%s WHERE id=%s",
        (name, email, age, id)
    )
    db.commit()
    return redirect(url_for('display_students'))
if __name__ == '__main__':
    app.run(debug=True)
```

The first route retrieves the student's existing data using the student's ID and displays it in a form. The second route receives the updated data after the form is submitted. The UPDATE SQL query modifies the existing record in the database. The %s placeholders are used for security. The commit() function saves the changes permanently.

5.5.2 Delete data from database

Deleting data is removing one or more records from a database table. This is done using the SQL DELETE statement. It is very important to carefully specify which record should be deleted, usually by using a unique identifier such as an **ID**. If not done correctly, important data may be lost.

To delete data using Python (Flask), follow these steps:

- The data stored in the database is displayed to the user, usually in a table format. Each record includes a delete option such as a button or link.
- The user clicks the delete button for a specific record.
- The request is sent to the Flask application along with the ID of the selected record.
- Flask receives the ID and connects to the database.
- A cursor is created to execute SQL commands.
- A DELETE query is executed using a placeholder.
- The commit () function is used to save the changes. Finally, the database connection is closed, and the updated list is displayed.

Below is an example to illustrate how delete is done:

```
import mysql.connector

from flask import Flask, redirect, url_for

app = Flask(__name__)

db = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="student_db"
)

@app.route("/delete/<int:ID>")
def delete(ID):
    cursor = db.cursor()
    # Delete record
```

```
cursor.execute("DELETE FROM students WHERE id = %s", (ID,))
```

```
db.commit()
```

```
return redirect(url_for('display_students'))
```

```
if __name__ == '__main__':
```

```
app.run(debug=True)
```

The Flask route receives the ID of the record to be deleted, then a database connection is used to create a cursor. The DELETE SQL statement removes the record with the specified ID, while the %s placeholder ensures safe execution. The commit() function saves the changes permanently, and finally, the user is redirected to the page displaying the updated list of records.

Application Activity:5.5

Section A: Choose the correct answer

1) What is the purpose of the SQL UPDATE statement?

- | | |
|-----------------------------------|---------------------------------------|
| a) To remove records from a table | c) To modify existing data in a table |
| b) To add new records | d) To display records |

2) Why is a condition important in an UPDATE query?

- | | |
|--------------------------|--|
| a) To increase speed | c) To specify which record should be updated |
| b) To select all records | d) To delete records |

3) Which Flask function is used to redirect a user after updating or deleting data?

- | | |
|----------------------|---------------|
| a) render_template() | c) redirect() |
| b) request.form() | d) commit() |

4) Which SQL statement is used to remove a record from a database?

- | | | | |
|-----------|-----------|-----------|-----------|
| a) Insert | b) Update | c) Select | d) Delete |
|-----------|-----------|-----------|-----------|

5) What is the role of commit() after executing UPDATE or DELETE queries?

- | | |
|--|-----------------------------|
| a) To cancel the operation | c) To display data |
| b) To save changes permanently in the database | d) To close the application |

Section B: Answer the following questions

1) Discuss the differences between Insert, Update and Delete

2) Why is it important to include a WHERE clause when performing Update and Delete?

5.6 End unit assessment

5.6.1 Project activity

Project Title: Student Registration Web Application Using Flask and MySQL Database

Guidance for the project

- Create a database named student_db and a table called students.
- Design an HTML registration form to collect student name, email, and age.
- Use CSS to make the registration form attractive and user-friendly.

- Develop a Flask application (app.py) to connect the frontend with the database.
- Use Flask routes to receive and process form data.
- Connect Python Flask to MySQL using mysql.connector.
- Insert student records into the database using the SQL INSERT statement.
- Retrieve and display stored student records using the SQL SELECT statement.
- Implement update and delete functionalities for managing records.
- Test the application to ensure all pages and database operations work correctly.

Questions on the project

- 1) What is the purpose of the Student Registration Web Application?
- 2) Explain the role of Flask in connecting the frontend and the database.
- 3) Why is a database important in this project?
- 4) Describe the steps followed when inserting form data into the database.
- 5) Explain the difference between inserting data and retrieving data in a web application.

5.6.2 Assessment questions

Section A: Choose the correct answer

- 1) Which of the following is a Python web framework primarily used for building web applications?
a) NumPy b) Flask c) Pandas d) Matplotlib
- 2) In a web application, the frontend is responsible for:
a) Storing data in the database c) Displaying forms, buttons, and text to the user
b) Processing business logic d) Managing server security
- 3) The POST method in an HTML form is used for:
a) Retrieving data from the server c) Displaying the home page
b) Sending data securely to the server d) Deleting records from the database
- 4) Which SQL statement is used to remove a record from a database table?
A) Insert B) Update C) Delete D) Select
- 5) In Flask, which object is used to retrieve data submitted via an HTML form?
a) Request.Get b) Request.Form c) Request.Data d) Request.Connect
- 6) Which SQL constraint ensures that no two records have the same value in a column?
a) Primary key b) Unique c) Not null d) Default
- 7) The WHERE clause in an UPDATE or DELETE statement is important because:
a) It makes the SQL query run faster
b) It specifies which records to update or delete
c) It automatically backs up the database
d) It generates a new table

Section B: Answer the following questions

- 8) Create a simple HTML form that allows students to enter name, email, and age.
- 9) Compare and contrast the following SQL operations: INSERT, UPDATE, DELETE, providing a real-life example for each in a school management system.
- 10) Your HTML form is submitting data but nothing appears in the database. List at least three possible reasons for this problem.
- 11) Discuss the importance of separating frontend and backend in web application development

EXPERIMENTAL VERSION

UNIT 6: COMPUTER NETWORKING



Key Unit Competence: Configure computer networks using Packet Tracer successfully

Learning objectives

At the end of this unit, you will be able to:

Knowledge and understanding

- Explain the concept and importance of computer networks, including wired and wireless connections.
- Identify the types of networks and their devices, and explain basic network configuration concepts.
- Apply basic network troubleshooting, including the use and role of ping commands.

Skills

- Configure network devices, including IP addresses and subnet masks.
- Test and troubleshoot network connectivity, using tools like ping commands.

Attitude and values

- Appreciate the importance of networking for communication, data sharing, and cybersecurity.
- Demonstrate responsibility and accuracy when performing network configuration tasks.
- Develop teamwork and collaboration when building and testing networks.

6.0. Introduction

Computer networking enables computers and other digital devices to communicate, share information, and access resources efficiently. Through networking, devices such as computers, smartphones, printers, and servers can exchange data, access the internet, and collaborate in real time. Networks support many everyday activities, including online learning, communication, banking, business operations, entertainment, and access to digital services.

Knowledge of networking helps users connect devices, share resources, configure network settings, and troubleshoot connectivity problems. It also provides a foundation for further studies in information technology, cybersecurity, cloud computing, and internet services.

As a student, competences in computer networking will help you use effectively network resources while understanding the logic behind it.

In this unit, you will explore the concepts of computer networking, including different types of networks and network connections, common networking devices, network configuration, and troubleshooting techniques. Using Cisco Packet Tracer, you will develop practical skills in designing, configuring, testing, and maintaining simple computer networks.

Introductory activity

A school has created a new computer laboratory with internet access. The lab has desktop computers connected using switches and routers. Teachers and students can also use Wi-Fi to connect their laptops, tablets, and smartphones from classrooms and the library. The school wants learners to understand how devices communicate, share files and printers, and use online services. Learners are asked to observe the school network and discuss how wired and wireless connections help communication and learning in the school.

- 1) Based on the school computer lab and Wi-Fi system, how would you define a computer network?
- 2) Identify the devices in the school that are connected using wired connections and those using wireless connections.
- 3) Explain how networking helps students and teachers in their daily activities.

6.1 Introduction to computer networks

Learning activity 6.1

The school has recently set up a computer lab where students can access the internet, share files, and use a shared printer. The school also provides Wi-Fi access so that students can connect their laptops, tablets, and smartphones from the library or classrooms.

- 1) What is Wi-Fi?
- 2) Which devices do you use to access the internet?
- 3) Explain how you connect your phone or laptop to the internet?
- 4) Why is it beneficial to connect school computers in a network?

6.1.1 Definition of a computer network

A computer network is a group of two or more interconnected computers and devices that can communicate, share resources, and exchange data with each other. The connection can be established through cables (wired) or wireless technologies like Wi-Fi or Bluetooth.

In networking, computers are connected using wired or wireless technologies so that they can communicate with each other efficiently. The connection may use devices such as switches, routers, and network cables or wireless signals.

6.1.2 Importance and benefits of networking

Computer networking plays a very important role in modern society. It allows people to communicate easily through emails, messaging, and video calls. Computer networking is essential across various sectors, improving efficiency, communication, and service delivery. Its importance can be understood better when examined in different areas:

a) Education

In schools and universities, networking enables sharing of learning resources, online classes, and collaboration between students and teachers. Platforms like Google Classroom support assignments, communication, and remote learning.

b) Business and commerce

Networking allows businesses to share data, manage operations, and communicate with clients and partners. Systems such as Enterprise Resource Planning integrate departments like finance, sales, and human resources, improving productivity and decision-making.

c) Healthcare

Hospitals use networking to store and access patient records, share medical information, and support telemedicine. This improves diagnosis, treatment, and coordination among healthcare professionals.

d) Banking and finance

Networking enables secure online transactions, ATM services, and real-time account access. It supports digital banking systems, making financial services faster and more accessible.

e) Government and public services

Governments use networks to deliver services such as online registration, tax systems, and e-governance platforms. This improves transparency, efficiency, and service delivery to citizens.

f) Communication and media

Networking powers communication tools such as email, messaging, and social media. Applications like WhatsApp allow instant communication and information sharing globally.

g) Transportation and logistics

Networks help manage traffic systems, track shipments, and coordinate logistics operations. This improves efficiency in transportation and delivery services.

h) Entertainment

Networking enables online gaming, streaming, and digital content distribution. Platforms like YouTube allow users to access and share multimedia content easily.

i) Research and development

Researchers use networks to share data, collaborate internationally, and access vast online databases, accelerating innovation and discoveries.

j) Security and surveillance

Networking supports security systems such as CCTV monitoring and access control, helping to enhance safety in organizations and public places.

6.1.3 Types of network connections

The main types network connections are wired and wireless connections.. Each type has its own characteristics, advantages, and limitations.

a) Wired network connection

A wired network connection uses physical cables to connect devices together. The most common type of cable used is the ethernet cable.

In a wired connection, data travels through cables from one device to another. Devices such as computers are connected to networking equipment like switches or routers using these cables.

One advantage of wired connections is that they are fast and reliable. They are less affected by interference and provide consistent performance.

However, wired connections have some disadvantages. They require installation of cables, which can be costly and complex, and they limit mobility since devices must remain physically connected.



Figure 6. 1. Devices and cables in a wired connection

b) Wireless network connection

A wireless network connection uses **radio waves** instead of cables to connect devices. The most common example of a wireless connection is **Wi-Fi**.

In a wireless network, devices such as laptops, smartphones, and tablets connect to a wireless router or access point without the need for physical cables. This allows users to move freely within the network coverage area.

The main advantage of wireless networks is mobility and convenience. Users can access the network from different locations.

However, wireless connections may be slower than wired connections and can be affected by interference, distance, or obstacles such as walls. They may also have security risks if not properly protected.



Figure 6. 2. Devices in a wireless connection connected via access point and gateway

c) Comparison of wired and wireless network connections

Feature	Wired connection (Ethernet)	Wireless connection (Wi-Fi)
Transmission Medium	Physical cables such as fiber optics	Radio frequency signals
Speed	Generally faster and consistent	Can be slower and varies with distance
Stability	Very high; no interference from walls	Susceptible to interference and signal drops
Mobility	Low; devices are fixed by cable length	High; allows users to move freely
Security	More secure, requires physical access	Less secure, signals can be intercepted
Installation	Harder, requires cable routing and drilling	Easier; requires an Access Point (AP)
Cost	Higher initial cost for cabling/labor	Lower cost for connecting many devices

Application activity 6.1

Section A: Choose the correct answer

- 1) A computer network is
 - a) A single computer used for calculations
 - b) A group of interconnected devices that can communicate and share resources
 - c) A type of software used for programming
 - d) A storage device for files
- 2) Which of the following is not a benefit of computer networking?
 - a) Resource sharing
 - b) Communication improvement
 - c) Data exchange
 - d) Slowing down internet access
- 3) Which device is commonly used in both wired and wireless networks to connect devices?
 - a) Printer
 - b) Router
 - c) Keyboard
 - d) Monitor
- 4) What is the main characteristic of a wired network connection?
 - a) Uses radio waves
 - b) Uses Bluetooth only
 - c) Uses physical cables like Ethernet
 - d) Has no security
- 5) Which of the following is an advantage of wireless networks?

- a) Requires cables for all devices
- b) Very low mobility
- c) Easy mobility and flexibility
- d) Always faster than wired networks

Section B: Answer the following questions

- 1) Explain the difference between a wired and a wireless network connection.
- 2) The school library allows students to connect their laptops and smartphones to the Internet without any cables.
 - a) Identify the type of network connection being used
 - b) State one advantage and one disadvantage of this connection.
- 3) In a computer lab, all computers are connected to a shared printer and the internet using ethernet cables.
 - a) Identify this type of network connection.
 - b) Explain why this type of connection is suitable for the lab.

6.2 Types of networks

Learning activity 6.2

- 1) Which of the following is not a type of network according to geographical area?

- a) Wide Area Network
- b) Personal Area Network (PAN)
- c) Local Area Network (LAN)
- d) World Local Network

- 2) A network connecting university campuses across the city of Kigali is an example of which type of network?

- a) LAN
- b) PAN
- c) MAN
- d) WAN

- 3) Why do some networks cover only a small area while others span cities or the whole world?
- 4) If you can print from your office computer using a shared printer how does your computer access that printer?

6.2.1 Personal Area Network

A PAN is a short-range network that connects personal devices like smartphones, tablets, and computers. It typically covers less than 10 meters and usually uses wireless technologies such as Bluetooth.

Example: Bluetooth connection between a phone and wireless earbuds

Advantages:

- Allows for easy communication between personal devices in close proximity.
- Use wireless technology, which eliminates the need for wires and cables.
- Easy to set up and low cost

Disadvantages:

- Limited coverage area.
- PANs have limited bandwidth, so they are not suitable for large-scale or high-speed data transfer.
- May experience interference from other wireless devices.



Figure 6. 3. Personal Area Network (PAN)

6.2.2 Local Area Network

A Local Area Network (LAN) is a network that connects computers and devices within a small geographical area such as a school, office, or home. It usually uses switches and routers. LANs are high-speed, inexpensive to set up, and easy to maintain.

Examples: Network found in school computer laboratory, office network and home Wi-Fi network.

Advantages:

- Provides fast data transfer rates and high-speed communication.
- Easy to set up and manage.
- Can be used to share peripheral devices such as printers and scanners.

Disadvantages:

- Limited geographical coverage.
- Limited scalability may require significant infrastructure upgrades to accommodate growth.
- May experience congestion and network performance issues with increased usage.

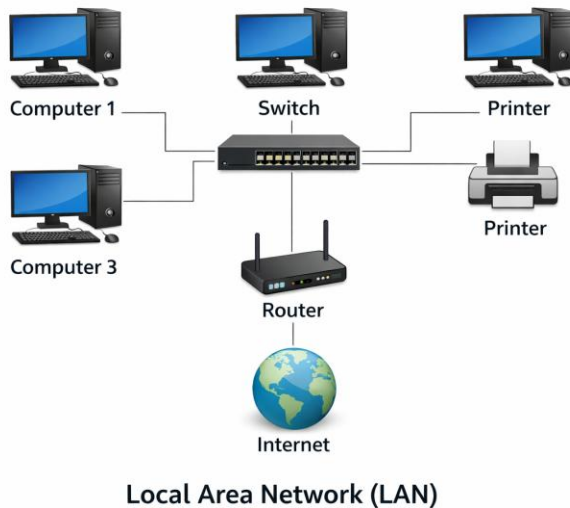


Figure 6. 4. Diagram illustrating a Local Area Network (LAN)

6.2.3 Metropolitan Area Network

A Metropolitan Area Network (MAN) is a network that spans a city or large town, connecting multiple LANs together. It connects computers across a city or between nearby cities. MANs provide high-speed connectivity (in Mbps), can act as ISPs and are useful for organizations needing fast communication.

Examples:

- Network connecting university campuses in Kigali
- City-wide internet service network
- Bank branch networks within one city

Advantages:

- Provides high-speed connectivity over a larger geographical area than LAN.
- Can be used as an ISP for multiple customers.
- Offers higher data transfer rates than WAN in some cases.

Disadvantages:

- Can be expensive to set up and maintain.
- May experience congestion and network performance issues with increased usage.
- May have limited fault tolerance and security compared to LANs.

6.2.4 Wide Area Network

A Wide Area Network (WAN) is a network that covers a very large geographical area, such as countries or continents. It can be private (for organizations) or public (like the internet). WANs offer high-speed communication but are costly to set up and maintain.

Example: internet

Advantages:

- Covers large geographical areas and can connect remote locations.
- Provides connectivity to the internet.
- Offers remote access to resources and applications.

Disadvantages:

- Can be expensive to set up and maintain.
- Offers slower data transfer rates than LAN or MAN.
- May have lower fault tolerance and security compared to LANs.

Application Activity: 6.2

Section A: Choose the correct answer

- 1) What is the main characteristic of a Local Area Network (LAN)?
a) Covers countries or continents c) Covers a small area like a school or office
b) Covers a city d) Covers one person only
- 2) Which network connects multiple LANs within a city?
a) PAN b) WAN c) MAN d) LAN
- 3) Which kind of internet is classified as
a) LAN b) MAN c) PAN d) WAN
- 4) Which of the following is a feature of a PAN?
a) Covers a large country c) Works within a few meters around a person
b) Connects cities d) Uses satellites only
- 5) Which technology is commonly used in a PAN?
a) Fiber optics c) Ethernet cables
b) Bluetooth d) Satellite communication

Section B: Answer the following questions

- 1) What is a Wide Area Network and what makes it different from MAN and LAN?
- 2) Which network covers the smallest area and which covers the largest?
- 3) If a school wants to connect all computers and printers in its building, which network should it use and why?

6.3 Network devices

Learning activity 6.3

You are an IT technician tasked with setting up a small computer lab network for a school. The lab has 3 computers, 1 server, 1 switch, 1 router, and an access point for Wi-Fi devices. Your goal is to

connect to all devices so that students can access the server, the internet, and communicate with each other.

- 1) Identify which devices you will use in this scenario.
- 2) Explain why each device is important for the lab network.

6.3.1 Types of networks devices

a) Switch

A **switch** is a hardware device used in a computer network to connect multiple devices such as computers, printers, and servers within a Local Area Network (LAN) and forward data to the correct destination using MAC addresses.



Figure 6. 5. A network switch

An network switch works by learning and storing the MAC addresses of all connected devices in a table, which allows it to identify where to send data. A switch forwards data only to the intended recipient, making communication more secure and efficient. By reducing unnecessary data traffic, switches significantly improve network performance and efficiency. They also support full-duplex communication, enabling devices to send and receive data simultaneously without collisions

b) Hub

A network hub is a basic networking device that connects multiple computers or devices in a network and broadcasts data to all connected devices without filtering or directing it.

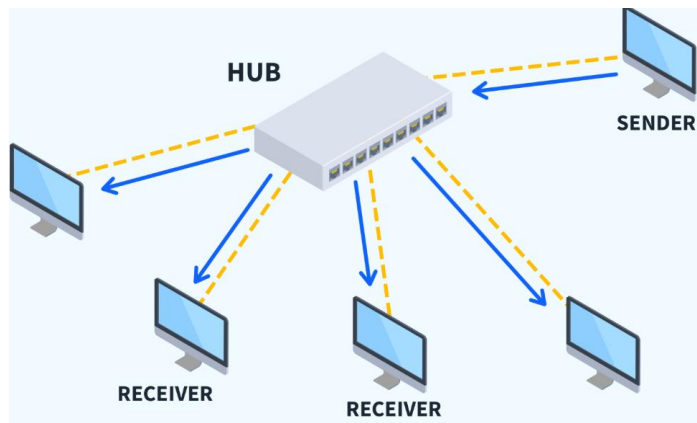


Figure 6. 6. Illustration of a network hub

A network hub functions as a simple connection point for multiple devices. It does not store or learn MAC addresses, which means it cannot identify specific destinations for data. Instead, whenever a hub receives data, it broadcasts it to all connected devices, regardless of the intended recipient. This approach often leads to network congestion and data collisions, since all devices share the same communication channel. Additionally, hubs support only half-duplex communication, meaning devices cannot send and receive data at the same time. Due to these limitations, hubs are less efficient and mostly outdated, having been replaced by switches in modern networks.

c) Router

A router is a networking device that connects different networks and directs data packets between them using IP addresses.

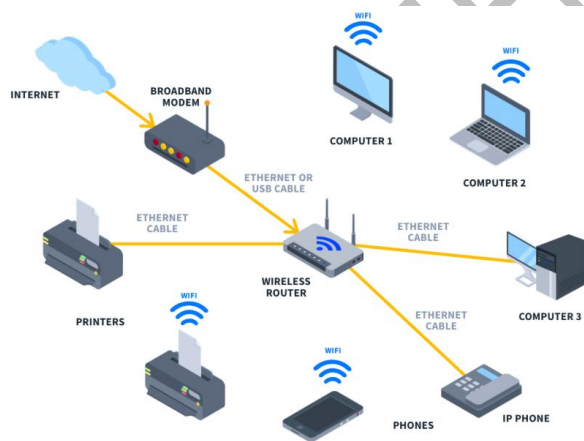


Figure 6. 7. A network diagram with a router at the center

A router is used to connect a local network (LAN) to the internet (WAN). It uses IP addresses to decide the best path for sending data from one network to another. A router can also provide security features, such as a firewall, to protect the network from unwanted access. In many cases, routers come with built-in Wi-Fi, allowing wireless devices like phones and laptops to connect easily to the network.

d) Modem

A modem (Modulator-Demodulator) is a device that converts digital signals from a computer into analog signals for transmission over communication lines, and vice versa.

A modem is used to connect a network to an Internet Service Provider (ISP) so that users can access the internet. It works by converting digital signals from computers into analog signals that can travel over communication lines and then converting them back to digital form. In most setups, a modem is connected to a router, which then distributes the internet connection to multiple devices.

e) Access point

An Access point is a device that allows wireless devices to connect to a wired network using Wi-Fi. It is used to extend wireless network coverage, allowing devices to connect to the network over a wider area. It is typically connected to a router using a cable, which provides it with internet access.

Access points are commonly used in schools, offices, and public places to provide reliable Wi-Fi connections. They can support multiple wireless users at the same time, making them useful in environments with many connected devices.

f) Network Interface Card (NIC)

A Network Interface Card is a hardware component that allows a computer to connect to a network. It can be wired using Ethernet or wireless using Wi-Fi, depending on how the device connects to the network. Each NIC has a unique MAC address, which helps identify the device on the network. It is usually installed inside a computer or attached externally, such as via USB. NIC enables the device to communicate with other devices and access the network, making it an essential component for network connectivity.

6.3.2 Connecting network devices using Packet Tracer

Packet Tracer is a network simulation tool developed by Cisco that allows users to create, configure, and test virtual computer networks without needing physical hardware.

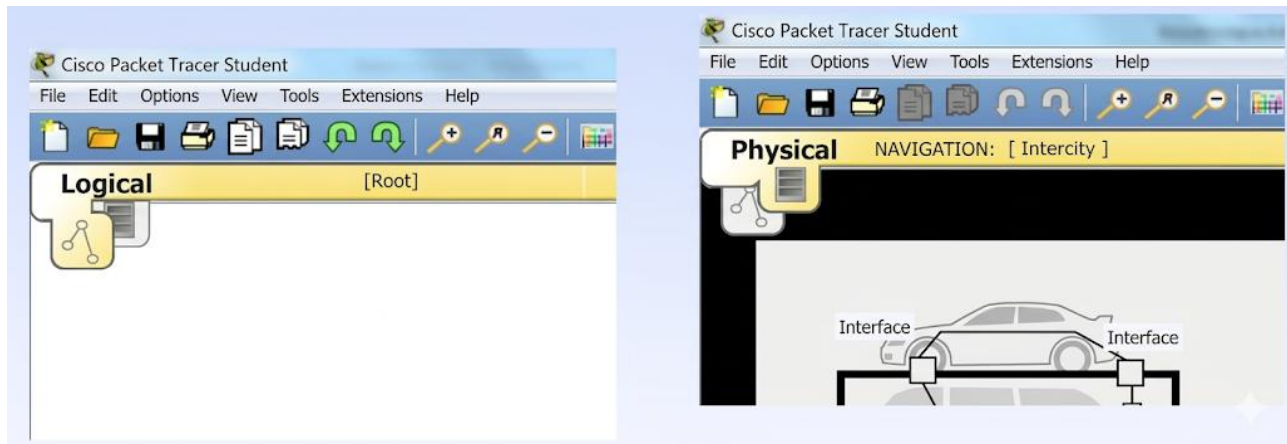


Figure 6. 8. Logical and Physical Workspace views in Cisco Packet Tracer

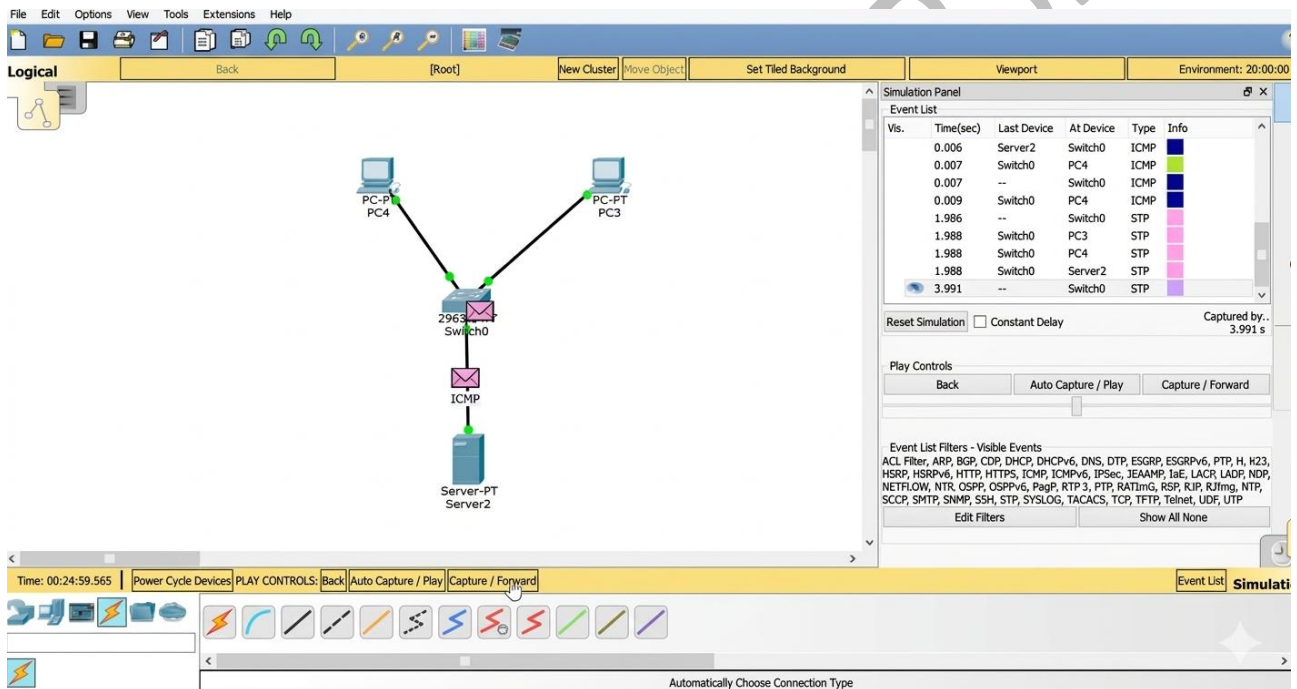


Figure 6. 9. Network Simulation in Cisco Packet Tracer

Step 1: Open Packet Tracer and prepare workspace

- Launch Cisco Packet Tracer. Then from the device toolbar, select the devices you need including End Devices namely PCs, laptops, servers or Network Devices such as switches (Layer 2), routers (Layer 3)
- Drag and drop the devices onto the workspace and arrange them logically.

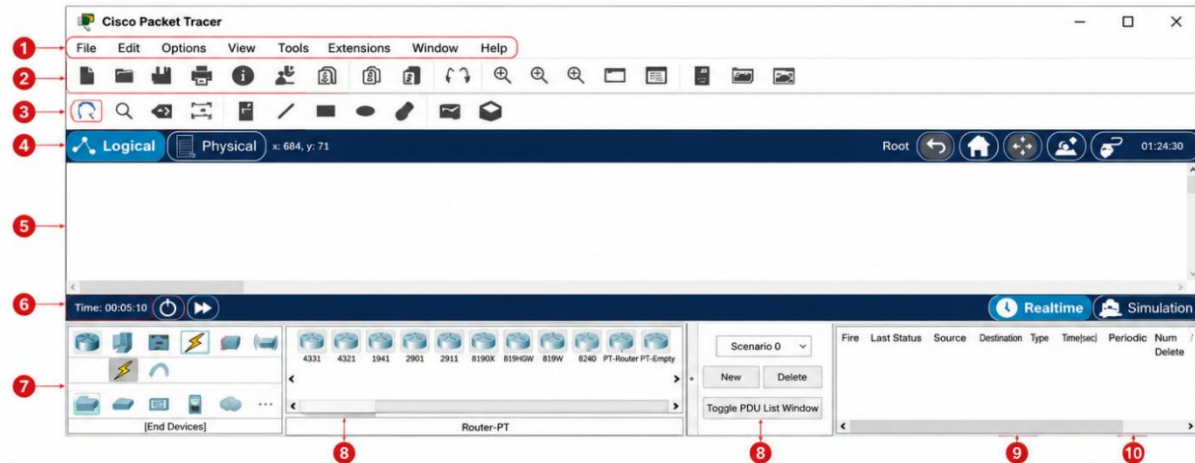


Figure 6. 10. Cisco Packet Tracer working environment

Below are the functions of each of the menu shown in the image above:

No.	Interface	Description
1	Menu Bar	This bar provides the File, Edit, Options, View, Tools, Extensions, Window and Help menus.
2	Main Tool Bar	This bar provides shortcut icons to most commonly used menu commands.
3	Common Tools Bar	This bar provides access to the most commonly used workspace tools such as Select, Inspect, Delete, Resize Shape, Place Note, Drawing Palette, Add Simple PDU, and Add Complex PDU.
4	Logical/Physical Workspace and Navigation Bar	You can toggle between the Physical Workspace and the Logical Workspace with the tabs on this bar. In Logical Workspace, this bar also allows you to go back to a previous level in a cluster, create a New Cluster, Move Object, Set Tiled Background, and Viewport. In Physical Workspace, this bar allows you to navigate through physical locations.
5	Workspace	This area is where you will create your network, watch simulation, and view many kinds of information and statistics.
6	Realtime/Simulation Bar	You can toggle between Realtime Mode and Simulation Mode with the tabs on this bar.
7	Network Component Box	This box is where you choose devices and connections to put into the workspace. It contains the Device-Type Selection Box and the Device-Specific Selection Box.
8	Device-Type Selection Box	This box contains the type of devices and connections available in Packet Tracer. This will change depending on which type of device you choose.
9	Device-Specific Selection Box	This box is where you choose specifically which devices you want to put in your network and which connections to make.
10	User Created Packet Window	This window manages the packets you put in the network during simulation scenarios.

Table 6. 1. Description of the Cisco Packet Tracer environment

Step 2: Select and connect devices using cables

The cables that can be used are:

- Copper straight through cables which are used to Connects computers (PC) with switches or routers with switches.
- Copper cross over cable which is used to connect identical devices such as two switches or two routers

Follow these steps to connect devices:

- Click the **Connections** icon (lightning bolt).
- Select the appropriate cable type.
- Click the first device then choose the correct port such as **Fa0** for PC, **Fa0/1** for Switch.
- Click the second device and select the port to connect.

Example: PC2 → Switch (Fa0/1), Switch → Router (Fa0/0)

Step 3: Assign IP Addresses to End Devices

- Click a PC then on Desktop then on IP Configuration.
- Assign IP addresses and subnet masks.
Example: PC2: 192.168.1.2 / 255.255.255.0, PC3: 192.168.1.3 / 255.255.255.0, default gateway (Router interface): 192.168.1.1
- Repeat for all PCs and the server.

Step 4: Configure router interfaces

- Click the router then click on CLI tab.
- Enter configuration mode by entering these commands:

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router# write memory
```

Step 5. Test the network connectivity: Open Command Prompt on any computer and test connectivity using the ping command. If replies are received, the network is properly connected.

```
ping 192.168.1.3 (other PC)
ping 192.168.1.100 (server)
```

Step 6. Optional simulation: Switch to Simulation Mode in Packet Tracer then send data packets between computers (PCs) and observe the path of packets through switches and routers.

Application Activity: 6.3

Section A: Choose the correct answer

- 1) What is the main function of a network switch?
 - a) To connect different networks using IP addresses
 - b) To broadcast data to all devices
 - c) To forward data only to the correct destination using MAC addresses
 - d) To convert digital signals to analog
- 2) Which OSI layer does a switch mainly operate on?
 - a) Physical Layer (Layer 1)
 - b) Data Link Layer (Layer 2)
 - c) Network Layer (Layer 3)
 - d) Application Layer (Layer 7)
- 3) What is the main disadvantage of a network hub?
 - a) It is too expensive
 - b) It routes data using IP addresses
 - c) It broadcasts data to all devices causing collisions
 - d) It supports full-duplex communication
- 4) What is the function of a router in a network?
 - a) Connect devices within a single LAN only
 - b) Connect different networks using IP addresses
 - c) Convert analog signals to digital
 - d) Store MAC addresses of devices
- 5) What is the purpose of Packet Tracer?
 - a) To design websites
 - b) To simulate and test computer networks virtually
 - c) To store network data
 - d) To replace routers and switches

Section B: Answer the following question

- 1) Explain the role of a NIC in a computer.
- 2) If you only had a hub instead of a switch, what problems could arise in the network?
- 3) a) Draw a simple network diagram showing how devices will be connected in a network. Include PCs, server, switch, router, modem, and access point.
 - b) Identify which cable types you will use for each connection and explain why.
 - c) Assign IP addresses to the 4 computers, server, and router interface. Use the subnet mask 255.255.255.0.
 - d) Specify the default gateway for the computers.
 - e) Explain why each device needs a unique IP address.

6.4 Network configuration

Learning activity 6.4

You have been hired as a network technician to set up a small office network with 3 PCs, a switch, a router connected to the internet via a modem and a wireless access point for mobile devices. Your task is to connect the devices, assign IP addresses and a default gateway, and ensure the network is fully functional.

How would you connect the devices to allow all computers and mobile devices to communicate and access the Internet?

In Packet Tracer, network configuration is the process of setting up a computer network to ensure that all devices on a computer network can communicate efficiently and securely. It defines how devices are connected, assigned addresses, and how they exchange data.

To configure a simple network, you need at least two PCs, a switch, a router and cables (usually Copper Straight-Through).

To configure a network successfully, follow these steps:

Step 1. Open Packet Tracer and place the required devices in the workspace. Select PCs from the End Devices section and a switch from the Switches section. Arrange them clearly in the workspace.

Step 2. Connect the devices using appropriate cables. Use copper straight-through cables to connect each PC to the switch. Ensure that the connections turn green, indicating that the devices are properly connected.

Step 3. Assign IP addresses to each PC then click on a PC. Go to the Desktop tab and open IP Configuration. Enter the IP address and subnet mask.

For example, PC1 can have IP address 192.168.1.2 and PC2 can have 192.168.1.3 both using subnet mask 255.255.255.0.

Step 4. If a router is used, configure its interface. Click on the router, go to the CLI or Config tab, and assign an IP address such as 192.168.1.1 to the interface connected to the switch.

Step 5. Test the network connection. Open the Command Prompt on one PC and use the **ping command** to test communication with another PC. If replies are received, the network is correctly configured.

Example of IP address configuration

Device	IP address	subnet mask
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
Router	192.168.1.1	255.255.255.0

Table 6. 2. Examples of IP addresses

6.4.1. Network addresses

In computer networking, devices must be identified in order to communicate effectively. Two important types of addresses used in networks are the MAC Address and the IP Address. These addresses work together to ensure that data is sent and received correctly between devices.

a) MAC address

A MAC address is made up of 12 hexadecimal digits (0–9 and A–F) and is usually written in pairs separated by colons or hyphens. A MAC address is unique for every device, permanent, and assigned during manufacturing. It is used for communication within a local network (LAN).

Example: 00:1A:2B:3C:4D:5E

The MAC address has two main parts:

- The first half identifies the manufacturer of the device also called OUI (Organizationally Unique Identifier).
- The second half uniquely identifies the specific device.

b) IP address

An IP address is a logical address assigned to a device to identify it on a network and enable communication across different networks. It can change depending on whether it is assigned manually or automatically. IP addresses are used by routers to route data between different networks, ensuring that information reaches the correct destination. There are two main types of IP addresses namely IPv4 and IPv6 which has a longer format.

IPv4 addresses are divided into five classes based on their range and usage summarized in the table below:

Class	Network size/ usage	IP address range	Default subnet mask
A	Large networks	0.0.0.0 – 127.255.255.255	255.0.0.0
B	Medium networks	128.0.0.0 – 191.255.255.255	255.255.0.0
C	Small networks	192.0.0.0 – 223.255.255.255	255.255.255.0
D	Multicast	224.0.0.0 – 239.255.255.255	N/A
E	Experimental	240.0.0.0 – 255.255.255.255	N/A

Table 6. 3. Classes of IP addresses

6.4.2 Subnet mask and default gate way

Beside having IP addresses, for device communication in a computer network, computers need subnet mask and the default gateway. These help devices determine whether to communicate within the same network or send data to another network.

a) Subnet mask

A subnet mask is a number used together with an IP address to identify the network part and the host part of that address. It helps a device know whether another device is in the same network or a different one.

- **Example:**

Consider an IP address 192.168.1.10 and the subnet mask 255.255.255.0. The subnet mask indicates that "192.168.1" represents the network portion of the address, while ".10" identifies the specific host or device on that network.

- **What is subnetting**

Subnetting is used in computer networking to divide a larger network into smaller, more manageable pieces called subnets. This is primarily done to improve network efficiency, security, and organization.

- **Key concepts in subnetting**

Concept	explanation
IP address	A unique identifier assigned to a device on a network
Subnet mask	Determines which part of the IP address is network and host
Network address	The first address in a subnet (identifies the subnet)
Host address	Addresses assigned to devices
Broadcast address	The last address in a subnet used to send data to all devices

- **Steps in subnetting**

Step 1. Identify the IP Address Class and determine whether the IP is Class A, B, or C.

Step 2. Determine the number of subnets needed and decide how many sub-networks are required.

Step 3. Borrow bits from host portion and take some bits from the host part to create subnets.

Step 4. Calculate the new subnet mask and add borrowed bits to the default subnet mask.

Step 5. Determine the number of subnets by using the formula "number of subnets = 2^n " where n is borrowed bits.

Step 6. Determine the number of hosts per subnet using the formula "hosts per subnet = $2^h - 2$ " where h is the remaining host bits.

Step 7: Determine the subnet ranges by finding the network address, the first usable host, the last usable host and broadcast address.

Example of subnetting:

Given the network address 192.168.1.0/24 which is to be divided into 4 subnets.

Solution:

This is a class C network to be divided into smaller pieces. You begin with a standard prefix length of /24. Because the objective requires creating 4 subnets, you must determine how many host bits to convert into network bits by using the formula $2^n = 4$. Solving this shows that you need to borrow exactly 2 bits (2^2), which changes your network prefix length from /24 to a new CIDR notation of /26 and changes the subnet mask to 255.255.255.192.

Now calculate the number of usable devices available for each subnetwork based on the 6 remaining host bits. By applying the standard host formula $2^6 - 2$, you find that each of the four subnets can accommodate exactly 62 usable hosts, as two addresses are automatically subtracted to serve as the network ID and the broadcast address.

Common subnet masks

Subnet mask	CIDR notation	Number of hosts
255.0.0.0	/8	Large networks
255.255.0.0	/16	Medium networks
255.255.255.0	/24	Small networks

Table 6. 4. CIDR notation of subnets

b) Default gateway

A default gateway is the device, usually a router, that connects a local network to other networks, including the internet. It acts as an exit point for data leaving the network.

Example: IP Address: 192.168.1.2; subnet mask: 255.255.255.0; default gateway: 192.168.1.1

In this case:

- The device communicates directly with devices in the same network (192.168.1.x).
- If it wants to communicate with another network, it sends data to 192.168.1.1 (router).
- If destination is outside such as 192.168.1.0 then send to 192.168.1.1 (router)

c) Configuration of devices in Packet Tracer

Packet Tracer is a network simulation tool where you can visually create and test network topologies without real devices.

For this case select devices that you might use namely PCs/ Laptops end devices where users connect, switches for LAN connectivity, routers for connecting multiple networks or the internet, access points for wireless connections and cables to connect devices physically.

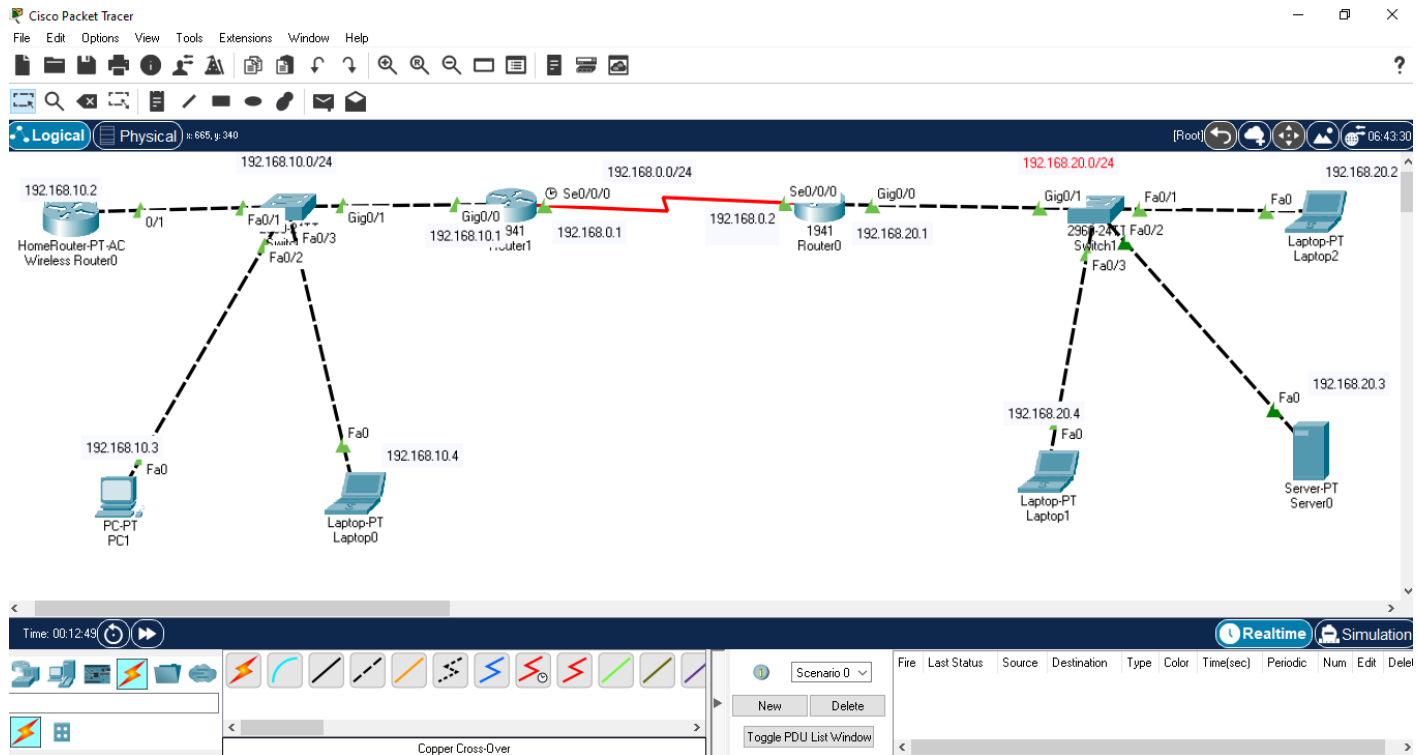
• Choose cables

Choose the right cable to connect different devices in Packet Tracer. Note that Packet Tracer auto-selects cables if you use the “Automatically Choose Cable” option.

Connection type	Cable type in Packet Tracer	When to use
PC to Switch	Copper Straight-Through	Standard LAN connection
Switch to Switch	Copper Cross-Over	To connect two switches directly
Router to Switch	Copper Straight-Through	Router LAN interface to switch
Router to Router	Copper Cross-Over	Router interconnection
PC to PC	Copper Cross-Over	Direct PC-to-PC connection

• Connect devices

The topology is divided into three distinct subnets:



In the Packet Tracer above there is:

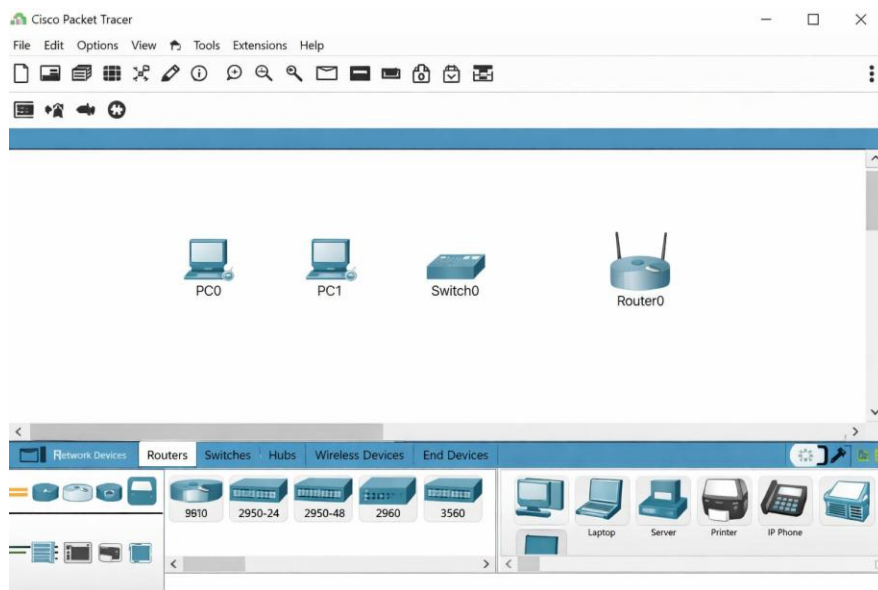
- The left LAN (192.168.10.0/24) consists of end devices like a PC and a laptop connected to a central switch, which also links to a wireless router and the default gateway interface (Gig0/0 at 192.168.10.1) on Router1.
- The WAN link (192.168.0.0/24) is a point-to-point serial connection (Se0/0/0) that bridges Router1 (192.168.0.1) and Router0 (192.168.0.2) to enable routing between the two separate local networks.
- Right LAN (192.168.20.0/24) which features two laptops and a local server connected via a switch to the default gateway interface (Gig0/0 at 192.168.20.1) on Router0.

6.4.3 Configuration of devices in Packet Tracer

This is the process of setting up network devices such as PCs, routers, and switches within the simulation environment by assigning network parameters so that they can communicate with each other. Follow these steps to configure the network:

Step 1. Add devices to the workspace

Open Packet Tracer and drag and drop devices such as PCs, switches and routers.



Step 2. Connect the devices

First select the Connections (cable) tool then choose the appropriate type of cable depending on the devices you want to connect. Use a Copper Straight-Through cable when connecting different devices, such as a PC to a switch, and use a cross-Over cable when connecting similar devices. After selecting the correct cable, connect the devices by clicking on the required ports on each device.

Step 3. Configure IP address on a PC

Click on the PC, then open the Desktop tab and select IP Configuration. In the configuration window, enter the required network details, including the IP Address (for example, 192.168.1.2), the Subnet Mask (for example, 255.255.255.0), and the Default Gateway (for example, 192.168.1.1).

Step 4. Configure the router

Click on the Router and open the CLI (Command Line Interface) tab then enter the commands needed to access configuration mode and set up the interface. Start by typing enable, followed by configure terminal. Next, select the required interface by entering interface fast ethernet 0/0, then assign the IP address and subnet mask using IP address 192.168.1.1 255.255.255.0. After that, activate the interface with the “no shutdown command”.

Step 5. Configure the switch

Click on the Switch and open the CLI tab then enter the required commands to access configuration mode and set the switch name. Start by typing enable, followed by configure terminal, and then enter hostname Switch1 to assign the name Switch1 to the switch.

Step 6. Save configuration

Save the device configuration so that the changes are kept even after restarting the device by typing the command “write memory” in the CLI.

Step 7: Test connectivity

Click on a PC, go to the Desktop tab and open Command Prompt. In the command window, type the IP address (for example ping 192.168.1.1) to test communication with the router, or ping another PC on the network. If replies are received, it means the network configuration has been done successfully.

Application activity 6.4

Section A: Choose the correct answer

- 1) What is the main purpose of a subnet mask in networking?
 - a) To connect to the internet
 - b) To identify the manufacturer of a device
 - c) To separate network and host parts of an IP address
 - d) To encrypt data
- 2) What is the default gateway used for?
 - a) To store IP addresses
 - b) To assign MAC addresses
 - c) To connect devices within the same LAN only
 - d) To send data to other networks outside the local network
- 3) Which device is responsible for assigning MAC addresses?
 - a) Router
 - b) Switch
 - c) NIC
 - d) Modem
- 4) Which Packet Tracer cable is used to connect a PC to a switch?
 - a) Cross-over cable
 - b) Fiber optic cable
 - c) Copper straight-through cable
 - d) Serial cable
- 5) What happens when you use the ping command in Packet Tracer?
 - a) Deletes network configuration
 - b) Tests connectivity between devices
 - c) Assigns IP addresses automatically
 - d) Configures routers

Section B

- 1) Draw a simple network diagram showing how you would connect the 3 PCs, switch, router, and wireless access point.
- 2) Assign IP addresses, subnet masks, and default gateways for all PCs and the router.
- 3) Explain the role of the switch, router, and access point in this network.
- 4) Describe how you would test connectivity between the devices and ensure Internet access.

6.5 Network troubleshooting

Learning activity 6.5

As a network technician you are asked to manage a small office network. The network consists of 3 PCs (PC1, PC2, PC3), 1 switch, 1 router connected to the internet via a modem, 1 wireless access point

After setting up the network, a report is showing that some PCs cannot access the internet, while others cannot communicate with each other.

- 1) How can you check if a computers are communicating to each other.
- 2) Which devices should be checked first when some PCs cannot access the internet?
- 3) How can you check whether a PC is communicating with the router?
- 4) What could happen if a PC has the wrong IP address?
- 5) How can an incorrect default gateway affect internet access?

- 6) What should you do if the ping command shows Request timed out?
- 7) Why is testing communication between devices an important step in network troubleshooting?

Network troubleshooting is the systematic process of detecting and resolving issues in a computer network. It involves checking hardware, software, and configuration settings to ensure that all devices can communicate effectively.

6.5.1 Testing connectivity

In computer networks, it is important to verify whether devices can communicate with each other. One of the simplest and most commonly used tools for testing network connectivity is the ping command. It helps users check if a device is reachable over the network.

a) How Ping works

When a user runs the ping command:

- A request message is sent to the target device
- The target device replies with a response message
- The system displays whether the request was successful or failed. If replies are received, the network is working. If not, there may be a problem.

b) Syntax of Ping command

The ping command is written followed by an IP address “ping [IP address]”

Example: ping 192.168.1.3

The ping command in the example above checks if the device with IP address **192.168.1.3** is reachable and gives a reply such as “Reply from 192.168.1.3: bytes=32 time<1ms TTL=128” meaning that the connection is working correctly.

However, if the reply of the ping command is “Request timed out” that means that the device is not reachable or there is a network problem.

6.5.3 Verifying connection

Verifying connection is the process of checking whether network devices are correctly connected and able to communicate with each other. It involves using tools and methods to confirm that data can be successfully transmitted across the network.

Methods of verifying network connection:

There are several methods used to verify network connections.

- Using the ping command tests whether a device can reach another device on the network.

- Checking physical connections, such as ensuring cables are properly connected and devices are powered on.
- Observing indicator lights (LEDs) on network devices to confirm connectivity. Depending on the device, LED are green or blue indicating that network connection is working correctly while a red may indicated that the network device has a error.

6.5.4 Correcting errors

Correcting errors is the process of fixing problems detected during network troubleshooting. These problems may be related to hardware, software, or configuration settings.

After identifying and verifying problems in a network, the next important step is correcting errors. This involves fixing any issues that prevent devices from communicating properly. Correcting errors ensures that the network works efficiently and reliably.

a) Common network errors

Here are the most common errors in a computer network:

- One common error is incorrect IP address configuration, where devices are assigned wrong or duplicate IP addresses.
- Another issue is wrong subnet mask or default gateway, which prevents communication between devices.
- Loose or damaged cables can also cause connectivity problems.
- Devices may be incorrectly connected or turned off.
- Misconfiguration of routers or switches may also lead to network failure.

Correcting errors in Packet Tracer

In Packet Tracer, errors can be corrected in the following ways.

- If the connection line is red, reconnect the cable or choose the correct cable type.
- If IP addresses are incorrect, open the PC settings and enter the correct values.
- If devices cannot communicate, ensure they are in the same network or properly connected through a router.
- Use Simulation Mode to identify where packets are failing and fix the issue accordingly.

Application activity 6.5

Section A: Choose the correct answer

1) What is network troubleshooting?

- | | |
|---------------------------------|--|
| a) Increasing internet speed | b) The process of detecting and solving network problems |
| c) Installing operating systems | d) Designing a new network |

- 2) What is the main purpose of the ping command?
- a) To delete network settings
 - b) To test network connectivity between devices
 - c) To install network drivers
 - d) To assign IP addresses
- 3) What does “Request timed out” in a ping result mean?
- a) The connection is successful
 - b) The device is printing data
 - c) The device is not reachable or there is a network problem
 - d) The IP address is correct
- 4) Which of the following is a method of verifying network connection?
- a) Formatting the computer
 - b) Checking cables and using ping command
 - c) Installing antivirus only
 - d) Changing the keyboard
- 5) What is a common cause of network errors?
- a) Correct IP configuration
 - b) Properly connected cables
 - c) Incorrect IP address or subnet mask
 - d) Using switches in the network

Section B: Answer the following questions

- 1) How can you verify that the physical connections between devices are correct?
- 2) What steps would you take to correct an incorrect IP address assigned to a PC?
- 3) Explain how the subnet mask and default gateway affect network communication.
- 4) If the router is misconfigured, how would it affect the network?
- 5) After correcting errors, how would you confirm that the network is fully operational?

6.6. End unit assessment

6.6.1. Project activity

Project title: Designing and configuring a small school computer network using Cisco Packet Tracer

Guidance for the project

- Create a simple school network using Cisco Packet Tracer.
- Add at least: 3 PCs, 1 Switch, 1 Router, 1 Wireless Access Point and 1 Server
- Connect the devices using the correct cable types.
- Use Copper Straight-Through cables to connect PCs and switches.
- Configure IP addresses for all devices.
- Assign subnet masks correctly.
- Configure the default gateway on each PC.
- Configure the router interface using CLI commands.
- Test the network using the ping command.
- Ensure all wired and wireless devices can communicate successfully.
- Save the Packet Tracer project file after completing the configuration.

- Work collaboratively and follow network safety and accuracy practices.

Questions on the project

- 1) What type of network connection was used in your project, and why?
- 2) Which network devices were used in the project and what is the role of each device?
- 3) Explain how IP addresses, subnet masks, and default gateways were configured in the network.
- 4) How did you test network connectivity between devices in Packet Tracer?
- 5) What problems did you face during the network setup, and how did you solve them?

6.6.2: Assessment questions

Section A: Choose the correct answer

1) Which networking device operates at the Data Link Layer and uses MAC addresses to forward data only to the intended device on a network?

- a) Hub b) Router c) Switch d) Modem

2) A school network connects all computers within a single building. Which type of network is this?

- a) WAN b) MAN c) LAN d) PAN

3) Given an IP address of 192.168.1.10 and a subnet mask of 255.255.255.0, what is the network part of this address?

- a) 192.168.1 b) 192.168 c) 192 d) 10

Section B: Answer the following questions

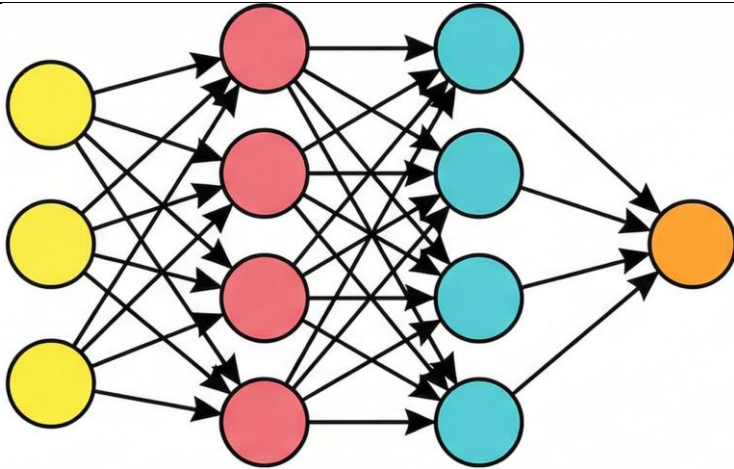
- 1) List two types of network connections.
- 2) Identify any four network devices.
- 3) State the difference between a MAC address and an IP address.
- 4) Explain the importance of computer networking in education.
- 5) Describe the difference between wired and wireless connections.
- 6) Explain how a switch differs from a hub.
- 7) Describe the role of a default gateway in a network.
- 8) Explain how the ping command helps in network troubleshooting.
- 9) A school laboratory has 4 computers connected using Ethernet cables:
 - a) Identify the type of network connection used.
 - b) Suggest one advantage of this connection.
- 10) Identify the network part and host part in the address with an IP address 192.168.1.5 and subnet mask of 255.255.255.0

- 11) A company wants a reliable and secure network. Evaluate whether they should use a wired or wireless network. Justify your answer.
- 12) A student configures a network but cannot access the internet. Evaluate three possible causes of this problem.
- 13) Assess the importance of assigning unique IP addresses to devices in a network.
- 14) Draw and design a simple network diagram that includes 3 PCs, 1 switch, 1 router and 1 access point

EXPERIMENTAL VERSION

EXPERIMENTAL VERSION

UNIT 7. ARTIFICIAL NEURAL NETWORK



Key unit competence: Explain the structure, functioning, and applications of Artificial Neural Networks

Knowledge and understanding

- Explain the fundamental concepts of Artificial Neural Networks (ANN), including their architecture, types, and training processes.
- Describe the importance and applications of ANN in Artificial Intelligence, including the role of transfer learning.
- Explain how bias can arise in neural networks and discuss its ethical implications in

Skills

- Design and build simple neural network models, including visualization of their structure.
 - Train and evaluate neural networks using Python libraries, analyzing their performance and structure.
 - Apply ANN models to solve practical problems such as image or speech recognition,
- Attitudes and values**
- Appreciate the role of Artificial Neural Networks in solving real-world problems and show curiosity in exploring AI concepts.
 - Demonstrate ethical responsibility in AI, including transparency, fairness, and ethical reasoning in neural network decisions.
 - Exhibit teamwork and collaboration skills when coding, analyzing, and solving problems using ANN models.

7.0 Introduction to neural networks

Artificial Neural Networks (ANNs) are a key concept in Artificial Intelligence inspired by the way the human brain works. Just as the brain is made up of interconnected neurons that process and transmit information, neural networks consist of layers of connected nodes also called artificial neurons, that work together to analyze data and make decisions.

Introductory activity

Eric uses his smartphone every day to unlock it using face recognition. The phone quickly identifies his face and unlocks automatically. Later, Eric uses a voice assistant to search for information, and the phone understands his speech correctly. Eric becomes curious and asks himself how a phone can recognize his face and understand his voice? He also asks himself if a computer can learn like a human brain?

During the ICT lesson, the teacher explains that these smart systems use Artificial Neural Networks which are computer models designed to learn patterns from data in a way similar to how the human brain works.

- 1) Which technology enables the smartphone to recognize Eric's face?
- 2) How can a computer learn to recognize different voices?
- 3) In what way is the human brain similar to Artificial Neural Networks?
- 4) Why are Artificial Neural Networks important in AI?

7.1 Introduction to Artificial Neural Networks

Learning Activity 7.1

A school installs a smart attendance system that uses face recognition. Each morning, students stand in front of a camera, and the system automatically marks attendance.

Eric is recognized quickly, but when a new student tries, the system fails to recognize him. After a few days, the system starts recognizing the new student correctly.

- 1) How does the system recognize Eric?
- 2) Why does it fail to recognize the new student at first?
- 3) What must happen for the system to learn the new student's face?
- 4) What does this tell you about how ANNs learn?

7.1.1 Concepts in ANN

An Artificial Neural Network (ANN) is an information processing model that is inspired by the way biological nervous systems, such as the human brain, process information. It is composed of a large number of highly interconnected processing elements (neurons) working together to solve specific problems. Like in humans, ANNs learn by example.

An ANN is configured for a specific application, such as pattern recognition or data classification, through a learning process. Learning in biological systems involves adjustments to the synaptic connections that exist between the neurons. This is the same for ANNs.

Here below are the most common terms used:

Key Terms	Meaning
ANN	Artificial Neural Network a computational model inspired by the human brain, made of connected artificial neurons.
Neuron	The basic processing unit in a neural network, modelled on a biological nerve cell.
Weight	A value that controls how much influence an input has on a neuron's output like synapse strength.
Synapse	The connection point between two biological neurons where signal strength is adjusted.
Training	The process of feeding examples to an ANN so it can learn and adjust its weights.
Pattern recognition	The ability of an ANN to identify patterns in data such as recognizing faces or handwriting.
Data classification	Sorting data into categories such as classifying emails as spam or not spam.

7.1.2 Importance of ANN in AI

Neural networks are important in AI because they provide a powerful way to model complex, non-linear relationships in data. Their significance lies in different key areas:

- **Pattern recognition and generalization:**

ANNs can identify patterns and make generalizations from noisy, complex, or incomplete data. They can learn to recognize images, sounds, and text with a high degree of accuracy.

- **Learning and adaptability:**

Unlike traditional algorithms that follow rigid instructions, ANN learn from data. They can adapt to new situations and improve their performance over time as they are exposed to more examples.

- **Modeling non linearity:**

The real world is fundamentally non-linear. ANNs, through their activation functions, can model these relationships, making them suitable for tasks like stock market prediction, speech recognition, and medical diagnosis.

- **Parallel processing:**

The structure of an ANN allows for parallel processing of information, enabling fast computation, especially when implemented on specialized hardware like GPUs.

7.1.2. Relationship between AI, Machine Learning, ANN and Deep Learning

Artificial Intelligence: AI is the general field of making computers perform tasks that normally require human intelligence, such as reasoning, problem solving, understanding language,

recognizing images, making decisions, etc. Therefore in this relationship AI can be considered the largest circle.

Machine Learning: This is a branch of AI in which computers learn from data instead of being programmed with every rule step by step. Instead of telling the computer exactly what to do in all situations, we give it examples or data, and it finds patterns by itself. Therefore Machine Learning is part of AI.

Artificial Neural Network: ANN is a method used in Machine Learning inspired by the way the human brain works. It is made up of connected processing units called neurons, which work together to process information. ANNs are used to detect patterns, classify information, and make predictions.

Deep Learning: This is a special type of ANN which is a more advanced form of machine learning that uses artificial neural networks with many layers.

Therefore Deep Learning is a specialized form of ANN, and ANN is within Machine Learning, which is within AI.

Application activity 7.1

Section A: Choose the correct answer

1) What is an Artificial Neural Network inspired by?

- a) Computer hardware systems
- b) Internet communication systems
- c) The human brain and biological nervous system
- d) Database systems

2) What is the basic processing unit in an ANN called?

- a) Node
- b) Synapse
- c) Neuron
- d) Layer

3) What does the weight in an ANN represent?

- a) The speed of the computer
- b) The size of the dataset
- c) The influence of an input on the output
- d) The number of neurons in the network

4) Which of the following is a key importance of ANNs in AI?

- a) They only store data
- b) They cannot learn from examples
- c) They can recognize patterns and learn from data
- d) They replace all programming languages

Section B: Answer the following questions

- 1) What is the role of neurons in ANN?
- 2) Compare ANN with traditional computer programs.
- 3) A hospital uses a computer system to help doctors detect diseases from patient data
 - a) Identify the input, processing, and output
 - b) Explain how the system uses ANN to make decisions

4) Give two real-life examples of pattern recognition powered by ANNs.

7.2 Network architecture

Learning activity 7.2

A school develops a smart grading system that predicts whether a student will pass or fail based on study hours, attendance and assignment scores. The system processes this data and gives a final prediction of Pass or Fail.

- 1) What are the three pieces of student data used in the system?
- 2) What do you think the output of the system will be?
- 3) If the system predicts Pass or Fail, how many possible outputs are there?
- 4) Why do you think a computer needs student data before making a prediction?
- 5) If a student studies more hours and has good attendance, what result might the system predict?
- 6) If each input is represented by one neuron, how many neurons will be in the input layer?
- 7) How is a neural network similar to the way a human learns from mistakes?

7.2.1 Input, hidden, and output layers

A typical ANN is organized in three layers which are the input layer, hidden layer and output layer.

a) **Input layer**

This is the first layer of the network. It receives the raw input data, analogous to the dendrites in a biological neuron. Each neuron in the input layer represents a feature from the dataset. The number of neurons in this layer equals the number of features in the input data.

b) **Hidden layer**

These layers are the processing engine of the network. They consist of neurons that perform computations on the inputs received from the previous layer. The hidden nature refers to the fact that their desired outputs are not directly observable in the training data. A network can have one or many hidden layers.

c) **Output layer**

This is the final layer of the network. It produces the final result or prediction. The number of neurons in the output layer depends on the problem. For binary classification, one neuron can represent the probability of the positive class. For multi-class classification, there is typically one neuron per class.

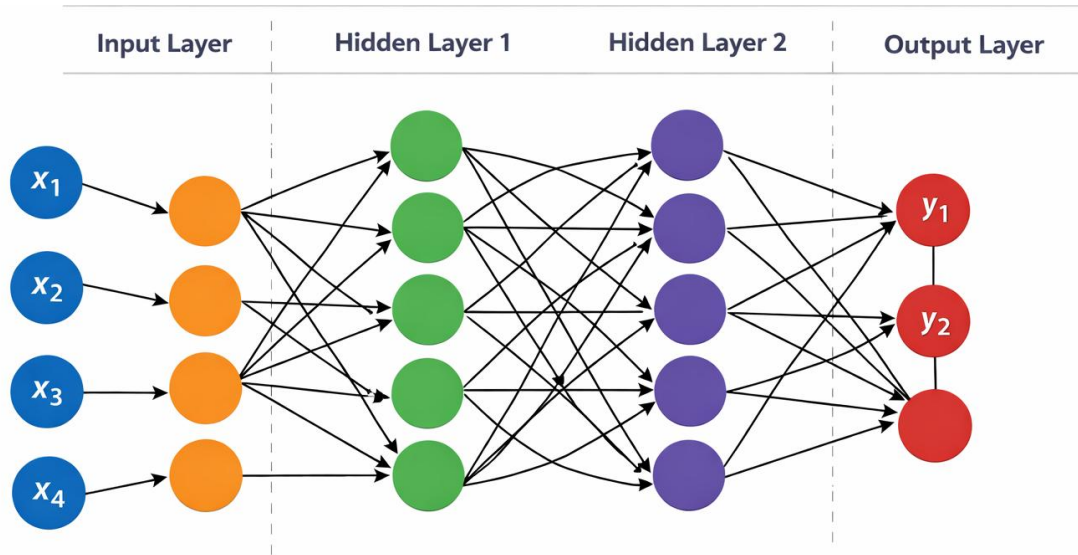


Figure 7. 1. A fully connected neural network with one input layer (4 neurons), two hidden layers (5 neurons each), and an output layer (3 neurons).

7.2.2 Training of Neural Networks

Training a neural network is the process of finding the optimal set of weights and biases that minimize the error between the network's predictions and the actual target values. This is done through a process called backpropagation, which is often combined with an optimization algorithm like gradient descent.

The training process can be summarized in these steps:

- b. **Forward propagation:** An input vector is fed into the network. It travels through the network, layer by layer, as each neuron calculates its weighted sum and applies an activation function. The process ends with a final output from the output layer.
- c. **Error calculation:** A loss function is used to calculate the error, which is the difference between the network's predicted output and the actual target output.
- d. **Backward propagation:** The error is then propagated backward through the network. Using calculus (the chain rule), the algorithm calculates the gradient of the loss function with respect to each weight and bias in the network. This gradient indicates how much each weight contributed to the total error.
- e. **Weight and bias update:** An optimization algorithm, most commonly Gradient Descent, uses these gradients to update the weights and biases in the direction that reduces the error. The size of the update is controlled by a hyperparameter called the learning rate.

This process of “forward pass, error calculation, backpropagation, and weight update”) is repeated for many iterations (epochs) over the training dataset until the network's performance is satisfactory.

Application activity 7.2

Section A: Choose the correct answer

- 1) What is the main role of the input layer in an ANN?
a) Produces final predictions c) Receives raw input data
b) Performs backpropagation d) Updates weights
- 2) Which layer is responsible for performing most of the computations in a neural network?
a) Input layer b) Output layer c) Hidden layer d) Data layer
- 3) The output layer in an ANN is mainly used to:
a) Store training data c) Produce final predictions
b) Perform feature extraction d) Initialize weights
- 4) What is the main purpose of backpropagation in neural networks?
a) To store input data c) To update weights by reducing error
b) To calculate network errors only d) To create new datasets

Section B: Answer the following questions

- 1) Explain how data flows through a neural network.
- 2) What would happen if a neural network had no hidden layer?
- 3) How does increasing hidden layers affect performance?
- 4) Why is backpropagation important in improving accuracy?
- 5) How does the learning rate affect training?

7.3 Types of ANN

Learning activity: 7.3

An agricultural technology company aiming to support farmers in the community, is building three smart systems: a Crop Identification System that uses photos of crops to identify plant types by detecting shapes, colors, and patterns, a Disease Detection System that analyzes leaf images to detect plant diseases and a Yield Prediction System that predicts crop yield based on factors such as rainfall, soil quality, temperature, and fertilizer.

- 1) Which of the three smart systems will use actual images or photos as inputs?
- 2) What specific data inputs will the Yield Prediction System use instead of images?
- 3) The Yield Prediction System needs to look at multiple hidden layers to process factors like rainfall, temperature, and soil quality together. What specific type of Artificial Neural Network (ANN) is it using?
- 4) What does the word "deep" refer to in the term "Deep Neural Network"?
- 5) CNNs are designed for processing structured grid data like images. Based on this, which two systems from the scenario match this description?
- 6) Name the three key components of a CNN mentioned in the text that will help the Crop Identification System detect shapes, colors, and patterns.

7.3.1 Deep Neural Networks

A Deep Neural Network (DNN) is an ANN with multiple hidden layers. The "deep" in deep learning refers to the depth of the network, which allows it to learn hierarchical representations of data. Early layers might learn simple features such as edges in an image, while deeper layers combine these simple features to recognize complex patterns such as a face.

On the other hand, a Shallow Neural Network has these characteristics:

- **Input layer:** Receives the raw data and does not perform any mathematical operations.
- **Single hidden layer:** The neurons in this layer apply weights to the input data, add a bias, and pass the result through an activation function to extract patterns.
- **Output layer:** Combines the features processed by the hidden layer to produce the final prediction or classification.

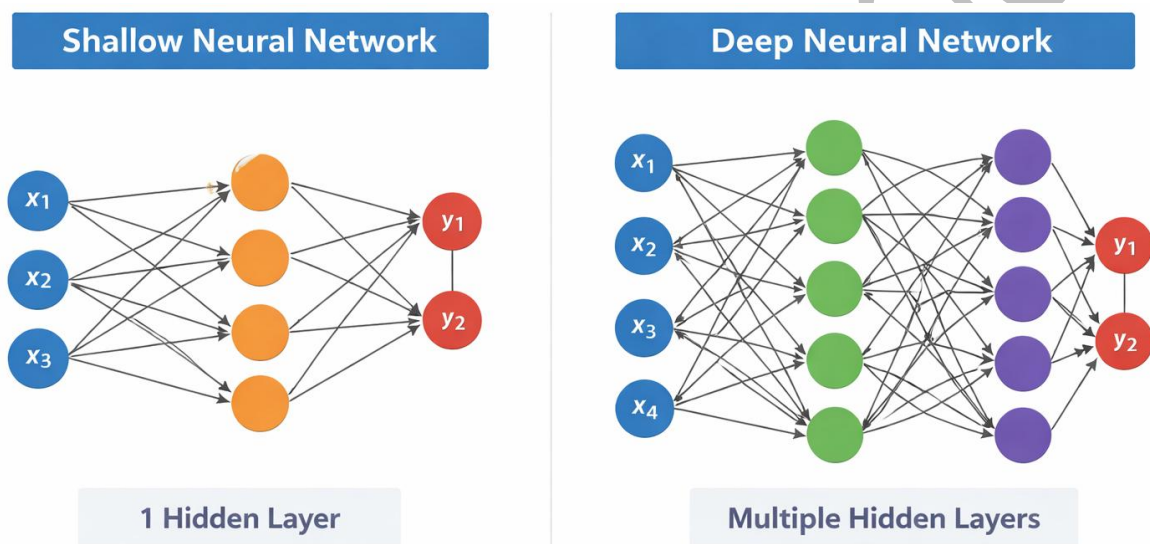


Figure 7. 2. Deep Neural Networks have multiple hidden layers, enabling them to learn complex, hierarchical features

7.3.2 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a specialized type of ANN designed for processing structured grid data, such as images. Their architecture is based on the mathematical operation of convolution.

Its key components include:

- **Convolutional layers:** These layers apply filters to the input to create feature maps. These filters learn to detect spatial features like edges, textures, and shapes.
- **Pooling layers:** These layers down sample the feature maps, reducing their spatial dimensions. This reduces the number of parameters and computational load, while also making the network more robust to variations in the position of features.

- **Fully connected layers:** At the end of the network, after the convolution and pooling layers have extracted high-level features, fully connected layers are used to perform the final classification or regression task.

The image below illustrates how a CNN architecture can be used for image classification:

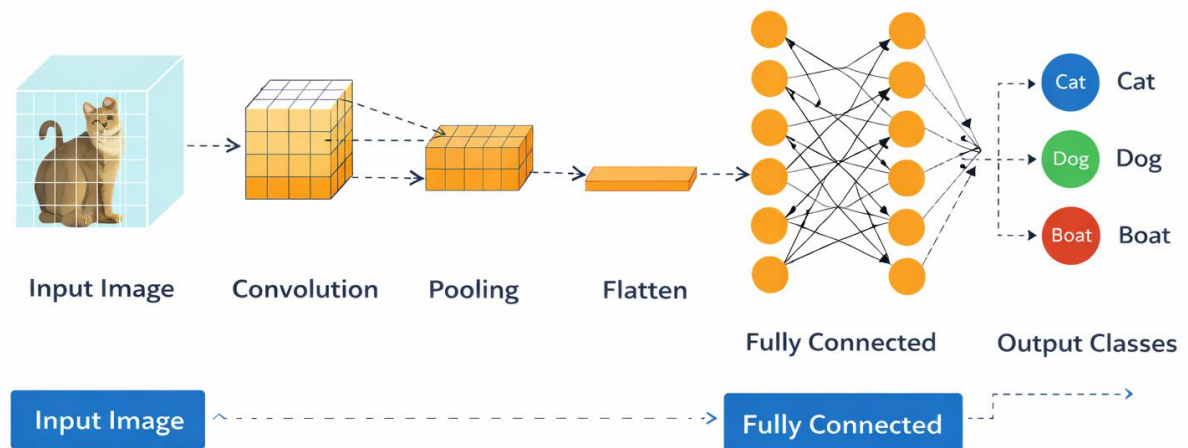


Figure 7. 3. Illustration of CNN use in image classification

7.3.4 Transfer learning in ANN

Transfer learning is a machine learning technique where a model that has already learned one task is reused and adapted for a new and related task. Instead of training a new model, you start with a pre-trained model and adapt it to your specific dataset. This saves time and requires less data.

Steps for transfer learning:

- **Choose a pre-trained model:** This model should be related to the new task you want to solve.
- **Prepare the new dataset:** Collect and organize the data for the new task. The dataset should be labeled according to the new classes or outputs required. The new data should consist of training data, validation data and test data sets so that the model can be trained and evaluated properly.
- **Load the pre-trained model:** The pre-trained model is loaded together with the knowledge it has already learned from the original task.
- **Modify the output layer:** The original output layer of the pre-trained model is then replaced with a new one that matches the requirements of the new task. For example, if the original model was trained to classify 1000 categories but the new task only has 4 categories, the final layer must be changed so that it produces 4 outputs only.
- **Freeze the earlier layers:** In transfer learning, the first layers of the model are often kept unchanged during the first training stage. This makes the model keep the useful knowledge it already learned such as basic patterns and features. Keeping these layers unchanged saves

training time and helps the model avoid losing important information from the original training.

- **Train the new layers:** After keeping the old layers unchanged, the new output layer and the other open layers are trained using the new data. In this step, the model learns the specific patterns and classes needed for the new task while still using the general knowledge it already had from the pre-trained model.
- **Evaluate the model:** After training, the model is tested using validation or test data to see how well it works. This helps to check whether the model has learned the new task correctly.

Application activity 7.3

Section A: Choose the correct answer

1) What defines a Deep Neural Network?

- a) A network with no hidden layers
- b) A network with a single input layer only
- c) A neural network with multiple hidden layers
- d) A network that does not learn

2) What is the main purpose of convolutional layers in CNNs?

- a) To store data permanently
- b) To detect features such as edges and shapes
- c) To calculate database queries
- d) To reduce learning rate

3) What is the role of pooling layers in CNNs?

- a) Increase image size
- b) Store weights permanently
- c) Reduce feature map size and computation
- d) Replace convolution layers

4) What is the main advantage of transfer learning?

- a) It requires more data than training from scratch
- b) It only works for small models
- c) It allows reuse of pre-trained models to save time and data
- d) It removes the need for neural networks

Section B: Answer the following questions

- 1) Differentiate between a shallow neural network and a deep neural network.
- 2) Give an example of how early layers and deeper layers work together in image recognition.
- 3) Why are deeper networks more powerful than shallow networks?
- 4) How does a convolutional layer extract features from an image?
- 5) What is the purpose of pooling layers?

7.4 Applications of ANN

Learning activity 7.4

A school prepared a graduation ceremony in which many pictures were taken and many people took speeches.

The school secretary now has a task to sort the pictures and arrange those that have a certain relationship. The secretary must also rename the speech recordings and take summaries of what was

said. The pictures are too many to sort manually and the recordings are too many to listen to one by one. This makes the work slow and tiring.

To save time, the school wants to use a smart computer system that can recognize and classify images by identifying what appears in a photo, detect faces or objects in an image and recognize spoken words in audio recordings and convert them into text.

- 1) What types of input data are given to the computer system in this scenario?
- 2) What output is expected when the system processes a voice recording?
- 3) What output is expected when the system processes a photo of people?
- 4) If the system locates a person's face in a group photo, what is it doing?
- 5) How does speech recognition reduce the secretary's work when working with many audio recordings?
- 6) How does image recognition reduce the teacher's work when handling many photos?
- 7) Which is more likely to use spoken language as input: speech recognition or face recognition?

7.4.1 Image recognition

Image recognition is an application of Artificial Neural Networks in which a computer system learns to identify, classify, or recognize objects, people, animals, scenes, or other features in digital images.

ANN are trained using many examples of images which then make it learn patterns such as shapes, colors, edges, and textures which will be applied when a new image is presented, the ANN analyzes these features and predicts what the image contains.

Here are different types of image recognition:

- **Image classification:** Assigning a label to an entire image such as cat or dog.
- **Object detection:** Locating and classifying objects within an image.
- **Image segmentation:** Classifying each pixel in an image to a specific object or region.
- **Face recognition:** Identifying or verifying a person from an image.

7.4.2 Speech recognition

Speech recognition is an application of Artificial Neural Networks (ANNs) that allows computers to recognize spoken words and convert them into text. It learns patterns in human speech from audio recordings and uses them to understand what is being said.

Speech recognition is applied in the following areas:

- **Voice assistants:** Programs such as Siri, Google Assistant use speech recognition to understand spoken language and answer questions.
- **Transcription services:** These systems automatically convert spoken words from meetings, interviews, lectures, or speeches into written text.

- **Voice controlled systems:** These are devices or applications that can be operated using voice commands. For example, a user can say "Turn on the lights" and the system will carry out the command.

7.4.3 Sample Python program implementing ANN

This practical example demonstrates building a simple ANN for classifying handwritten digits using the MNIST dataset.

Sample Python Program for a Simple ANN (using Keras/TensorFlow)

1. Import necessary libraries

```
import tensorflow as tf
from tensorflow import keras
import numpy as np
import matplotlib.pyplot as plt
```

2. Load the MNIST dataset (handwritten digits)

This dataset contains 70,000 grayscale images of digits (0-9), each 28x28 pixels.

```
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()
```

3. Preprocess the data

Normalize pixel values from 0-255 to 0-1.

```
x_train = x_train.astype('float32') / 255.0
```

```
x_test = x_test.astype('float32') / 255.0
```

Flatten the 28x28 images into a 1D vector of 784 pixels.

This is necessary for a basic feedforward network.

```
x_train = x_train.reshape(-1, 784)
```

```
x_test = x_test.reshape(-1, 784)
```

4. Define the model architecture

```
model = keras.Sequential([
```

```
    # Input layer: 784 neurons (one for each pixel)
```

```
    # Hidden layer 1: 128 neurons, using ReLU activation function
```

```
    keras.layers.Dense(128, activation='relu', input_shape=(784,)),
```

```
    # Hidden layer 2: 64 neurons, using ReLU activation function
```

```
    keras.layers.Dense(64, activation='relu'),
```

```
    # Output layer: 10 neurons (one for each digit 0-9), using Softmax activation
```

```
    # Softmax converts the outputs into a probability distribution that sums to 1.
```

```
    keras.layers.Dense(10, activation='softmax')
```

```
])
```

5. Compile the model

```
model.compile(optimizer='adam',      # Optimization algorithm
```

```
            loss='sparse_categorical_crossentropy', # Loss function for classification
```

```
            metrics=['accuracy'])    # Metric to track during training
```

6. Display a summary of the model architecture

```
model.summary()
```

```

# 7. Train the model
print("\n--- Training the model ---")
model.fit(x_train, y_train,          # Training data
          epochs=5,                  # Number of times to iterate over the entire dataset
          batch_size=32,             # Number of samples per gradient update
          validation_split=0.2)      # Use 20% of training data for validation

# 8. Evaluate the model on the test dataset
print("\n--- Evaluating the model on test data ---")
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=2)
print(f"\nTest accuracy: {test_acc:.4f}")

# 9. Make a prediction on a single test image
# Let's predict the first image from the test set
prediction = model.predict(x_test[0:1]) # Predict for one sample
predicted_class = np.argmax(prediction) # Get the index with the highest probability
actual_class = y_test[0]

print(f"\n--- Prediction for a sample image ---")
print(f"Predicted digit: {predicted_class}")
print(f"Actual digit: {actual_class}")

# Optional: Display the image
# sample_image = x_test[0].reshape(28,28)
# plt.imshow(sample_image, cmap='gray')
# plt.title(f"Predicted: {predicted_class}, Actual: {actual_class}")
# plt.show()

```

a. Explanation of the code steps:

- **Import libraries:** tensorflow provides the tools.
- **Load dataset:** MNIST is a standard benchmark dataset.
- **Preprocess data:** Normalization speeds up training. Flattening converts 2D image data into a 1D format suitable for a standard feedforward network.
- **Define the model:** Use a sequential model, which stacks layers. Dense layers are fully connected. The input_shape defines the size of the input data. relu (Rectified Linear Unit) is a common activation function for hidden layers. softmax is used for multi-class classification output.
- **Compile model:** We specify the optimizer (adam), the loss function (sparse_categorical_crossentropy), and the metric to track (accuracy).
- **Train Model:** The fit function does the learning. It iterates for 5 epochs over the data in batch_size chunks. validation_split holds back some data to evaluate performance during training.
- **Evaluate the model:** We check the final accuracy on the unseen x_test data to estimate how well the model generalizes.

- **Make prediction:** The model uses its learned knowledge to predict the digit for a new, unseen image.

b. Exercise

A fruit processing company installs an automated system that uses sensors to measure the weight and size of fruits on a conveyor belt. A dataset of different fruits namely apple, banana, orange, grape, and mango is collected and split into training and testing sets. A Decision Tree Classifier is trained to predict the type of fruit based on these features.

- Train the model and evaluate its performance using appropriate metrics.
- Based on your results, interpret how well the model classifies each fruit type
- Explain any misclassifications observed
- Discuss whether the model is suitable for real-world deployment

Answer:

```
# Creating simple data
import numpy as np
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
np.random.seed(42)

# 1. Data - 500 samples (100 per fruit) with realistic noise
data = pd.DataFrame({
    "weight": np.concatenate([
        np.random.normal(180, 8, 100), # Apple
        np.random.normal(120, 8, 100), # Banana
        np.random.normal(130, 8, 100), # Orange
        np.random.normal(50, 8, 100),  # Grape
        np.random.normal(200, 8, 100)  # Mango
    ]),
    "size": np.concatenate([
        np.random.normal(7.5, 0.4, 100), # Apple
        np.random.normal(18, 0.4, 100),  # Banana
        np.random.normal(7.2, 0.4, 100),  # Orange
        np.random.normal(3.0, 0.4, 100),  # Grape
        np.random.normal(12, 0.4, 100)    # Mango
    ])
```

```

]),
"fruit": ([ "Apple" ] * 100 + [ "Banana" ] * 100 +
          [ "Orange" ] * 100 + [ "Grape" ] * 100 +
          [ "Mango" ] * 100)
})

```

c. split data into test and train

2. Split

```

X = data[["weight", "size"]] y = data["fruit"]
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2,
                                                    random_state=42, stratify=y
)

```

d. Train model

3. Train

```

model = DecisionTreeClassifier(max_depth=4, random_state=42) model.fit(X_train, y_train)

```

e. Model evaluation

4. Results

```

y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred)) print(f"Test samples:
{len(y_test)}\n")
print(classification_report(y_test, y_pred))

```

Accuracy: 0.99

Test samples: 100

	precision	recall	f1-score	support
Apple	1.00	0.95	0.97	20
Banana	1.00	1.00	1.00	20
Grape	1.00	1.00	1.00	20
Mango	1.00	1.00	1.00	20
Orange	0.95	1.00	0.98	20
Accuracy			0.99	100
macro avg	0.99	0.99	0.99	100
weighted avg	0.99	0.99	0.99	100

Application activity 7.4

Section B: Choose the correct answer

1) Which of the following is not an application of image recognition?

- a) Face recognition
- b) Object detection
- c) Image segmentation
- d) Text file compression

2) What is the main role of Convolutional Neural Networks in image recognition?

- a) Encrypting images
- b) Detecting features like edges and shapes
- c) Storing images in databases
- d) Converting images into audio

5) What does the SoftMax activation function do in the output layer of the ANN?

- a) Converts outputs into a probability distribution
- b) Removes hidden layers
- c) Increases image resolution
- d) Sorts training data

Section B: Answer the following questions

You are part of a team of young innovators in Rwanda working on a smart agriculture project. Farmers in your community are facing challenges such as crop diseases, theft of produce, and difficulty in monitoring large farms.

To solve these problems, your team designs a smart system using cameras and AI. The system is installed on the farm and performs several tasks:

- It takes images of crops and automatically tells the farmer the type of crop growing in the field.
- It scans plant leaves and highlights diseased spots to help farmers take early action.
- It separates different parts of the image such as soil, healthy leaves, and infected areas.
- It uses a camera at the farm gate to recognize the farmer's face and allow access only to authorized people.

1) Which tasks involve analyzing images?

2) How does AI help the farmer in each case?

3) Explain the difference between the following image recognition tasks using examples from the scenario.

- a) Image classification
- b) Object detection
- c) Image segmentation
- d) Face recognition

4) Explain how ANNs can convert speech to text

7.5 Ethics in Neural Networks

Learning activity 7.5

A technology company develops an AI system that is used in different sectors. In the healthcare sector, the system helps doctors diagnose diseases in hospitals. In the banking sector, it is used to

approve or reject loan applications. In the security sector, the system uses face recognition to allow access to authorized users. It is also used in self-driving cars to make driving decisions.

After deployment, some issues are reported: Some patients receive incorrect diagnoses, certain groups of people are unfairly denied loans, personal data is leaked from the system and it is unclear who is responsible when the system makes mistakes

- 1) Identify at least four ethical issues from the scenario.
- 2) How can bias affect decisions in AI systems?
- 3) What could be done to make the loan approval system fair to all people?
- 4) Why should personal data collected by AI systems be protected?
- 5) Who should be held responsible when a self-driving car controlled by AI causes harm or makes a wrong decision?

Ethics in Neural Networks refers to the principles that guide the responsible development and use of neural network systems. Neural networks are used in areas such as healthcare, banking and security where their decisions can affect people's lives. Because of this, it is important to ensure that these systems are fair, accurate and safe. Ethical issues in neural networks may include bias, lack of privacy, unfair decisions and unclear responsibility when mistakes happen.

- **Bias and fairness:** ANNs trained on biased data can spread and even amplify societal biases related to race, gender, and socioeconomic status. For example, a hiring algorithm trained on historical data might learn to discriminate against certain groups. Solution strategies include careful data curation, algorithm auditing, and fairness-aware modeling.
- **Transparency and explainability:** Many neural networks, it is often impossible to understand why a model made a specific decision. This is a major problem in high-stakes domains like healthcare, finance, and criminal justice. The field of Explainability is developing techniques to make these models more interpretable.
- **Privacy:** Training large models often requires vast amounts of data, which can include sensitive personal information. There is a risk that models can memorize training data, potentially leaking private information. Techniques like differential privacy are being developed to protect data privacy during training.
- **Accountability:** When an autonomous system powered by a neural network causes harm, the developer, the manufacture and the owner should be made accountable.
- **Job displacement:** Automation driven by AI and neural networks has the potential to displace workers in various sectors, from manufacturing to customer service. The society and education systems must adapt to manage these economic shifts.

Security and intentional attacks: This refer to the risks where ANNs are intentionally manipulated to produce incorrect or harmful outputs. Attackers may change or trick the input data so that the model makes wrong decisions, even if it was trained correctly.

Application activity 7.5

Part A. Answer by True or False

- 1) Biased training data can lead to unfair decisions by neural networks.
- 2) Explainability helps users understand how a neural network makes decisions.
- 3) Privacy is not a concern when training neural networks.

Part B. Choose the correct answer

- 4) What is the main purpose of ethics in neural networks?
 - a) To increase processing speed
 - b) To guide the responsible development and use of neural network systems
 - c) To reduce the cost of computers
 - d) To replace human workers entirely
- 5) Which of the following is an example of bias in a neural network?
 - a) A system processing data quickly
 - b) A hiring algorithm discriminating against certain groups
 - c) A model using large datasets
 - d) A network storing information
- 6) Why is transparency important in neural networks?
 - a) It reduces memory usage
 - b) It helps understand how decisions are made
 - c) It increases internet speed
 - d) It eliminates the need for data

Part C. Answer these questions

- 7) What is bias in neural networks?
- 8) Why is explainability important in healthcare and finance?
- 9) A hospital uses a neural network to diagnose patients, but doctors cannot understand how it reaches its conclusions. Which ethical challenge is involved?
- 10) You are developing a neural network for a national healthcare system. Identify the major ethical challenges you would face and explain

End unit assessment:

Section A: Choose the correct answer

- 1) Artificial Neural Networks are inspired by:
 - a) Computer hardware
 - b) Human brain
 - c) Internet
 - d) Database systems
- 2) The first layer in a neural network is called:
 - a) Output layer
 - b) Hidden layer
 - c) Input layer
 - d) Processing layer
- 3) The main purpose of training a neural network is to:
 - a) Increase data size
 - b) Reduce error
 - c) Delete neurons
 - d) Store information
- 4) Which ANN is best for sequential data like speech?
 - a) CNN
 - b) RNN
 - c) DNN
 - d) ANN

Section B: Answer the following questions

- 5) State two differences between ANN and traditional programs.
- 6) Explain the role of the following layers:
 - a) Input layer
 - b) Hidden layer
 - c) Output layer
- 7) A school develops a system to predict student performance based on: Study hours, Attendance and Assignment scores

- a) Identify the inputs
 - b) How many neurons are in the input layer?
 - c) What is the output of the system?
 - d) Explain the role of hidden layers in this system.
- 8) A system is designed to identify crops from images, detect plant diseases, segment parts of images and recognize farmer's face
- a) Identify tasks that involve image recognition.
 - b) Explain how AI helps farmers in this system.

EXPERIMENTAL VERSION

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